Motorsport Hotel Complex

Qiddiya, Kingdom of Saudi Arabia

Facade Performance Specifications

100% Detailed Design Stage - Rev. 0A



QC00912012-SSH-TEC-SPC-FAC-EVS-Z00-00001

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SECTION 05 7313.01 - GLAZED DECORATIVE METAL RAILING

PART 1 - GENERAL

Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.1 SECTION INCLUDES

A. Furnish all labor, materials, tools and equipment and perform all services and operations necessary for the complete supply and installation of pre-engineered, component-based metal and glass railing systems.
   1. Free standing laminated (structural interlayer) safety glass balustrade (glazed railing system) with SS 316 edge handrail; bottom fixed on the continuous U-channel steel bracket assembly.

B. Design to follow the architectural intent indicated in the drawings.

C. This section to be read in conjunction with the related wall types/ exterior wall system drawings submitted as a part of the tender package, not limited to:
   1. The vertically free-standing glass balustrade consists of structural interlayer or SGP laminated glass captured in an extruded alum. profile anchored to the Slab, the perimeter of balustrade is provided with proper sealing by gaskets and sealants.
      Locations: Tower H32, above saw tooth façade and below.
      Building H30, roof floor/ balcony/ terrace Balustrade.
      Building H27, roof floor/ balcony/ terrace Balustrade.

1.2 RELATED SECTIONS

1.	Section 01 4500	-	Quality Control
2.	Section 03 3000	-	Cast-in-Place Concrete
3.	Section 04 2200	-	Concrete Unit Masonry
4.	Section 05 5000	-	Metal Fabrications
5.	Section 07 6200.01	-	Sheet Metal Flashing and Trims
6.	Section 07 2500.01	-	Weather Barrier
7.	Section 07 9200.01	-	Joint Sealants
8.	Section 08 8000.01	-	Glazing

1.3 PERFORMANCE REQUIREMENTS

A. Structural Performance: Provide metal and glass railing system capable of withstanding the following structural loads without exceeding allowable design working stress of materials for handrails, railings, anchors, and connections:
   1. Horizontally uniformly distributed line load to act on the handrail (min. 1100mm above the finished floor level): 1.5kN/m.
   2. A uniformly distributed load applied to the infill: 1.5kN/m².
   3. A point load applied to part of the infill: 1.5kN.

B. For Impact:
   1. A hard body impact test with the hard body impact energy not less than 10Nm.
   2. A soft body impact test to glazed elements with the soft body impact energy not less than 500Nm.

3. A soft body impact test to non-glazed elements with the soft body impact energy not less than 500Nm.

C. Thermal Movements: Provide metal and glass railing that allow for thermal movements resulting from maximum change (range) in ambient and surface temperatures by preventing buckling, opening of joints, overstressing of components, failure of connections, and other detrimental effects. Base engineering calculation on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.
   1. Temperature Change (Range): 70 deg. C, ambient; 100 deg. C, material surfaces.
   2. Night time or lower range +10 deg. C.

D. Control of Corrosion: Prevent galvanic action and other forms of corrosion by insulating metals and other materials from direct contact with incompatible materials.

E. Comply with the specification document Glass Types Overview at its latest revision.
   1. Tower H32: QC00912012-SSH-DRW-GEN-FAC-EV-H32-00003 and QC00912012-SSH-DRW-GEN-FAC-EV-H32-00004.
   2. Building H30: QC00912012-SSH-DRW-GEN-FAC-EV-H30-00003 and QC00912012-SSH-DRW-GEN-FAC-EV-H30-00004.
   3. Building H27: QC00912012-SSH-DRW-GEN-FAC-EV-H27-00003 and QC00912012-SSH-DRW-GEN-FAC-EV-H27-00004.

1.4 DESIGN REQUIREMENTS

A. Design, fabrication and installation of guardrails as detailed on drawings.

B. Each handrail or guard section shall be supported by a minimum of three glass balusters, or shall be otherwise supported to remain in place if one baluster panel fails, and one of the following criteria shall be met.
   1. Provided with hand/ guard rail.
   2. Shall be laminated glass with two or more glass plies of equal thickness and the same glass type.

C. Provide metal and glass railing complying with performance requirements specified, based on testing performed according to ASTM E 894 for anchorage and ASTM E 935 for performance.

D. Meet code requirements for height, size of openings and restrictions of climb-ability.

E. Glass and Glazing: Refer to Section 08 80 00.

1.5 SUBMITTALS:

A. Product Data for metal and glass railing system; with timber handrail.

B. Shop Drawings: Show fabrication and installation of metal and glass railing system. Include plans, elevations, sections, details, and attachments to other Work.

C. Calculations: Submit calculations, under seal and signature of the sub-contractor's professional Engineer, showing the following but not limited to:
   1. Design load overview according to specifications and wind tunnel analysis or project wind analysis report, as available.
   2. Detailed engineering of all structural system members of metal and glass railing system.
   3. Detailed engineering of anchorage hardware, sub structure, brackets, clip angles, washers, and anchor bolts, welds, torque pressures.

4. Materials proposed and their allowable shear and bending stresses.
5. Deflections and expansion and contractions in accordance with all thermal and structural impacts.
6. Comprehensive glass calculation (dead load, live load, thermal load etc.).
7. Prepare calculations in a clear comprehensive manner to permit easy review together with drafts of the unit and location at the project and the applicable specific wind load according to the project wind tunnel analysis report or project wind report as applicable. Incomplete calculations will be rejected un-reviewed.

D. Samples: For each type of exposed finish required, prepared on components indicated below and of same thickness and metal indicated for the Work.
   1. 300 mm long sections of each different linear railing member.
   2. 300 mm square samples of glass.
   3. Fittings, connections and locking devices for glazing.
   4. Accessories.
   5. Assembled Samples of railings, made from full-size components, including rail, and infill. Show method of finishing members at intersections. Samples need not be full height.

E. Qualification Data: For firms and persons specified in “Quality Assurance” Article to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.

F. Product Test Reports: Indicating products comply with requirements, based on comprehensive testing of current products.

1.6 QUALITY ASSURANCE

A. Professional Engineer Qualifications: A professional engineer who is legally qualified to practice where Project is located and who is experienced in providing engineering services for installation that are similar to those indicated for this Project in material, design, and extent.

B. Fabricator: A firm with minimum 10 (Ten) years’ experience in successfully producing work similar to that indicated for this Project, with a record of successful in-service performance, and with sufficient production capacity to produce required units without causing delay in the Work.

C. Installer: An installer with minimum 10 (Ten) years’ experience in the use of the materials and equipment to be employed in the Work. Install by same firm which fabricates the Work.

D. Mockups: Before installing, build mockups to verify selections made under sample Submittals and to demonstrate aesthetic effects and qualities of materials and execution. Build mockups in location as directed to comply with the following requirements, using materials indicated for the completed Work:
   1. Build mockups for each form and finish of railing consisting of railing, infill area, and anchorage system components that are full height and are not less than 1.5 m in length.
   2. Build Mock-up to the satisfaction of the Engineer. If the required standard is not achieved, the Contractor shall replace (at the contractor’s cost) until the required standard is achieved as judged by the Engineer
   3. Demonstrate the proposed range of aesthetic effects and workmanship.
   4. Obtain written approval of mock-up from the Engineer before starting fabrication.
   5. Maintain mock-ups during construction in an undisturbed condition as a standard for judging the completed Work.

6. Approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

E. All components and fittings are furnished by the same manufacturer.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Delivery: Deliver materials to site in manufacturers original packaging, with labels clearly identifying product name and manufacturer.
   1. Deliver materials properly protected against damage to finished surfaces during transit.
   2. Inspect materials upon delivery for damage. Unless minor defects can be made to meet the Architect's specifications and satisfaction, damaged parts should be removed and replaced.

B. Storage: Store metal and glass railing at project site under cover in a dry, well-ventilated, weathertight place in accordance with manufacturer's instructions.

C. Handling: Product materials and finish from damage during handling and installation.

1.8 PROJECT CONDITIONS

A. Field Measurements: Verify dimensions by field measurements before fabrication and indicate measurements on Shop Drawings.

B. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

1.9 COORDINATION

A. Coordinate installation of anchorages. Furnish Setting Drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

B. Schedule installation so that the metal and glass railing systems are mounted only on completed substrate. Do not support temporarily by any means that do not satisfy structural performance requirements.

1.10 WARRANTY

A. General Warranty: The Contractor is to provide a written agreement to indemnify the Employer against any defects in the design, fabrication, workmanship, and quality of materials or performance of the Works included in this Part and to repair or replace defective design, fabrication, workmanship or materials during the warranty period.
   1. Warranty Period: Manufacturer's standard but not less than 10 (Ten) years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS / FABRICATOR

A. Fabricators/Assemblers/System manufacturers:
   1. Technical Glass, Middle East.
   2. Glassline Industries, Middle East.
   3. Alico, Middle East.
   4. Alumco, Middle East.
   5. Al Abbar, Middle East.
   6. Reem Emirates Aluminium LLC, Middle East.
   7. Permasteelisa Gartner, Middle East.

   Or Equivalent, Subjected to Engineer's acceptance.

   Conditions to qualify "equivalent": Qualification exercise consists of assessment of technical adaptability of the design and the material, quality procedures maintained, minimum 10 (Ten) years of its usage in similar project and credibility of its performance record and other project references. All cost related to this exercise shall be borne by the proposer. The proposer shall agree to use above listed pre-approved system manufactures, should the proposed system be not acceptable to the Engineer, after the completion of qualification exercise.

2.2 STAINLESS STEEL

A. Metal Surfaces, General: Provide materials with smooth surfaces, without seam marks, roller marks, rolled trade names, stains, discolorations, or blemishes.

B. Tubing: ASTM A 554, Grade MT 316L for Exterior and MT 304 for Interior.

C. Pipe: ASTM A 312/A 312M, Grade TP 316L for Exterior and MT 304 Interior.

D. Brackets, Flanges, and Anchors: Stainless steel ASTM A 666, Type 316L for Exterior and MT 304 for Interior.

E. Finish: Unless otherwise indicated; polished directional satin finish.

2.3 GLASS PRODUCTS AND GLAZING MATERIALS

A. Glass Type as indicated on drawings and comply with Section 08 8000.

B. Glazing Cement and Accessories: Provide glazing cement, setting blocks, shims, and related accessories as recommended or supplied; by railing manufacturer for installing structural glazing in metal subrails.
   1. Glazing Cement: Provide nonshrinking organic cement designed for curing by passing an electric current through metal subrail holding glass panel, as standard with manufacturer.

C. Glazing Gaskets for Glass Infill Panels: Provide glazing gaskets and related accessories recommended or supplied by railing manufacturer for installing glass infill panels in post-supported railings.

D. Glazing Accessories: Provide glazing cement and related accessories recommended or supplied by metal and glass railing system manufacturer for bonding glass to metal subrails.

2.4 FASTENERS

A. Fasteners for Anchoring Railings to Other Construction: Select fasteners of type, grade, and class required to produce connections suitable for anchoring metal and glass railing system to

other types of construction indicated and capable of withstanding design loads. Use Type 316 stainless steel fasteners.

B. Fasteners for Interconnecting Railing Components: Use fasteners fabricated from same basic metal as fastened metal, unless otherwise indicated. Do not use metals that are corrosive or incompatible with materials joined.
   1. Provide concealed fasteners for interconnecting railing components and for attaching them to other Work, unless exposed fasteners are unavoidable or are standard fastening method for handrail and railing indicated.

C. Anchors fabricated from corrosion-resistant materials with capability to sustain, without failure, a load equal to six times the load imposed when installed in unit masonry and equal to four times the load imposed when installed in concrete.

2.5 GROUT AND ANCHORING CEMENT

A. Nonshrink, Nonmetallic Grout: Premixed, factory-packaged, nonstaining, noncorrosive, nongaseous grout complying with ASTM C 1107. Provide grout specifically recommended by manufacturer for interior and exterior applications.

2.6 MISELANEOUS COMPONETNS

A. Aluminum Components: Conforming to ASTM B 221/ASTM B221M, Alloy 6063- T52.

B. Balustrade Cap Setting Materials: Nylon or Neoprene setting block and silicone sealant.

C. Bolts, Screws and Nuts: Stainless steel with finish as scheduled for base material.

D. Internal Handrail Cap Connection Sleeves: Metal tube, material compatible with handrail cap material.

E. Setting Blocks: Elastomeric material with a Shore A durometer hardness of 85 + 5.

2.7 FABRICATION

A. Assemble railings in shop to greatest extent possible to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations. Clearly mark units for reassembly and coordinated installation. Use connections that maintain structural value of joined pieces.

B. Form changes in direction of railing members as indicated.

C. Form curves by bending members in jigs to produce uniform curvature for each repetitive configuration required; maintain profile of member throughout entire bend without buckling, twisting, cracking, or otherwise deforming exposed surfaces of handrail and railing components.

D. Welded Connections: Cope components at perpendicular and skew connections to provide close fit, or use fittings designed for this purpose. Weld connections continuously to comply with the following:
   1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals. Obtain fusion without undercut or overlap. Remove flux immediately.

2. At exposed connections, finish exposed surfaces smooth and blended so no roughness shows after finishing and welded surface matches contours of adjoining surfaces.

E. Brackets, Flanges, Fittings, and Anchors: Provide manufacturer's standard wall brackets, flanges, miscellaneous fittings, and anchors to connect handrail and railing members to other construction.

F. Provide inserts and other anchorage devices to connect metal and glass railing system to substrate. Fabricate anchorage devices capable of withstanding loads imposed by metal and glass railing system. Coordinate anchorage devices with supporting structure.

G. Shear and punch metals cleanly and accurately. Remove burrs from exposed cut edges.

H. Ease exposed edges to a radius of approximately 1/32 inch (1 mm), unless otherwise indicated. Form bent-metal corners to smallest radius possible without causing grain separation or otherwise impairing the Work.

I. Close exposed ends of railing members with prefabricated end fittings.

J. Isolate dissimilar metals to prevent electrolytic action by applying primer to concealed surfaces of metal components.

2.8 GLAZING PANEL FABRICATION

A. Glass Panels: Provide thickness indicated, but not less than that required to support structural loads specified in this specification.

B. Comply with Section 08 8000 for glazing manufacturing. preprocessing and fabrication requirements.

2.10 REFERENCED STANDARDS

A.	ASTM C 1107:	Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
B.	ASTM C 1172	Standard Specification for Laminated Architectural Flat Glass
C.	ASTM E 894:	Test Method for Anchorage of Permanent Metal Railing Systems and Rails for Buildings
D.	ASTM E 935:	Test Methods for Performance of Permanent Metal Railing Systems and Rails for Buildings
E.	ASCE 7-10	American Society of Civil Engineers: Minimum Design Loads for Buildings and Other Structures
F.	ASTM C 864	Specification for Dense Elastomeric Compression Seal Gaskets, Setting Blocks, and Spacers
G.	ASTM C 920	Specification for Elastomeric Joint Sealants
H.	ASTM C 1048	Specification for Heat-Treated Flat Glass - Kind HS, FT Coated and Uncoated Glass

I. ASTM C 1172 — Specification for Laminated Architectural Flat Glass

J. ASTM E 1300 — Practice for Determining Load Resistance of Glass in Buildings

K. 16 CFR 1201 — Safety Standard for Architectural Glazing Materials

L. The minimum standards for products specified in this section shall be including as under but not limited to the following. Except as otherwise specified herein, perform work in accordance with specifications, codes and standards cited therein, and their latest editions, revisions, applicable addenda and supplements. Where there is conflict between the referenced standards the most stringent of the conditions/requirements shall be applicable.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verification of Conditions: Examine areas and conditions under which Work is to be performed and identify conditions detrimental to proper and or timely completion.
   1. Fully inspect the existing structure to verify a structurally sound base for anchoring railing system.

B. Do not begin installation until substrates have been properly prepared.

C. If substrate preparation is the responsibility of another installer, notify Architect of unsatisfactory preparation before proceeding.

3.2 PREPARATION

A. Surface Preparation: Coordinate and furnish anchorages and setting drawings, diagrams, templates, instructions, and directions for the installation of items having integral anchors which are to be embedded in concrete or masonry construction. Coordinate delivery of such items to the Project Site.

B. Clean surfaces thoroughly prior to installation.

C. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

D. Clean glazing channels and other framing members receiving glass immediately before glazing. Remove coatings not firmly bonded to substrates.

3.3 INSTALLATION, GENERAL

A. Install handrails in accordance with manufacturer’s recommended installation instructions and approved shop drawings.

B. Fit exposed connections together to form tight, hairline joints.

C. Cutting, Fitting, and Placement: Perform cutting, drilling, and fitting required for installing metal and glass railing system. Set metal and glass railing accurately in location, alignment, and elevation, measured from established lines and levels and free from rack.

1. Do not weld, cut, or abrade surfaces of railing components that have been coated or finished after fabrication and that are intended for field connection by mechanical or other means without further cutting or fitting.
2. Set posts plumb within a tolerance of 2 mm in 1 m.

D. Adjust handrails or railings before anchoring to ensure alignment at abutting joints. Space posts at interval indicated, but not less than that required by structural loads.

E. Fastening to In-Place Construction: Use anchorage devices and fasteners where necessary for securing metal and glass railing system and for properly transferring loads to in-place construction. All embedded anchor plates and supporting steel shall be separately installed and coordinated with the railing Supplier

3.4 INSTALLING GLASS PANELS

A. Glazed Railing System and Post Supported railing System: Install assembly to comply with railing manufacturer's written instructions.

3.5 FIELD QUALITY CONTROL

A. The design engineer responsible for the preparation of shop drawings will perform regular field reviews during installation and submit reports to the Consultant.

3.6 CLEANING

A. Clean glazing surfaces after installation, complying with requirements contained in the manufacturer’s instructions. Remove excess glazing sealant compounds, dirt or other substances.

B. Clean metal by washing thoroughly with clean water and soap and rinsing with clean water.

C. Do not use harsh cleaning materials or methods that would damage glass or finish. Clean with warm water.

3.7 PROTECTION

A. Protect finishes of metal and glass railing system from damage during construction period with temporary protective coverings approved by railing manufacturer. Remove protective coverings at the time of Substantial Completion.

B. Repair components damaged by subsequent construction activities in accordance with manufacturer’s recommendations; replace damaged components that cannot be repaired to Architect’s acceptance.

END OF SECTION 05 7313.01

SECTION 05 75 00.01 - DECORATIVE FORMED METAL (SOLID ALUM SHEET CLADDING)

PART 1 - GENERAL

Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.1 SECTION INCLUDES

A. Design, engineer and furnish all labor, materials, tools and equipment and perform all services and operations necessary for the complete supply and installation of the following, decorative formed solid aluminium sheet metal cladding shapes including but not limited to:
   1. Preformed solid aluminium metal sheet cladding panels, including aluminium sub framing assembly, insulation, sealant and waterproofing works.
   2. Tower H32, unitized curtain wall integrated metal sheet cladding architectural ~~C~~ features at slab, and wall cladding at event box and lower levels.
   3. Building H30, stick curtain wall integrated metal sheet cladding architectural feature at slab.
   4. Building H27, masonry/ concrete wall cladding, making a weathertight interface with the adjacent stick curtain wall.
   5. Ground floor free standing shading canopy cladding.
   6. Provide finish compatible Nano/Protective coating for reducing the façade cleaning frequency.

B. Design to follow the architectural intent indicated in the drawings.

C. This section to be read in conjunction with the related wall types/ exterior wall system drawings submitted as a part of the tender package, not limited to the scope stated in point 1.1.A above.

1.2 RELATED SECTIONS

1.	Section 01 4500	-	Quality Control
2.	Section 07 2100.01	-	Thermal Insulation
3.	Section 07 2500.01	-	Weather Barrier
4.	Section 07 6200.01	-	Sheet Metal Flashing and Trim
5.	Section 07 8400.01	-	Fire Stopping
6.	Section 07 9200.01	-	Joint Sealants
7.	Section 08 4000	-	Aluminium Works
8.	Section 08 4413.01	-	Glazed Aluminium Curtain Wall
9.	Section 08 8000.01	-	Glazing
10.	Section 09 9100	-	Painting

1.3 REFERENCES

A. The minimum standards for products specified in this section shall be including as under but not limited to the following. Except as otherwise specified herein, perform work in accordance with specifications, codes and standards cited therein, and their latest editions, revisions, applicable addenda and supplements. Where there is conflict between the referenced standards the most stringent of the conditions/requirements shall be applicable.

1. ASTM A153 : Standard Specification for Zinc Coating (Hot: Dip) on Iron and Steel Hardware.

2. ASTM A307 : Standard Specification for Carbon Steel Bolts and Studs, 60 000 PSI Tensile Strength.

3. ASTM A325 : Standard Specification for Structural Bolts, Steel, Heat Treated 120/105 ksi Minimum Tensile Strength.

4. ASTM A325 : Standard Specification for High-Strength Bolts for Structural Steel Joints.

5. ASTM B209 : Standard specification for Aluminium and Aluminium-Alloy Sheet and Plate.

6. ASTM B210 : Standard Specification for Aluminium and Aluminium-Alloy Drawn Seamless Tubes.

7. ASTM B221 : Standard specification for Aluminium and Aluminium-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes.

8. NAAMM : Metal Finishes Manual.

9. NAAMM : Welding Specifications for Fabrication of Steel, Aluminium and Stainless-Steel bar Grating.

10. National Ornamental and Miscellaneous Metals Association (NOMMA) : Metal Rail Manual and Metal Finishes Manual.

11. Natural Fire Protection Association (NFPA) : 101 Life Safety Code.

1.4 PERFORMANCE REQUIREMENTS

A. The design and performance specified are minimum requirements. Methods which may be suggested by these specifications are not intended to limit methods of achieving the required design and performance criteria. All the drawings must be reviewed, approved, stamped and certified for the design of this project by the system supplier company (not manufacturer) and a Third-Party professional Engineer.

B. General: Provide decorative preformed solid aluminium sheet metal cladding panel system; including anchorage, capable of withstanding, without failure, the effects of the following:
   1. Structural loads.
   2. Thermal Movements.
   3. Movements of supporting structure indicated on Drawings including, but not limited to, story drift, twist, column shortening, long-term creep, and deflection from uniformly distributed and concentrated live loads.
   4. Dimensional tolerances of building frame and other adjacent construction.
   5. Failure includes the following (not limited to):
      a. Deflection exceeding specified limits.
      b. Thermal stresses transferred to building structure.
      c. Framing members transferring stresses, including those caused by thermal and structural movements, to glazing.

d. Noise or vibration created by wind and thermal and structural movements.
e. Loosening or weakening of fasteners, attachments, and other components.

C. Structural Loads:
1. Wind load: Refer to the available wind tunnel analysis or project wind report at its latest revision for the design loads.
2. For center of aluminium cladding sheet panels or aluminium sandwich panel: Maximum deflection (ver./hor.); L/120 or 15mm, whichever is smaller; including for the panel stiffeners.
3. The panel has to be reinforced/designed to avoid any kind of oil canning.
4. Seismic Performance: Metal cladding system shall withstand the effects of earthquake motions in accordance to ASCE 7-16 as acceptable to the local authorities.
5. Seismic load – As per ASCE 7-16 as acceptable to the local authorities.

D. Thermal requirements: Framing systems and its associated structure like anchoring structures shall accommodate expansion and contraction movement due to surface temperature differential without causing buckling, stress on glass, failure of joint seals, excessive stress on structural elements, reduction of performance or other detrimental effects.

E. Base engineering calculation on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.
1. Temperature Change (Range): 70 deg. C, ambient; 100 deg. C, material surfaces.
2. Night time or lower range: +10 deg. C.

F. Control of Corrosion: Prevent galvanic action and other forms of corrosion by insulating metals and other materials from direct contact with incompatible materials.

G. The construction has to ensure the possibility of unhindered expansions. Noises resulting from movements have to be avoided by means of intermediate layers which are permanently elastic. The metal sheets have to be fixed in such a way that they are secure from storm or demounting.

H. Delegated Design: Design decorative formed metal items, including comprehensive engineering analysis by a qualified professional engineer registered at the place of work, using performance requirements and design criteria indicated.

1.5 SUBMITTALS

A. Product Data: For each product used including finishing materials and methods.

B. Shop Drawings: Show fabrication and installation of work. Include plans, elevations, component details, and attachments to other Work. Indicate details of materials and profiles of metal member, fittings, joinery, finishes, fasteners, anchorages, and accessory items.

C. Structural Calculations: Submit all relevant structural calculations for dimensioning of all load bearing elements and parts including profiles, fixings, anchorages etc.

D. Samples for Verification: For each profile and pattern of fabricated metal and for each type of metal finish required, prepared on metal of same thickness indicated for the Work. If finishes involve normal color and texture variations, include sample sets, consisting of two or more units, showing the full range of variations expected.
1. Include 300mm square samples of sheets and 150mm long samples of strips.

E. Welding Certificates for Aluminium.

F. Qualification Data: For Installer, Fabricator, Professional Engineer and Testing Agency.

G. Contractor shall submit Method Statement for erection sequences.

1.6 QUALITY ASSURANCE

A. Fabricator Qualifications: A firm experienced in producing decorative formed metal similar to that indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units without delaying the work.

B. Installer Qualifications: Arrange for installation of unit specified in this Section by the same firm engaged in the fabrication of the unit.

C. Coating Applicator Qualifications: A firm experienced in successfully applying coatings type indicated to metals of and that employs competent control personnel to conduct continuing, effective quality-control program to ensure compliance with requirements.

D. Welding Qualifications: Qualify procedures and personnel according to the following:

1. AWS D1.2/D1.2M, "Structural Welding Code - Aluminum."

E. Mock-Up's: Built mock-up for the decorative preformed solid aluminium sheet metal cladding system to set quality standard for fabrication and installation.

1. Approved mock-up may become part of the completed work if undisturbed at the time of substantial completion upon Engineer's acceptance.

1.7 PROJECT CONDITIONS

A. Field Measurements: Verify dimensions of units by field measurements before fabrication and indicate measurements on Shop Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

1.8 COORDINATION

A. Coordinate installation of anchorages for decorative metal items. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

1.9 WARRANTY

A. Special Warranty: Submit a written warranty executed by the Fabricator, Installer and Contractor agreeing to repair or replace panel and components of the decorative units that fail in materials or workmanship within the specified warranty period. Failures include, but are not limited to the following:

1. Failure of system to meet performance standards.
2. Deterioration of materials, metal finishes and other materials beyond natural wear and tear.

3. Failure of support system to retain panels in place.

F. Warranty Period: Not less than 10 (Ten) years from date of Substantial Completion.

PART 2 PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS/FABRICATORS

1. Technical Glass, Middle East.
2. Glassline Industries, Middle East.
3. Alico, Middle East.
4. Alumco, Middle East.
5. Al Abbar, Middle East.
6. Permasteelisa Gartner, Middle East.
7. Hidayath Group, Middle East.
8. Or Equivalent, Subjected to Engineer's acceptance.

Condition to qualify "equivalent": Qualification exercise consists of assessment of technical adaptability of the design and the material, quality procedures maintained, minimum 10 (Ten) years of its usage in similar project and credibility of its performance record and other project references. All cost related to this exercise shall be borne by the proposer. The proposer shall agree to use above listed pre-approved system manufactures, should the proposed system be not acceptable to the engineer, after the completion of qualification exercise.

2.2 SHEET METAL

A. General: Provide metals free from surface blemishes where exposed to view in finished unit. Exposed-to-view surfaces exhibiting pitting, seam marks, roller marks, stains, discolorations, or other imperfections on finished units are not acceptable.

A. ALUMINUM

1. Aluminum, General: Provide alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated, and with strength and durability properties for each aluminum form required not less than that of alloy and temper designated below.

   a. Custom fabricated to the design with concealed fasteners and as indicated on drawings.

   b. Thickness: As indicated on drawings.

   c. Finish/Colour: As per approved RAL color sample.

   d. Panel Type: Factory cut and fabricated preformed solid aluminium sheet panels.

2. Extruded Bars and Shapes: ASTM B 221, Alloy 6063-T5.

3. Extruded Structural Pipe and Round Tubing: ASTM B 429, Alloy 6063- T6.

4. Drawn Seamless Tubing: ASTM B 210 or ASTM B 483/B 483M, Alloy 6063-T832.

5. Plate and Sheet: ASTM B 209, Alloy 6061-T6.

6. Die and Hand Forgings: ASTM B 247, Alloy 6061-T6.

7. Castings: ASTM B 26, Alloy A356.0-T6.

2.3 MISCELLANEOUS MATERIALS

A. Filler Metal and Electrodes: Provide type and alloy of filler metal and electrodes as recommended by producer of metal to be welded or brazed and as necessary for strength, corrosion resistance, and compatibility in fabricated items.

B. Fasteners: Fabricated from same basic metal and alloy as fastened metal unless otherwise indicated. Do not use metals that are incompatible with materials joined.

C. Structural Anchors: For applications indicated to comply with certain design loads, provide chemical or torque-controlled expansion anchors with capability to sustain, without failure, a load equal to six times the load imposed when installed in unit masonry and four times the load imposed when installed in concrete, as determined by testing per ASTM E 488 conducted by a qualified independent testing agency.

D. Nonstructural Anchors: For applications not indicated to comply with design loads, provide powder-actuated fasteners metal expansion sleeve anchors or metal-impact expansion anchors of type, size, and material necessary for type of load and installation indicated, as recommended by manufacturer, unless otherwise indicated.

E. Anchor Materials:

1. Material for Interior Locations: Carbon-steel components zinc plated to comply with ASTM B 633 or ASTM F 1941 (ASTM F 1941M), Class Fe/Zn 5, unless otherwise indicated.

F. Sound-Deadening Materials:

1. Anti-Drumming coating on horizontally installed hollow cladding metal panels; if not filled with thermal insulation.

G. Backing Materials: Provided or recommended by decorative formed metal manufacturer.

2.4 GENERAL FINISH REQUIREMENTS

A. Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.

B. Complete mechanical finishes of flat sheet metal surfaces before fabrication where possible. After fabrication, finish all joints, bends, abrasions, and other surface blemishes to match sheet finish.

C. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

D. Finish items indicated on Drawings after assembly unless otherwise approved by Engineer.

2.5 FINISHES

A. All aluminium visible surfaces exposed to the building interior and exterior: Aluminium surfaces shall be finished with Super Durable Powder Coating complying with AAMA 2604 and that are thermoset, free from VOCs, with minimum dry film thickness of 60 microns.
   1. Colour and Finish: To match approved colour sample.

B. Prepare, pretreat and apply coating (including protective coat) to exposed metal surface to comply with AAMA 2604.

C. Concealed Aluminum Surfaces: Any aluminum sections which are completely concealed and therefore do not require the facing colour shall be anodized to a minimum standard of AMMA 611 Class II.

D. Identification marks shall not appear on finished surfaces.

E. Adhesion: No removal of finish after 1.5mm cross hatching to base metal, impacting to the point of metal rupture, and subjecting to application and quick removal of cellophane tape.

PART 3 EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions, with Installer present, for compliance with requirements indicated for conditions affecting performance of installation.

3.2 INSTALLATION, GENERAL

A. Provide anchorage devices and fasteners where necessary for securing work to in-place construction.

B. Set products accurately in location, alignment, and elevation; measured from established lines and levels. Provide temporary bracing or anchors in formwork for items to be built into concrete, masonry, or similar construction.

C. Fit exposed connections accurately together to form tight, hairline joints. Where cutting, welding, and grinding are required for proper shop fitting and jointing of units, restore finishes to eliminate any evidence of such corrective work.

D. Do not cut or abrade finishes that cannot be completely restored in the field. Return items with such finishes to the shop for required alterations, followed by complete refinishing, or provide new units as required.

E. Restore protective coverings that have been damaged during shipment or installation. Remove protective coverings only when there is no possibility of damage from other work yet to be performed at same location.
   1. Retain protective coverings intact; remove coverings simultaneously from similarly finished items to preclude non uniform oxidation and discoloration.

F. Erection tolerances for panel assemblies relate to the structural grid of the

building, and apply to each individual assembly as follows:

1. Vertical position: plus / minus 2 mm
2. Horizontal Position: plus / minus 2 mm
3. Racking in elevation: 1 mm
4. Deviation from true plumb over full height: maximum 3 mm
5. Deviation from true straight in plane over full length of each building face: maximum 3 mm
6. Tolerances shall not be cumulative.

3.3 PROTECTION

A. Protect finishes of units from damage during construction period with temporary protective coverings approved by ornamental metal fabricator. Remove protective covering at the time of Substantial Completion.

B. Restore finishes damaged during installation and construction period so no evidence remains of correction work. Return items that cannot be refinished in the field to the shop; make required alterations and refinish entire unit, or provide new units.

END OF SECTION 05 7500.01

SECTION 07 2100.01 - THERMAL INSULATION

PART 1 - GENERAL

Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.1 SECTION INCLUDES

A. Furnish all labor, materials, tools and equipment and perform all services and operations necessary for the complete supply and installation of the following, but not limited to:
   1. Thermal insulation behind glass spandrel and wall cladding panels and similar as indicated in the drawings.
   2. Thermal insulation at flashing and trims.
   3. Acoustic insulation at curtain wall and interfaces as required to meet the performance requirements.

B. Design to follow the architectural intent indicated in the drawings.

C. This section to be read in conjunction with the related wall types/ exterior wall system drawings submitted as a part of the tender package.

1.2 RELATED SECTION

A. This schedule is intended to be used as a helpful indication of the related sections within the Project Specifications. It is not necessarily comprehensive or complete and it is the Contractor's responsibility to ascertain all applicable sections required to understand the full scope of Works intended.

B. Work to be coordinated and used in conjunction with this specification includes but is not restricted to:

1.	Section 01 4500	-	Quality Control
2.	Section 03 3000	-	Cast-in-Place Concrete
3.	Section 04 2200	-	Concrete Unit Masonry
4.	Section 05 7500.01	-	Decorative Formed Metal
5.	Section 07 2500.01	-	Weather Barrier
6.	Section 07 6113.01	-	Standing Seam Sheet Metal Roof
7.	Section 07 6200.01	-	Sheet Metal Flashing and Trim
8.	Section 07 8400.01	-	Fire Stopping
9.	Section 07 9200.01	-	Joint Sealants
10.	Section 08 4000	-	Aluminium Works
11.	Section 08 4413.01	-	Glazed Aluminium Curtain Wall
12.	Section 08 8000.01	-	Glazing

1.3 DEFINITIONS

A. Thermal Conductivity (K value).

B. FS: Foil Scrim; jacketing, single-sided foil facing that is comprised of aluminum and reinforced with tri-directional fiberglass scrim, The FS moisture barrier does not contain

combustible Kraft paper which improves the insulations resistance to fire.

C. Glass Mineral Wool: Interchangeable with fiber glass, but replacing the term in the attempt to disassociate and differentiate Glass Mineral Wool from the potential health and safety risk of special purpose or reinforcement products that do not meet the bio solubility criteria of insulation made from glass.

D. Recycled Content: materials such as bottled glass collected at curbside or other collection sites after consumer use and/or materials used or created from one manufacturing process which are collected as scrap and placed back into another manufacturing process rather than being placed in a landfill or incinerated.

E. UL Classified: UL has tested and evaluated samples of the product with respect to certain properties of the product. UL Classifies products to:
   1. Applicable UL requirements.
   2. Standards for safety.
   3. Standards of other National and International organizations.

F. EPD: Environmental Product Declaration. A third party verified document that reports environmental data of products based on the Life Cycle Assessment (LCA) and other relevant information, and in accordance with the International Standard ISO 14025 (Type III Environmental Declarations).

G. Rigid Insulation: Rigid board material that provides thermal resistance. Foam glass, phenolic boards, including foam plastics such as expanded polystyrene (EPS), extruded polystyrene (XPS), and polyisocyanurate are commonly used examples of rigid insulation material.

H. Semi-Rigid Insulation: A type of insulation that is sold in compressed formed boards composed of mineral fibers (typically glass or rock wool). Normally used for its non-combustible properties.

I. Batt: Thermal insulation, glass fiber or rock fiber wool in roll form.

1.4 PERFORMANCE REQUIREMENTS

A. The U-values for walls shall be as indicated.
   1. The contractor must demonstrate compliance with these U-values prior to the procurement of the envelope components, through provision of test certificates, calculations and drawings.
   2.

Component	U-value
Walls (Including, façade spandrel, overall)	0.30 W/m^2K (As provided by Sustainability specialist)

   2. Meet minimum U-Value requirements for the thickness specified and indicated in the drawings. Resist growth of mold and mildew.
   3. The thickness and density of the insulation shall comply with the thermal performances required.
   4. The insulation shall not sag, delaminate or otherwise deteriorate after installation, and shall not deteriorate under or be reduced in performance by exposure to sunlight or water or any other conditions to which it might be subjected during or after construction.
   5. Insulation shall be non-combustible class.

1.5 SUBMITTALS:

A. Product data for each type of product indicated. Submit:
   1. Complete list of all items proposed to be installed under this Section.
   2. Manufacturer's Specifications and other data necessary to demonstrate compliance with the Specification.
   3. Environmental conditions, required for storage and installation.
   4. Manufacturer's Certificates meeting the specified requirements.

B. Samples: 300mm x 300mm samples of all kinds of insulation and all fixing accessories.

C. Manufacturer's installation instructions: Submit preparation and installation requirements and techniques.

D. Product Test Reports: Based on evaluation of comprehensive tests performed by a qualified testing agency acceptable to the authorities having jurisdiction, for each product indicated.

1.6 QUALITY ASSURANCE

A. Manufacturer's Qualification: Manufacturer shall be ISO 9001:2000 certified or one certified as employing an equivalent or higher system of quality management; and capable of providing field service representation, as required by Engineer, during installation.

B. Installer's Qualifications: The installer with minimum 5 years' experience in handling rigid insulation materials and shall be fully conversant with current manufacturer's published data.

C. Surface-Burning Characteristics: As determined by testing identical products according to ASTM E 84 by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.
   1. Surface-Burning Characteristics: To ASTM E 84.
   2. Combustion Characteristics: To ASTM E 136.
   3. Fire Classification (reaction to fire): To BS EN 13501.1.
   4. Most stringent of the above point 1, 2 or 3 and as required by the local authorities shall be adopted/followed.

D. For vapor barriers, perform work in accordance with SWRI-Sealant and Caulking Guide Specifications requirements for material and installation.

E. Regulatory Requirement for Fire-Resistance-Rated and Fire-Rated Materials: Products/assemblies specified in this Part shall comply with fire-rating and fire-resistance-rating requirements of the Projects; and shall be approved for use and certified in writing as such by the local Authorities.

F. Comply with all requirements of local Authorities and BS EN 13501.1, A1 building product requirements.

G. Comply with most stringent of the local code or acceptable national/international building standard requirements for VOC's limits.

1.7 DELIVERY AND PROTECTION

A. Protect insulation materials from physical damage and from deterioration by moisture, soiling, UV and other deleterious materials.

B. Store inside and in a dry location.

C. Provide temporary coverings or enclosures where insulation is subject to abuse and cannot be concealed and protected by permanent construction immediately after installation.

D. Comply with Manufacturer's written instruction for handling, storage and protecting products during installation.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. Thermal Insulation
   1. Rock Mineral Fiber
      a. Kimmco.
      b. Knauf Insulations.
      c. Foamglass (for cellular glass insulation).
      d. Fujairah Rockwool.
      e. Saudi Rockwool.
      f. Or Equivalent, Subjected to Engineer's acceptance.

   2. Glass Mineral Fiber
      a. Kimmco.
      b. Knauf Insulations.
      c. Afico.
      d. Or Equivalent, Subjected to Engineer's acceptance.

   3. Extruded Polystyrene
      a. SSO Foam.
      b. ESSCO Foam.
      c. Arabian Chemical Co.
      d. Or Equivalent, Subjected to Engineer's acceptance.

B. Acoustic Insulation
   1. Mineral Fiber.
      a. Kimmco.
      b. Knauf Insulations.
      c. Foamglass.
      d. Fujairah Rockwool.
      e. Saudi Rockwool.
      f. Or Equivalent, Subjected to Engineer's acceptance.

   Conditions to qualify "equivalent": Qualification exercise consists of assessment of technical adaptability of the design and the material, quality procedures maintained, minimum 10 (Ten) years of its usage in similar project and credibility of its performance record and other project references. All cost related to this exercise shall be borne by the proposer. The proposer shall agree to use above listed pre- approved system manufactures, should the proposed system be not acceptable to the Engineer, after the completion of qualification exercise.

2.2 POLYSTYRENE RIGID THERMAL INSULATION (ONLY IF, APPROVED FOR THE USE BY THE LOCAL AUTHORITY/ CIVIL DEFENCE FOR SPECIFIC CONCEALED AREAS)

A. The insulation shall be a non-absorbent, foam extruded, closed cell polystyrene board or phenolic board specifically manufactured for use in roofing and as low-rise wall insulation. Comply with ASTM C578 and shall meet all requirements of authorities having jurisdiction.

B. Edges of boards shall be rebated (Shiplap).

C. Physical properties: For external wall and roof

1.	Thermal conductivity at 25°C:		0.016 W/m^{o}K (ASTM C518)
	– 5 years aged value	:	0.032 W/m^{o}K (ASTM C177)
2.	Water Absorption	:	1% by volume to ASTM D2842
3.	Capillarity	:	Nil
4.	Density	:	32-35 kg/m^{3}.
5.	Compressive strength	:	220-360 KPa to DIN ISO-844:2009 or 42-65 PSI to ASTM D-1621
6.	Fire	:	Class B1 to BS 476 or Pass DIN 4102.
7.	Thicknesses	:	As indicated on the drawings and to meet the minimum U-Value and Acoustic requirement specified.

D. Physical properties for infill between concrete slab

1.	Thermal conductivity at 25 ^{o}C:		0.016 W/m^{o}K (ASTM C518)
	5 Years aged value	:	0.032 W/m^{o}K (ASTM C177)
2.	Water absorption	:	1% by volume to ASTM D 2842.
3.	Capillarity	:	Nil
4.	Density	:	38-42 kg/m3
5.	Compressive strength	:	330-470 KPa (ISO-844:2009) 45-70 PSI (ASTM D-1621)
6.	Fire	:	Class B1 to BS 476 or Pass DIN 4102.
7.	Thickness	;	As indicated on the drawings and to meet the minimum U-Value and Acoustic requirement specified.

E. Vapor barrier: 400-micron thick clear polyethylene conforming to ASTM C171 lapped and taped 100mm on sides and ends. Physical properties for materials shall conform to ASTM E154.

2.3 MINERAL FIBER THERMAL INSULATION, HIGH DENSITY

A. Mineral fiber insulation manufactured from stable glass or rock fibers bonded with thermosetting resins. Comply with ASTM C 612. Edges of boards shall be square. Installation shall be as per manufacturer's recommendations.

B. Physical Properties: For external wall and soffit.

1.	Thermal conductivity at 25°C	:	0.035~0.041 W/m^{o}K (ASTM C518)
2.	Density	:	64~110kg/m^{3}.
3.	Thickness	:	As indicated on the drawings and to

meet the minimum U-Value and Acoustic requirement specified.

4. Facing : FS, glass reinforced aluminium foil scrim as vapor retarder.

C. Fire Characteristics: To ASTM E84.
   1. Flame Spread: 15 (Max).
   2. Smoke Development: 0.

D. Combustion Characteristics: To ASTM E 136.
   1. Non-combustible.

E. Fire Classification (reaction to fire): To BS EN 13501.1
   1. Class A1 building product.

2.4 MINERAL FIBER THERMAL INSULATION, LOW DENSITY

A. Mineral fiber insulation manufactured from stable glass or rock fibers bonded with thermosetting resins. Comply with ASTM C 612. Installation shall be as per manufacturer's recommendations.

B. Physical Properties: For external wall.
   1. Thermal conductivity at 25°C : 0.038~0.045 W/m^{o}K (ASTM C518)
   2. Density : 24~64kg/m^{3}.
   3. Thickness : As indicated on the drawings and to meet the minimum U-Value and Acoustic requirement specified.
   4. Facing : FS, glass reinforced aluminium foil scrim as vapor retarder.

C. Fire Characteristics: To ASTM E84.
   1. Flame Spread: 15 (Max).
   2. Smoke Development: 0.

D. Combustion Characteristics: To ASTM E 136.
   1. Non-combustible.

E. Fire Classification (reaction to fire): To BS EN 13501.1.
   1. Class A1 building product.

2.5 SLAG-WOOL MINERAL FIBER BLANKET INSULATION

A. Unfaced, Slag-Wool-Fiber/Rock-Wool-Fiber Blanket Insulation: Manufactured from stable glass or rock fibers. Comply with ASTM C 665, Type I (blankets without membrane facing). Installation shall be as per manufacturer's recommendations.

B. Physical Properties:
   1. Nominal Density : 36kg/m^{3}.
   2. Facing : None
   3. Thickness : As indicated on the drawings and to meet the minimum U-Value and Acoustic

requirement specified.

C. Combustion Characteristics: To ASTM E 136.
   1. Non-combustible.

D. Faced, Slag-Wool-Fiber/Rock-Wool-Fiber Blanket Insulation: Manufactured from stable glass or rock fibers. Comply with ASTM C 665 Type III (blankets with reflective membrane facing); Installation shall be as per manufacturer's recommendations.

E. Physical Properties:

1.	Nominal Density	:	36kg/m^3.
2.	Facing	:	FS, glass reinforced aluminium foil scrim as vapor retarder.
3.	Thickness	:	As indicated on the drawings and to meet the minimum U-Value and Acoustic requirement specified.

F. Fire Classification (reaction to fire): To BS EN 13501.1.
   1. Class A1 building product.

2.6 ADHESIVE

A. As recommended by insulation manufacturer for specific application with demonstrated capacity to bond insulation securely to substrate indicated without damaging insulation and substrates.

B. Do not install adhesive when temperature or weather conditions are detrimental to successful installation.

C. Adhesive type 1: Type recommended by insulation manufacturer for application.

D. Adhesive type 2: Vapor retarder type, trowel consistency; fire retardant compatible with insulation and substrate, as recommended by the manufacturer.

2.7 ACCESSORIES

A. Fasteners and accessories required for the installation shall be as recommended by the manufacturer.

B. Insulation Fasteners: Impale clip type of plain finished steel; of type to be adhered to surface to receive rigid insulation; length to suit insulation thickness; capable of securely and rigidly fasting insulation in place.

C. Approved Fasteners are:
   1. Hilti, USA
   2. MKT, Germany
   3. Ancon, UK
   4. Fischer, Germany
   5. Or Equivalent, Subjected to Engineer's acceptance.

   Conditions to qualify "equivalent": Qualification exercise consists of assessment of technical adaptability of the design and the material, quality procedures maintained, minimum 10 (ten) years of its usage in similar project and credibility of its performance

record and other project references. All cost related to this exercise shall be borne by the proposer. The proposer shall agree to use above listed pre- approved system manufactures, should the proposed system be not acceptable to the Engineer, after the completion of qualification exercise.

D. Tape: Bright Aluminium, Polyethylene and /or Polyester self-adhering type, mesh reinforced, minimum 50 mm wide.

E. Protective Boards: As recommended by the manufacturer.

F. Thinner and cleaner: As recommended by sheet material manufacturer.

G. Attachments: Stainless steel type 316L, bars and anchors.

2.8 OTHER MATERIAL

A. Provide other materials, not specifically described or indicated but required for the complete and proper installation of insulation materials, as recommended in writing by the manufacturer.

2.9 REFERENCED STANDARDS

A. The minimum standards for products specified in this section shall be including as under but not limited to the following. Except as otherwise specified herein, perform work in accordance with specifications, codes and standards cited therein, and their latest editions, revisions, applicable addenda and supplements. Where there is conflict between the referenced standards the most stringent of the conditions/requirements shall be applicable.

1. ASTM C 171 Standard Specifications for Sheet Materials for Curing Concrete.
2. ASTM C 177 Standard Test Methods for Steady State Heat Flux Measurements and Thermal Transmission Properties.
3. ASTM C240 Standard Test Methods of Testing Cellular Glass Insulation Block.
4. ASTM C552 Standard Specification for Cellular Glass Thermal Insulation.
5. ASTM C 578 Standard Specifications for Rigid Cellular Polystyrene Thermal Insulation.
6. ASTM C 612 Standard Test Method for Compressive Properties of Rigid Cellular Plastics.
7. ASTM C1013 Standard Specification for Faced Rigid Cellular Polyisocyanurate Roof Insulation.
8. ASTM D 1621 Standard Test Method for Compressive Properties of Rigid Cellular Plastics.
9. ASTM D 2842 Standard Test Method for Water Absorption of Rigid Cellular Plastics.
10. ASTM E 84 Standard Test Method for Surface Burning Characteristics of Building Materials.
11. ASTM E 136 Standard Test Method for Behavior of Material in a Vertical Tube Furnace.
12. ASTM E 154 Standard Test Methods for Water Vapor Retarders used in Contact with Earth under Concrete Slab, on Walls or as Ground Cover.

13. ASTM E90-97 Standard Test Method for Laboratory Test for Airborne Sound Transmission Loss of Building Partitions and Elements.
14. ASTM C423-90a Standard specification for testing noise reduction coefficient.
15. BS 476 Fire tests on Building Materials and Structures Non-combustibility test for materials.
16. BS 2972 Methods of Test for Inorganic thermal insulating materials.
17. NFPA 285 Standard Fire Test Method for Evaluation of Fire Propagation Characteristics of Exterior Non-Load-Bearing Wall Assemblies Containing Combustible Components
18. ASTM E119 Standard Test Methods for Fire Tests of Building Construction and Materials
19. BS EN 13501-1 Fire classification of construction products and building elements. Classification using test data from reaction to fire tests

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and condition, with Installer present, for compliances with requirements and for any unsatisfactory work which may adversely affect performance of insulation.

B. Do not proceed with installation until unsatisfactory conditions have been corrected

C. Ensure that surfaces to receive insulation are clean, dry, and suitable to provide a complete and satisfactory installation.

3.2 PREPARATION

A. Clean substrates of substances harmful to insulation materials, including removing projections capable of puncturing insulation or interfere with the proper installation of insulation.

B. Close off openings in cavities receiving pored-in-place or semi-rigid insulation material to prevent subsequent displacement.

3.3 INSTALLATION, GENERAL

A. Comply with Manufacturer's written instructions applicable to products and applications indicated.

B. Install only undamaged, dry and unsoiled insulation material.

C. Bond units to substrate with adhesive or use mechanical anchorage to provide permanent placement and support of units.

D. Extend insulation in thickness indicated to envelop entire area to be insulated. Cut and fit tightly around obstructions and fill voids with insulation. Remove projections that interfere with proper placement.

E. Provide sizes to fit applications indicated and selected from manufacturer's standard

thicknesses, widths, and lengths. Apply single layer of insulation units to produce thickness indicated unless multiple layers are otherwise shown or required to make up total thickness.

3.4 INSTALLATION OF INSULATION

A. General: Apply insulation units to substrates, complying with manufacturer's written instructions. Bond units to substrate with adhesive or use mechanical anchorage to provide permanent placement and support of units.

B. Rigid Insulation Boards
   1. Install in accordance with manufacturer's recommendations and staggering joints in both directions.
   2. Cut clearly to fit closely around projections, up stands, rainwater outlets etc.
   3. Protect from damage and moisture after installation.

C. Batt Insulation
   1. Install batt insulation without gaps or voids in accordance with manufacturer's instructions.
   2. Trim insulation neatly to fit spaces. Use batt free of damage.
   3. Place blankets in cavities formed by framing members to produce a friction fit between edges of insulation and adjoining framing members.
   4. Fit insulation tight in spaces and tight to exterior side of mechanical and electrical s e r v i c e s within the plane of insulation. Leave no gaps or voids.

D. Insulation Fasteners
   1. Apply fasteners to surfaces in accordance with manufacturer's instructions. Check for compatibility of adhesive with any concrete form release agent used.
   2. Provide minimum of four fasteners per 600mmx1200mm board, allowing for a maximum load per fastener of 1.5kg.

E. Install separation fabric where required according to manufacturer's written instructions, overlapping sides and ends.

F. Place insulation in right orientation and with a minimum of 33% compression at perimeter fire-safing areas.

G. Install and fit insulation completely in such a way that no convection behind the insulation is possible.

3.5 INSTALLATION OF CURTAIN WALL (CLADDING PANEL/SPANDREL) INSULATION

A. Install board insulation in aluminium framed curtain wall construction, where indicated on Drawings according to manufacturer's/fabricator's written instructions.
   1. Hold insulation in place by securing metal clips and straps or integral pockets within curtain wall frames, spaced at intervals recommended in writing by insulation manufacturer to hold insulation securely in place.
   2. Fill curtain wall cladding panel with the high-density insulation; apply waterproofing and weather sealant as indicated in the drawings.
   3. Install insulation where it contacts perimeter fire-containment system to prevent insulation from bowing under pressure from perimeter fire- containment system.
   4. Compress respective insulation minimum up to 33% between curtain wall and vertical slab edge as indicated in the drawing.

5. Install insulation with right orientation.

3.6 INSTALLATION OF VAPOR BARRIERS

A. General: Extend vapor barriers to extremities of areas to be protected from vapor transmission. Secure in place with adhesives or other anchorage systems as indicated.

B. Seal joints caused by pipes, conduits and similar items penetrating vapor barrier with vapor retarder tape to crate air tight seal.

C. Repair tears or punctures in vapor barriers immediately before concealment by other work.

3.7 SEQUENCING

A. Sequence work to ensure fireproofing, fire stopping, vapor retarder and air barrier materials (if any) are in place before beginning work of this position.

3.8 COORDINATION

A. Coordinate work with specification section “Vapor Retarders” for installation of vapor barriers and specification section “Air Barrier” for air seal material.

3.9 ENVIRONMENTAL CONTROL AND PROTECTION

A. Protection:
   1. Protect all insulation from unnecessary construction traffic after installation. Protection shall be approved and shall prevent any damage likely to result in a decrease in the efficiency of the insulation.
   2. Protect from long periods of exposure before being concealed, by proper sequencing of the work.

END OF SECTION 07 2100.01

SECTION 07 2500.01 - WEATHER BARRIER

PART 1 - GENERAL

Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.1 SECTION INCLUDES

A. Furnish all labor, materials, tools and equipment and perform all services and operations necessary for the complete supply and installation of the fully adhered sheet membrane waterproofing to external joints around the aluminium and glass curtain wall system, cladding system and interfaces, to form a weathertight junction/ interface.

B. Design to follow the architectural intent indicated in the drawings.

C. This section to be read in conjunction with the related wall type/ exterior wall system drawings submitted as a part of the tender package.

1.2 RELATED SECTIONS

A. This schedule is intended to be used as a helpful indication of the related sections within the Project Specifications. It is not necessarily comprehensive or complete and it is the Contractor's responsibility to ascertain all applicable sections required to understand the full scope of Works intended.

B. Work to be coordinated and used in conjunction with this specification includes but is not restricted to:

1.	Section 01 4500	-	Quality Control
2.	Section 03 3000	-	Cast-in-Place Concrete
3.	Section 04 2200	-	Concrete Unit Masonry
4.	Section 05 7500.01	-	Decorative Formed Metal
5.	Section 07 2100.01	-	Thermal Insulation
6.	Section 07 6113.01	-	Standing Seam Sheet Metal Roofing
7.	Section 07 6200.01	-	Sheet Metal Flashing and Trim
8.	Section 07 9200.01	-	Joint Sealants
9.	Section 08 1116.01	-	Alum. Doors and Frames
10.	Section 08 4000	-	Aluminium Works
11.	Section 08 4413.01	-	Glazed Alum. Curtain Wall

1.3 PERFORMANCE REQUIREMENTS

A. General: Weather barrier shall be capable of performing as a continuous vapor-retarding air barrier and as a liquid-water drainage plane. The edges of the weather barrier should be flashed to any external envelop penetrations to form a single continuous membrane, discharging to the exterior. The membrane must prevent moisture ingress from either incidental condensation or water penetration. Weather barrier assemblies shall be capable of accommodating substrate movement and of sealing substrate expansion and control joints, construction material changes,

penetrations, and transitions at perimeter conditions without deterioration and air leakage exceeding specified limits.

1. Resist air leakage in accordance with ASTM E283, when tested as a part of the curtain wall assembly.
2. Resist water penetration in accordance with ASTM E331, when tested as a part of the curtain wall assembly.
3. Provide continuity of air seal materials and assemblies in conjunction with interfacing and adjacent materials.

1.4 SUBMITTALS

A. Product Data: Include manufacturer's written instructions for evaluating, preparing, and treating substrate; technical data; and tested physical and performance properties of weather barrier.

B. Shop Drawings: Show locations and extent of weather barrier. Include details for substrate joints and cracks, counter flashing strip, penetrations, inside and outside corners, terminations, and tie-ins with adjoining construction.
   1. Include details of interfaces with other materials that form part of weather barrier.
   2. Include details of mockups.

C. Product Certificates: For weather barriers, certifying compatibility of weather barrier and accessory materials with Project materials that connect to or that come in contact with weather barrier; signed by product manufacturer.

D. Qualification Data: For Applicator.

E. Samples, 150mm x 300mm minimum size, weather barrier material to be provided; sample for joints of such material shall be submitted for Engineer’s approval.

F. Product test reports from a qualified independent testing agency evidencing compliance of weather barrier with requirements and other physical properties reported by manufacturer based on comprehensive testing of products according to standard test methods within previous 5 (five) years.

G. Method statement for the installation, maintenance and repair of weather barrier assemblies.

H. Sustainable Design Submittals: Submit manufacturer’s printed statement of VOC content, manufacturing location relative to project site and recycled content for product used.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: Engage an installer with minimum 5 (five) years’ experience who has completed similar works to that indicated for this Project and who is acceptable to manufacture.

B. Mockups: Before beginning installation of weather barrier, build mockups of exterior wall assembly, incorporating backup wall construction, external cladding, door frame and sill, insulation, and flashing to demonstrate surface preparation, crack and joint treatment, and sealing of gaps, terminations, and penetrations of membrane.

1. If Engineer determines mockups do not comply with requirements, reconstruct mockups and apply air barrier until mockups are approved.
2. Approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.
3. Mockup approval is to be well recorded, documented and circulated among all the parties involved

C. Perform Work in accordance with SWRI – Sealant and Caulking Guide Specification requirements for materials and installation.

D. Applicable Regulations: Comply with local code and requirements of authorities having jurisdiction. Do not exceed VOC regulations as established by the State in which they are being installed; including total VOC content, in grams per liter, for all system components (i.e., primers, adhesives, coatings, and similar items).

E. Sourcing: Components used shall be sourced from one manufacturer, including sheet membrane, sealants, primers, mastics, and adhesives.

1.6 PRE-INSTALLATION CONFERENCE

A. Prior to scheduled commencement of the installation and associated work, conduct a meeting at the project site with the Installer, General Contractor, Engineer, Employer, Manufacturer's representative and any other persons directly involved with the performance of the Work.

B. The Installer shall record conference discussions and to include decisions and agreements reached (or disagreements), and furnish copies of recorded discussions to each attending party. The main purpose of this meeting is to review foreseeable methods and procedures related to the Work, including transitions to adjacent materials and materials in contact with air barrier.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Deliver materials to Project site in original packages with seals unbroken, labeled with manufacturer's name, product, and date of manufacture, and directions for storage.

B. Store materials in their original undamaged packages in a clean, dry, protected location and within temperature range required by manufacturer. Protect stored materials from direct sunlight.

C. Store all material as per manufacturer's requirements.

1.8 PROJECT CONDITIONS

A. Environmental Conditions: Apply weather barrier within range of ambient and substrate temperatures recommended by manufacturer.

1.9 WARRANTY

A. Special Warranty: Submit a written warranty signed by Contractor, manufacturer and installer agreeing to repair or replace weather barrier that does not meet requirements or that does not perform during the specified warranty period.
   1. Warranty Period: 10 (ten) years after date of Substantial Completion.

B. Warranty shall include manufacturer's statement that materials in contact with another have been tested and verified to be compatible. Include written testing documentation and test reports if requested by Engineer.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. Sika,
   Middle East
B. EFFISUS,
   Middle East
C. BASF Construction Chemicals,
   Middle East
D. Tremco,
   Middle East
E. Carlisle Coating & Waterproofing Inc.,
   USA
F. Grace WR,
   USA
G. Renolite Ltd.,
   France
   Or Equivalent, Subjected to Engineer's acceptance.

Conditions to qualify "equivalent": Qualification exercise consists of assessment of technical adaptability of the design and the material, quality procedures maintained, minimum 10 (ten) years of its usage in similar project and credibility of its performance record and other project references. All cost related to this exercise shall be borne by the proposer. The proposer shall agree to use above listed pre- approved system manufactures, should the proposed system be not acceptable to the Engineer, after the completion of qualification exercise.

2.2 SELF-ADHERING SHEET WEATHER BARRIER

A. EPDM membrane indicating on drawing as waterproofing membrane, self-adhering with release liner on adhesive side. Primer, adhesive and sheet thickness as per manufacturer's recommendation. Uniform thickness sheet membrane providing moisture vapor and air barrier with movement capability.

B. OR Modified Bituminous Sheet, (Only If, Approved for The Use By the local authority): Minimum 1mm thick, self-adhering sheet consisting of rubberized asphalt laminated to a high density cross laminated polyethylene film with release liner on adhesive side. Uniform thickness sheet membrane providing moisture vapor and air barrier with movement capability.

1. Physical and Performance Properties, (Approved Equivalent):
   a. Water Vapor Transmission: ASTM E 96, Method B – 2.9 ng/m2sPa (0.05 perms) maximum.
   b. Water Absorption: ASTM D 570 – Max 0.12% by weight.
   c. Puncture Resistance: ASTM E 154 – 222 N (50 lbf).
   d. Lap Adhesion: ASTM D 1876 – 880 N/M (5.0 lbs/in.) of width.
   e. Tensile Strength: ASTM D 412, Die C Modified – Min. 3.44 Mpa (500 psi).
   f. Elongation, Ultimate Failure of Rubberized Asphalt: ASTM D 412, Die C – Min. 200%.
   g. Air Permeability/Leakage: To meet the façade performance requirements when tested in accordance to EN or equivalent ASTM standard.

C. Capable of withstanding ground and water pressure.

2.3 AUXILIARY MATERIALS

A. General: Auxiliary materials recommended by weather barrier manufacturer for intended use and compatible with weather barrier. Liquid-type auxiliary materials shall comply with VOC limits of local authorities.

B. Primer: Liquid waterborne primer recommended for substrate by manufacturer of weather barrier material.

C. Termination Mastic: Cold fluid-applied elastomeric liquid; trowel grade.

D. Substrate Patching Membrane: Manufacturer's standard trowel-grade substrate filler.

E. Adhesive and Tape: Weather barrier manufacturer's standard adhesive and pressure-sensitive adhesive tape.

F. Elastomeric Flashing Sheet: ASTM D 2000, 2BC415 to 3BC620, minimum 1.3- 1.6mm thick, cured sheet neoprene with manufacturer's recommended contact adhesives and lap sealant with aluminum termination bars and stainless-steel fasteners.

G. Preformed Silicone: Sealant Extrusion: Manufacturer's standard system consisting of cured low-modulus silicone extrusion, sized to fit opening widths, with a single- component, neutral-curing, Class 100/50 (low-modulus) silicone sealant for bonding extrusions to substrates.

H. Joint Sealant: ASTM C 920, single-component, neutral-curing silicone; Class 100/50 (low-modulus), Grade NS, Use NT related to exposure, and, as applicable to joint substrates indicated, Use O. Comply with Division 07 Section "Joint Sealants."

2.4 STANDARD REFERENCED

A. The minimum standards for products specified in this section shall be including as under but not limited to the following. Except as otherwise specified herein, perform work in accordance with specifications, codes and standards cited therein, and their latest editions, revisions, applicable addenda and supplements. Where there is

conflict between the referenced standards the most stringent of the conditions/requirements shall be applicable.

1. ASTM E 96: Test Methods for Water Vapor Transmission of Materials.
2. ASTM D 570: Test Method for Water Absorption of Plastics.
3. ASTM E-154: Test Method for Water Vapor Retarders contact with Earth Under Concrete Slabs, on Walls or as Ground Cover.
4. ASTM D 1004: Test Method for Initial Tear Resistance of Plastic Film and Sheeting.
5. ASTM D 1938: Test Method for Tear Propagation Resistance of Plastic Film and Thin Sheeting by a Single Tear Method.
6. ASTM D 1876: Test Method for Peel Resistance of Adhesives.
7. ASTM D 412: Test Methods for Vulcanized Rubber & Thermoplastic Rubbers and Thermoplastic Elastomers Tension.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements and other conditions affecting performance.
   1. Verify that substrates are sound and free of oil, grease, dirt, excess mortar, or other contaminants.
   2. Verify that concrete has cured and aged for minimum time period recommended by air barrier manufacturer.
   3. Verify that concrete is visibly dry and free of moisture. Test for capillary moisture by plastic sheet method according to ASTM D 4263.
   4. Verify that masonry joints are flush and completely filled with mortar.
   5. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 SURFACE PREPARATION

A. Clean, prepare, and treat substrate according to manufacturer's written instructions. Provide clean, dust-free, and dry substrate for air barrier application.

B. Mask off adjoining surfaces not covered by weather barrier to prevent spillage and overspray affecting other construction.

C. Remove grease, oil, form-release agents, paints, curing compounds, and other penetrating contaminants or film-forming coatings from concrete.

D. Remove fins, ridges, mortar, and other projections and fill honeycomb, aggregate pockets, holes, and other voids in concrete with substrate-patching membrane.

E. Remove excess mortar from masonry ties, shelf angles, and other obstructions.

F. Prepare, fill, prime, and treat joints and cracks in substrates. Remove dust and dirt from joints and cracks according to ASTM D 4258.
   1. Install EPDM weather membrane and center over treated construction and

contraction joints and cracks exceeding a width of 1.6mm.

G. At changes in substrate plane, apply sealant or termination mastic beads at sharp corners and edges to form a smooth transition from one plane to another.

H. Cover gaps in substrate plane and form a smooth transition from one substrate plane to another with stainless-steel sheet mechanically fastened to structural framing to provide continuous support for weather barrier.

3.3 INSTALLATION

A. Install EPDM membrane sheets according to weather barrier manufacturer's written instructions

B. Corners: Prepare, prime, and treat inside and outside corners according to manufacturer's recommendations.
   1. Install EPDM membrane strips centered over vertical inside corners. Install 19mm fillets of termination mastic on horizontal inside corners.

C. Prepare, treat, and seal vertical and horizontal surfaces at terminations and penetrations with termination mastic and according to manufacturer's written instructions.

D. Apply primer to substrates at required rate and allow to dry. Limit priming to areas that will be covered by air barrier sheet in same day. Re-prime areas exposed for more than 24 hours.

E. Wall Openings: Prime concealed perimeter frame surfaces of curtain walls, claddings, and doors. Apply EPDM membrane transition strip so that a minimum of 75mm of coverage is achieved over both substrates. Maintain 75mm of full contact over firm bearing to perimeter frames with not less than 25mm of full contact.
   1. EPDM Transition Strip: Roll firmly to enhance adhesion.
   2. EPDM/Elastomeric Flashing Sheet: Apply adhesive to wall, frame, and flashing sheet. Install flashing sheet and termination bars, fastened at 150mm O.C. Apply lap sealant over exposed edges and on cavity side of flashing sheet.
   3. Preformed Silicone-Sealant Extrusion: Set in full bed of manufacturers recommended silicone sealant applied to walls, frame, and membrane.

F. Fill gaps in perimeter frame surfaces of curtain walls, claddings, doors, and miscellaneous penetrations of weather barrier membrane with foam sealant. Surface of any repair work is made good to match the surrounding areas.

G. At the end of each working day, seal top edge of membrane to substrate with termination mastic.

H. Apply joint sealants forming part of weather barrier assembly within manufacturer's recommended application temperature ranges. Consult manufacturer when sealant cannot be applied within these temperature ranges. Eventually, the contractor is responsible for proper application of the sealant.

I. Repair punctures, voids, and deficient lapped seams in weather barrier. Slit and flatten fishmouths and blisters. Patch with air barrier sheet extending 150mm beyond repaired areas in all directions.

J. Do not cover weather barrier until it has been inspected and tested by Employer's testing agency.

3.4 APPLICATION OF PENETRATION SEALANT

A. Seal membrane terminations, heads of mechanical fasteners, masonry tie fasteners, around penetrations, duct work, electrical and other apparatus extending through the barrier membrane and around the perimeter edge of membrane terminations at door frames with specified penetration sealant.

3.5 FIELD QUALITY CONTROL

A. Make notification when sections of work are complete to allow review prior to covering water, vapor and air resistive weather barrier system.

B. Cooperate with Employer’s independent testing agency, which will observe substrate and membrane installation prior to placement of cladding systems and provide written documentation of observations.

3.6 CLEANING AND PROTECTION

A. Protect weather barrier system from damage during application and remainder of construction period, according to manufacturer's written instructions.

B. Clean spills, stains, and soiling from adjacent construction that would be exposed in the completed work using cleaning agents and procedures recommended by manufacturer of affected construction.

END OF SECTION 07 2500.01

SECTION 07 6113.01 - STANDING SEAM SHEET METAL ROOFING

PART 1 - GENERAL

Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.01 SECTION INCLUDES

A. Design, engineer and furnish all labor, materials, tools and equipment and perform all services and operations necessary for the complete supply and installation of field installed standing seam roof panels including insulation, vapor/ water control layer, gutters and all accessory components and support system.
   1. At Tower H32 Event Box roof, interfaced with the adjacent skylight and façade systems.
   2. Integrated with fall arrest/ life-line system as indicated in the drawings, for the cleaning and safe access.

B. Design to follow the architectural intent indicated in the drawings.

C. This section to be read in conjunction with the related wall types/ exterior wall system drawings submitted as a part of the tender package.

D. Delegated Design Responsibility: Notwithstanding the drawings and other information provided, the Contractor shall be fully responsible for the design including comprehensive engineering analysis by a qualified professional engineer, detailing of the works, described herein, and shall provide a complete, and secure installation, which shall meet the specified performance requirements and the and design criteria indicated.

E. Installers of standing seam roof panel system shall be subject to approval.

1.02 RELATED SECTIONS

1.	Section 01 4500	-	Quality Control
2.	Section 05 5000	-	Metal Fabrication
3.	Section 07 2100.01	-	Thermal Insulation
4.	Section 07 2500.01	-	Weather Barrier
5.	Section 07 6200.01	-	Sheet Metal Flashing and Trim
6.	Section 07 9200.01	-	Joint Sealants
7.	Section 08 4000	-	Aluminium Works
8.	Section 08 4413.01	-	Glazed Aluminium Curtain Wall
9.	Section 11 8123.01	-	Façade Cleaning and Maintenance Equipment

1.03 REFERENCES

A. Product manufacture, testing and installation shall comply with the following references unless otherwise stated in the specification or otherwise approved.

1. AAMA 2605: Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels.
2. ASTM A 653: Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvanized) by the Hot-Dip Process.
3. ASTM A 755: Specification for Steel Sheet, Metallic Coated by the Hot-Dip Process and Prepainted by the Coil-Coating Process for Exterior Exposed Building Products.
4. ASTM B 32: Specification for Solder Metal.
5. ASTM B 209: Specification for Aluminum and Aluminum-Alloy Sheet and Plate.
6. ASTM C 920: Specification for Elastomeric Joint Sealants.
7. ASTM D 2244: Test Method for Calculation of Color Differences from Instrumentally Measured Color Coordinates.
8. ASTM D 4214: Test Methods for Evaluating Degree of Chalking of Exterior Paint Films.
9. ASTM D 4397: Specification for Polyethylene Sheeting for Construction, Industrial, and Agricultural Applications
10. NAAMM: Metal Finishes Manual for Architectural and Metal Products.
11. SSPC-Paint 12: Paint Specification No.12: Cold Applied Asphalt Mastic (Extra Thick Film).
12. UL 580: Tests for Uplift Resistance of Roof Assemblies.
13. Sheet Metal and Air Conditioning Contractors' National Association: Architectural Sheet Metal Manual. 1993 (Addendum 1997).

B. All applicable local codes, regulations, etc. of Authorities having Jurisdiction. Note: Contractor shall note that he shall keep copies of all of the referenced standards on the site and shall provide the Engineer copy of the requested referenced standard as and when requested.

1.04 PERFORMANCE REQUIREMENTS

A. General: Design and engineer metal roof panel assemblies that comply with performance requirements specified as determined by testing manufacturers' standard assemblies similar to those indicated for this Project, by a qualified testing and inspecting agency. Roofing System should be designed, fabricated and installed to withstand local weather conditions.

B. Air Infiltration: Air leakage through assembly of not more than 0.3 L/s per sq. m of roof area when tested according to ASTM E 1680 at the following test-pressure difference:
   1. Positive Preload Test-Pressure Difference: Greater than equal to 720 Pa and the greater of 75 percent of building live load or 50 percent of building design positive wind-pressure difference.
   2. Test-Pressure Difference: Negative 75 Pa.
   3. Negative Preload Test-Pressure Difference: 50 percent of design wind-uplift-pressure difference.

C. Water Penetration: No water penetration when tested according to ASTM E 1646 at the following test-pressure difference:
   1. Test-Pressure Difference: 137 Pa.
   2. Positive Preload Test-Pressure Difference: Greater than or equal to 720 Pa or 50 percent of building design positive wind-pressure difference.
   3. Negative Preload Test-Pressure Difference: 50 percent of design wind-uplift-pressure difference.

D. Wind-Uplift Resistance: Provide custom-fabricated sheet metal roofing capable of resisting the following design negative uplift pressure. Provide clips, fasteners, and clip spacing’s of type

indicated and with capability to sustain, without failure, a load equal to 3 times the design negative uplift pressure.

E. Wind-Uplift Resistance: Provide portable roll-forming equipment capable of producing sheet metal roofing assemblies that comply with UL 580 for Class 90 wind-uplift resistance.

F. Structural Performance: Provide metal roof panel assemblies capable of withstanding the effects of gravity loads and applicable wind loads and stresses within limits and under conditions indicated, based on testing according to ASTM E 1592 or equivalent standards.
   1. Wind load: Determine design wind loads applicable to Project based on basic wind speed for Project as indicated on drawings and calculated according to requirements of local authorities or the American Society of Civil Engineers' ASCE 7-10, "Minimum Design Loads for Buildings and Other Structures," 6 Method-2, "Analytical Procedure".
   2. Deflection Limits: Engineer metal roof panel assemblies to withstand design loads with vertical deflections no greater than 1/180 of the span.

G. U-Value of the complete standing seam sheet metal roofing system including thermal insulation (and structural slab underneath, if any) shall not be more than 0.20 W/m^2K as indicated on the design documents.

H. Comply with acoustic requirement as per project acoustic report at its latest revision.

I. Thermal Movements: Provide metal roof panel assemblies that allow for thermal movements resulting from maximum change (range) in ambient and surface temperatures by preventing buckling, opening of joints, overstressing of components, failure of joint sealants, failure of connections, and other detrimental effects.

J. Base engineering calculation on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.
   1. Temperature Change (Range): 70 deg. C, ambient; 100 deg. C, material surfaces.
   2. Night time or lower range: +10 deg. C.

1.05 SUBMITTALS:

A. Product Data: Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for each type of metal roof panel and accessory.

B. Manufacturer's Instructions: Submit Data books, manufacturers printed installation instructions, Operating Instructions Manual for work of this section and as specified.

C. Shop Drawings: Show fabrication and installation layouts of metal roof panels; details of edge conditions, joints, panel profiles, corners, anchorages, trim, flashings, closures, and accessories; and special details. Distinguish between factory- and field-assembled work.
   1. For installed products indicated to comply with design loads, include structural analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

D. Coordination Drawings: Roof plans drawn to scale and coordinating penetrations and roof-mounted items. Show the following:
   1. Roof panels and attachments.
   2. Purlins and rafters.
   3. Roof-mounted items including pipe supports and penetrations.

E. Samples for Verification: For each type of exposed finish required, prepared on Samples of size indicated below.
   1. Metal Panels: 300 mm long by actual panel width. Include fasteners, clips, closures, and other metal roof panel accessories.
   2. Trim and Closures: 300 mm long. Include fasteners and other exposed accessories.
   3. Accessories: 300 mm long Samples for each type of accessory.
   4. Vapour barriers: 150 mm square samples.

F. Sample shall represent physical and chemical properties, finishes and colors to be supplied.

G. Qualification Data: For qualified testing agency, Manufacturer, and Installer.

H. Field quality-control inspection reports.

I. Welding certificates.

J. Product Test Reports: Based on evaluation of comprehensive tests performed by a qualified testing agency, for the following:
   1. Metal Panels: Include reports for air infiltration, water penetration, thermal performance and structural performance.

K. Maintenance Data: For metal roof panels to include in maintenance manuals.

L. Warranties: Special warranties specified in this Section.

1.06 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as minimum.

B. Installer Qualifications: Engage an experienced installer with minimum 10 years experience to assume engineering responsibility and perform work of this Section who has specialized in installing systems similar to those required for this Project and who is acceptable to manufacturer.
   1. Installer's responsibilities include fabricating and installing metal roof panel assemblies and providing professional engineering services needed to assume engineering responsibility.
   2. Engineering Responsibility: Preparation of data for metal roof panels, including Shop Drawings, based on testing and engineering analysis of manufacturer's standard units in assemblies similar to those indicated for this Project.

C. Testing Agency Qualifications: To qualify for approval, an independent testing agency must demonstrate to the Engineer satisfaction, based on evaluation of agency-submitted criteria conforming to ASTM E699, that it has the experience and capability to satisfactorily conduct the testing indicated without delaying the Work.

D. Professional Engineer Qualifications: A professional engineer who is legally qualified to practice in the jurisdiction where Project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for installations of standing seam roofing system that are similar to those indicated for this Project in material, design and extent.

E. Source Limitations: Obtain roofing system from one source and by a single manufacturer confirming the availability of components during the warranty period.

F. Mockups: Build mockups to verify selections made under sample Submittals and to demonstrate aesthetic effects and qualities of materials and execution.
   1. Locate mock-up in the location and of the size indicated or, if not indicated, as directed by the Engineer.
   2. Mockup shall be complete with adjoining material and support system to the satisfaction of the Engineer. If the required standard is not achieved, the Contractor shall replace (at the contractor's cost) until the required standard is achieved as judged by the Engineer.
   3. Demonstrate the proposed range of aesthetic effects and workmanship.
   4. Obtain written approval of mock-up from the Engineer before starting fabrication.
   5. Maintain mock-ups during construction in an undisturbed condition as a standard for judging the completed Work.
   6. Mock up shall become part of finished work if approved by the Engineer at the time of Substantial Completion.

1.07 DELIVERY, STORAGE AND HANDLING

A. Delivery, storage and handling shall be as per recommendations of the manufacturer. Deliver components, sheets, metal roof panels, and other manufactured items so as not to be damaged or deformed. Package metal roof panels for protection during transportation and handling.

B. Unload, store, and erect metal roof panels in a manner to prevent bending, warping, twisting, and surface damage.

C. Stack metal roof panels on platforms or pallets, covered with suitable weathertight and ventilated covering. Store metal roof panels to ensure dryness. Do not store metal roof panels in contact with other materials that might cause staining, denting, or other surface damage.

D. Protect strippable protective covering on metal roof panels from exposure to sunlight and high humidity, except to extent necessary for period of metal roof panel installation.

A. Deliver components in the manufacturer's original protective packaging.

1.08 PROJECT CONDITION

A. Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit assembly of metal roof panels to be performed according to manufacturers' written instructions and warranty requirements.

B. Field Measurements: Verify locations of roof framing and roof opening dimensions by field measurements before metal roof panel fabrication and indicate measurements on Shop Drawings.

C. Ensure ambient and surface temperatures and joint conditions are suitable for the installation of materials such as sealants, waterproofing membrane.

1.09 COORDINATION

A. Coordinate metal panel roof assemblies with other adjoining work to provide a leakproof, secure, and noncorrosive installation.

1.010 WARRANTY

A. Special Warranty: Submit a written warranty executed by Contractor, Installer and Manufacturer agreeing to repair or replace components of metal roof panel assembly system that fail in materials, finish or workmanship within the specified warranty period. Failures include, but are not limited to, the following:
   1. Structural failures including, but not limited to rupturing, cracking, or puncturing.
   2. Noise or vibration caused by thermal movement.
   3. Failure of system to meet performance requirements.
   4. Deterioration of metals, metal finishes, and other materials beyond normal weathering.
   5. Air infiltration and water penetration.

B. Warranty Period: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.01 ACCEPTABLE MANUFACTURERS

A. Manufacturers and products specified hereinafter are for the purpose of establishing minimum quality standards. Manufacturers which are equal to or better than those specified, and which conform to the design requirements and color selections, may be acceptable subject to approval provided the Contractor requests in writing within 18 days prior to bid date. This request shall be accompanied by:
   1. A certificate of compliance from the material manufacturer stating that the proposed products meet or exceed the requirements of this specification.
   2. Documented proof that the proposed material has a proven record of successful performance for more than 10 years, confirmed by at least five local projects.

B. Section Manufacturers/ Installer
   1. AK Cladding LLC, Dubai.
   2. Kalzip (Corus), Germany.
   3. Bemo Systems, Germany.
   4. Rigidal Industries, Dubai.
   5. Technical Supplies & Services Co.L.L.C, Dubai.
   6. Or Equivalent, subjected to engineer's acceptance.

Condition to qualify "equivalent": Qualification exercise consists of assessment of technical adaptability of the design and the material, quality procedures maintained, minimum 10 years of its usage in similar project and credibility of its performance record and other project references. All cost related to this exercise shall be borne by the proposer. The proposer shall agree to use above listed pre-approved system manufactures, should the proposed system is not acceptable to the engineer, after the completion of qualification exercise.

2.02 MATERIALS (ROOF PANELS)

A. The roofing system shall be Aluminum standing seam roof system according to certification: Z-14.1-182 manufactured from Aluminum alloy in accordance to EN- AW-3004; material thickness of minimum 1.0 mm and PVDF finish on the exposed surface. The color to be selected and approved by the Engineer. The material properties of the non-profiled coil are as follows:
   1. Ultimate tensile strength: minimum 215N/mm^2.

2.	0.2% Proof Stress	:	minimum 195 N/mm^2.
3.	Elongation at failure	:	3%

B. The panels shall be supplied in single lengths with no overlaps in length direction and cover-width to be between 250 mm and 600 to suit the lay-out of the roof. Welding should be avoided as much as possible. The standing seam height shall be 65mm or as indicated on drawings. The sheets are to be laid to meet the requirements of the building geometry. The aluminum roof sheets will be fixed by concealed extruded aluminum halter clips.

C. The aluminum roof elements shall be joined longitudinally with other complementary aluminum roof elements and sealed via an electrically operated seam closure machine over the concealed extruded aluminum halter clips. The final assembled system shall be capable of allowing for the possibility of transmitting reaction (lateral and vertical) forces resulting from positive and/or negative loads and also accommodating unobstructed thermal movements resulting from the specified ambient temperature range.

2.03 THERMAL INSULATION FOR FIELD-ASSEMBLED METAL ROOF PANELS

A. Mineral-Fiber-Blanket Insulation (Unfaced): ASTM C 665, Type I (blankets without membrane facing); consisting of fibers, with maximum flame-spread and smoke developed indexes of 25 and 50, respectively, passing ASTM E 136 for combustion characteristics.

B. Mineral-Fiber-Blanket Insulation (Faced): ASTM C 665 Type III (blankets with reflective membrane facing); Class A (membrane-faced surface with a flame spread of 25 or less); Category 1 (membrane is a vapor barrier) faced with foil-scrim-kraft, foil-scrim, or foil-scrim-polyethylene vapor retarder membrane on 1 face.

C. Thermal Insulation: Compression protected, water repellant mineral wool insulating board with a special fixing system. The insulation material shall be able to provide the necessary thermal and acoustic requirements of the Project.

D. Thermal Spacer Blocks: Fabricated from extruded polystyrene, 25 mm thick or as indicated on drawings.

E. Vapor-Retarder Facing: As recommended by the manufacturer.

F. Retainer Strips: 0.5mm thick, formed, galvanized steel colored to match insulation facing.

2.04 UNDERLAYMENT MATERIALS

A. Vapor control and air-lock layer: As standard with the manufacturer.

2.05 TRAPEZOIDAL SHEET (ROOF DECK SHEET)

A. As standard with the manufacturer.

2.06 MISCELLANEOUS MATERIALS

A. General: Provide materials and accessories required for a complete roof panel assembly and as recommended by panel manufacturer, unless otherwise indicated.

B. Thermal Spacers: Where panels attached directly to purlins, provide thermal spacers recommended by panel manufacturer.

C. Fasteners: Self-tapping screws, bolts, nuts, self-locking rivets and bolts, end-welded studs, and other suitable fasteners designed to withstand loads.
   1. Use stainless-steel fasteners for exterior applications and galvanized steel fasteners for interior applications.
   2. Provide exposed fasteners with heads matching color of panel by means of plastic caps or factory applied coating.
   3. Provide metal-backed neoprene washers under heads of exposed fasteners bearing on weather side of panels.
   4. Locate and space exposed fasteners in true vertical and horizontal alignment. Use proper tools to obtain controlled uniform compression for positive seal without rupture of neoprene washer.

D. Accessories: Unless otherwise specified, provide components required for a complete roof panel assembly including trim, copings, fasciae, mullions, sills, corner units, ridge closures, clips, seam covers, battens, flashings, gutters, sealants, gaskets, fillers, closure strips, and similar items. Match materials and finishes of panels.
   1. Closure Strips: Closed-cell, self-extinguishing, expanded, cellular, rubber or cross-linked, polyolefin-foam flexible closure strips. Cut or premold to match configuration of panels. Provide closure strips where indicated or necessary to ensure weathertight construction.
   2. Sealing Tape: Pressure-sensitive, 100 percent solids, polyisobutylene compound sealing tape with release paper backing. Provide permanently elastic, nonsag, nontoxic, nonstaining tape.
   3. Elastormeric Joint Sealant: ASTM C 920, of base polymer. Type, grade, class, and use classifications required to seal joints in panel roofing and remain weathertight. Provide sealant recommended by panel manufacturer.

E. Expansion-Joint Sealant: For hooker-type expansion joints that must be free to move, provide nonsetting, nonhardening, heavy-bodied polyisobutylene sealant.

F. Primer: Rust-inhibitive primer recommended by panel manufacturer for finish coat.

2.07 FABRICATION

A. General: Fabricate and finish metal roof panels and accessories at the factory to greatest extent possible, by manufacturer's standard procedures and processes, as necessary to fulfill indicated performance requirements demonstrated by laboratory testing. Comply with indicated profiles and with dimensional and structural requirements.

B. Provide factory-formed metal roof panels designed to be field assembled by lapping and interconnecting raised side edges of adjacent panels with joint type indicated and mechanically attaching panels to supports using concealed clips in side laps. Include clips, cleats, pressure plates, and accessories required for weathertight installation.

2.08 FINISHES, GENERAL

A. All metal surfaces exposed to the building exterior, including framing members shall receive PVDF High performance organic finish:
   1. Finish shall be applied by an applicator licensed by coating manufacturer and in strict compliance with manufacturer’s specification for cleaning, priming, coating application and quality control.

2. Three coats of polyvinylidine fluoride (PVDF) resin for high performance thermoplastic organic coating system containing a minimum 70% Kynar 500 resin by weight combined with proprietary pigments for spray system. Thickness of coating 30-35 microns minimum.
3. Prepare, pretreat and apply coating to exposed metal surface to comply with AAMA 2605.
4. Color: To match approved sample; SRI: 3-year aged SRI of 0.64.
5. Preferred manufacturers: AkzoNobel/ Jotun or Equivalent as per Engineers approval.

B. Concealed Finish: Apply pretreatment and manufacturer's standard finish, consisting of prime coat and wash with a minimum total dry film thickness of 0.5 mil.

C. Identification marks shall not appear on finished surfaces.

D. Adhesion: No removal of finish after 1.5mm cross hatching to base metal, impacting to the point of metal rupture, and subjecting to application and quick removal of cellophane tape.

E. Appearance of Finished Work: Variations in appearance of abutting or adjacent pieces are acceptable if they are within one-half of the range of approved Samples. Noticeable variations in the same piece are not acceptable. Variations in appearance of other components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

F. Protect mechanical and painted finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

G. In order to minimize appearance differences, all requirements shall be processed in one production run using the same batch of paint.

2.09 PANEL SUPPORTS AND ANCHORAGE

A. Secondary Framing: Unless indicated on drawings, provide components complying with the Light Gage Structural Institute's "Guide Specifications,"
1. Sub-Purlins: C- or Z-shaped sections fabricated from minimum 1.5mm thick, shop-painted, roll-formed steel. Purlin spacers fabricated from minimum 2mm thick, cold-formed, galvanized steel sections.
2. Secondary structural members, shall be manufacturer's standard sections fabricated from minimum 2mm thick, cold-formed galvanized steel.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, metal roof panel supports, and other conditions affecting performance of work.
1. Verify that structural panel support members and anchorages have been installed within alignment tolerances required by metal roof panel manufacturer.
2. For the record, prepare written report, endorsed by Installer, listing conditions detrimental to performance of work.

B. Examine roughing-in for components and systems penetrating metal roof panels to verify actual locations of penetrations relative to seam locations of metal roof panels before metal roof panel installation.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.02 PREPARATION

A. Clean substrates of substances harmful to insulation, including removing projections capable of interfering with insulation attachment.

B. Miscellaneous Framing: Install subpurlins, eave angles, furring, and other miscellaneous roof panel support members and anchorage according to metal roof panel manufacturer's written recommendations.

3.03 PANEL INSTALLATION

A. General: Comply with panel manufacturer's written instructions and recommendations for installation, as applicable to project conditions and supporting substrates. Anchor panels and other components of the Work securely in place, with provisions for thermal and structural movement.
   1. Cut sheets and flashings to give clean true lines, with no distortion. Remove burrs and any lubricant.
   2. Cut openings in sheets for outlets, vent pipes, etc to the minimum size necessary and as per the panel manufacturer’s recommendations.
   3. Lay sheets with exposed joints of side laps away from prevailing wind unless shown otherwise on drawings.
   4. Ensure that ends of sheets, and end laps are fully supported with fixings at top of lap.
   5. Ensure that raking cut edges of hips and valleys are fully supported.
   6. Drill all holes and install fasteners perpendicular to the surface of the cladding-position fasteners at regular intervals in straight lines, centered on support bearings.
   7. Locate fasteners centrally within holes that are oversized.
   8. Remove dust and any other foreign matter before finally fixing sheets into position.
   9. Protect sheets adequately during fixing and up to practical completion against mechanical damage and disfigurement. Rectify any defects as quickly as practicable to minimize damage and nuisances.
   10. Install fasteners to correct tightness using only special tools recommended by the fastener manufacturer. When used, screw guns must be fitted with depth sensitive devices and used on the correct speed.
   11. Check fastening on completion and adjust as necessary to ensure that they are watertight, and sheeting is secure but no buckled or distorted.
   12. Paint all cut edges to match face finish.

B. Standing seam roof sheets to be rolled from coil on site and installed by mechanically seaming sheets to support clips with zipping machine as per manufacturer’s instructions. The standing seam roof and the liner should be thermally isolated throughout.

C. Structural Movement Joints: Leave space between sheets to coincide with structural movement joints. Fix movement joint cover to sheets on one side only. Movement joint cover details to be as per panel manufacturer’s recommendations.

D. Separate dissimilar metals by painting each metal surface in area of contact with primer or by applying sealant or tape recommended by manufacturer for this purpose or by other permanent separation as recommended by manufacturers of dissimilar metals.

E. Coat back side of metal panels with acrylic based protective paint/ coating where it will contact wood, ferrous metal, or cementitious construction.

F. Joint Sealers: Install gaskets, joint fillers, and sealants where indicated and where required for weatherproof performance of panel assemblies. Provide types of gaskets, fillers, and sealants indicated or, if not otherwise indicated, types recommended by panel manufacturer.
   1. Seal panel end laps with double beads of tape or sealant, full width of panel. Seal side joints where recommended by panel manufacturer.

G. Installation Tolerances: Shim and align panel units within installed tolerance of 6 mm in 6 m on slope and location lines as indicated and within 3-mm offset of adjoining faces and of alignment of matching profiles.

3.04 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified independent testing and inspecting agency to perform inspections and prepare reports.

B. Manufacturer's Field Service: Engage a factory-authorized service represent to inspect completed metal roof panel installation, including accessories. Report results in writing.

C. Remove and replace applications of metal roof panels where inspections indicate that they do not comply with specified requirements.

D. Additional inspections, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.

3.05 CLEANING AND PROTECTING

A. Damaged Units: Replace panels and other components of the Work that have been damaged or have deteriorated beyond successful repair by finish touchup or similar minor repair procedures.

B. Cleaning: Remove temporary protective coverings and strippable films, if any, as soon as each panel is installed. On completion of panel installation, clean finished surfaces as recommended by panel manufacturer and maintain in a clean condition during construction.

END OF SECTION 07 6113.01

SECTION 07 6200.01 - SHEET METAL FLASHING AND TRIMS

PART 1 - GENERAL

Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.1 SECTION INCLUDES

A. Furnish all labor, materials, tools appliances and equipment, and perform all operations and services necessary for the complete installation of the metal sheet flashing and all related components as required and indicated on the drawings; including but not limited to the following:
   1. Building envelope elements, curtain wall, wall cladding and interfaces.
   2. Standing seam sheet metal roofing and interfaces.
   3. Sealant in conjunction with sheet Metal Work specified herein.

B. Design to follow the architectural intent indicated in the drawings.

C. This section to be read in conjunction with the related wall types/ exterior wall system drawings submitted as a part of the tender package.

1.2 RELATED SECTIONS:

A. This schedule is intended to be used as a helpful indication of the related sections within the Project Specifications. It is not necessarily comprehensive or complete and it is the Contractor's responsibility to ascertain all applicable sections required to understand the full scope of Works intended.

B. Work to be coordinated and used in conjunction with this specification includes but is not restricted to:

1.	Section 01 4500	-	Quality Control
2.	Section 03 3000	-	Cast-in-Place Concrete
3.	Section 04 2200	-	Concrete Unit Masonry
4.	Section 05 7500.01	-	Decorative Formed Metal
5.	Section 07 2100.01	-	Thermal Insulation
6.	Section 07 2500.01	-	Weather Barrier
7.	Section 07 9200.01	-	Joint Sealants
8.	Section 08 1116.01	-	Alum. Doors and Frames
9.	Section 08 4000	-	Aluminium Works
10.	Section 08 4413.01	-	Glazed Alum. Curtain Wall
11.	Section 08 9100.01	-	Louvers

1.3 PERFORMANCE REQUIREMENTS

A. General: Install sheet metal flashing and trim to withstand wind loads, structural movement, thermally induced movement, and exposure to weather without failure due to defective manufacture, fabrication, installation or other defects in construction. Completed sheet metal flashing and trim shall not rattle, leak, or loosen, and shall remain watertight.

B. Water Infiltration: Provide sheet metal flashing that do not allow water infiltration to building interior.

C. Thermal Movements: Provide sheet metal flashing and trim that allow for thermal movements resulting from the following maximum change (range) in ambient and surface temperatures by preventing buckling, opening of joints, hole elongation, overstressing of components, failure of joint sealants, failure of connections, and other detrimental effects. Provide clips that resist rotation and avoid shear stress as a result of sheet metal and trim thermal movements. Base engineering calculation on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.

D. Base engineering calculation on surface temperatures of materials due to both solar heat gain and nighttime heat loss.
   1. Temperature Change (Range): 70^{0}C, ambient; 100^{0}C, material surfaces.
   2. Night time or lower range +10 deg. C.

1.4 SUBMITTALS

A. Product Data including manufacturer's material and finish data, installation instructions, and general recommendations for each specified flashing material and fabricated product.

B. Shop Drawings of each item specified showing layout, profiles, and methods of joining, sealing and anchorage details.

C. Samples of sheet metal flashing, trim, and accessory items, in the specified finish. Where finish involves normal color and texture variations, include Sample sets composed of 2 or more units showing the full range of variations expected.

D. Method statement for the installation, maintenance and repair of sheet metal flashing/trims assemblies.

E. Schedule of flashing types and locations.

1.5 QUALITY ASSURANCE

A. Submit document proof that the material has a proven record of successful performance for more than 10 (ten) years, confirm by at least 5 (five) local projects.

B. Installer Qualifications: Engage an experience Installer with minimum 5 (five) years' experience who has completed sheet metal flashing and trim work similar in material, design, and extent to that indicated for this Project and with a record of successful in- service performance.

C. Sheet Metal Standard for Flashing and Trim: Comply with NRCA's "The NRCA Roofing Manual" and SMACNA's "Architectural Sheet Metal Manual" requirements for dimensions and profiles shown unless more stringent requirements are indicated.

D. Comply with most stringent of the local code or acceptable national/international building standard requirements for VOC's limits.

E. Mock-ups: Prior to installing sheet metal flashing and trim, construct mock-ups in a location as directed to verify selections made under Sample submittals and to demonstrate aesthetic

effects as well as qualities of materials and execution.

1. Locate mock-up in the location and of the size indicated or, if not indicated, as directed by the Engineer.
2. Built Mock-Up to the satisfaction of the Engineer. If the required standard is not achieved, contractor shall replace (at the contractor's cost) until the required standard is achieved as judged by the Engineers.
3. Demonstrate the proposed range of aesthetic effect and workmanship.
4. Obtain written approval of mock-up from the Engineer before starting fabrication.
5. Maintain mock-ups during construction in an undisturbed condition as a standard for judging the completed work.
6. Approved mock-ups in an undisturbed condition at the time of substantial completion may become part of the completed work if approved by the Engineer.
7. Mock-up approval is to be well recorded, documented and circulated among all the parties involved.

1.6 DELIVERY, STORAGE AND HANDLING

A. Deliver sheet metal flashing materials and fabrications undamaged. Protect sheet metal flashing and trim materials and fabrications during transportation and handling.

B. Unload, store, and install sheet metal flashing materials and fabrications in a manner to prevent bending, warping, twisting and surface damage.

C. Stack materials on platforms or pallets, covered with suitable weather tight and ventilated covering. Do not store sheet metal flashing and trim materials in contact with other materials that might cause staining, denting, or other surface damage.

1.7 PROJECT CONDITIONS

A. Coordinate Work of this Section with interfacing and adjoining Work for proper sequencing of each installation. Ensure best possible weather resistance, durability of Work, and protection of materials and finishes.

PART 2 - PRODUCTS

2.1 METALS

A. Aluminium Sheet: BS 1470 alloy and/or ASTM B 209.

B. Aluminium Flashing (not cladding): Sheet Aluminium, mill finish or finish as required per exposure category (as per Section "Finish"), having a minimum thickness as specified herein below, for the applications indicated:
   1. Exposed to weather flashings and trim: 1.3mm thick
   2. Apron flashing, 1.3mm
   3. Base flashing, 1.3mm
   4. Coping covers, cap flashings, 1.3mm thick
   5. Roof penetrations: 1.3mm
   6. Gutters: 1.5mm thick
   7. Endplates, 2mm thick and 3mm thick with large effective spans

C. Semi-Exposed to view aluminium (roof flashing behind parapet walls, below wall cladding and above roof level, and other such locations): pre-finished aluminium sheets, minimum 1.3mm thick.

D. Semi-Exposed to view steel (roof flashing behind parapet walls, below wall cladding and above roof level, and other such locations): pre-coated galvanized steel sheets; ASTM A653/A653M, minimum 0.85mm thick.

E. Prefabricated sleeve flashing: Prefinished aluminium ASTM B221, alloy 6061-T6, pre-manufactured seamless spun aluminium complete with deck flange, mastic tape and heat shrink rubber collar.

2.2 MISCELLANEOUS MATERIALS AND ACCESSORIES

A. Reglets: Units of type, material and profile indicated, formed to provide secure interlocking of separate reglet and counterflashing pieces, and compatible with flashing indicated with factory-mitered and welded corners and junctions.
   1. Material: Aluminium, 0.6mm thick.
   2. Surface-Mounted Type: Provide with slotted holes for fastening to substrate, with neoprene or other suitable weather proofing washers and with channel for sealant at top edge.
   3. Flexible Flashing Retainer: Provide resilient plastic or rubber accessory to secure flexible flashing in reglet where clearance does not permit use of standard metal counterflashing or where Drawings show reglet without metal counterflashing.
   4. Counterflashing Wind-Restraint Clips: Provide clips to be installed before counterflashing to prevent wind uplift of counterflashing lower edge.

B. Fasteners: Same metal as sheet metal flashing or other non-corrosive metal as recommended by sheet metal manufacturer. Match finish of exposed heads with material being fastened.

C. Elastomeric Sealant: ASTM C 920, elastomeric polyurethane, polysulfide, silicone, polymer sealant; of type, grade, class, and use classifications required to seal joints in sheet metal flashing and trim and remain airtight and watertight.

D. Mastic Sealant: Polyisobutylene; non-hardening, non-skinning, non-drying, non- migrating sealant.

E. Protective Coating: One component, plasto-elastic coating based on acrylic dispersion with resistance against carbonation and ingress of chloride ions, sulphates and oxygen. Suitable for use in hot climate conditions, Class A when tested according to ASTM E84.

2.3 FABRICATION, GENERAL

A. General: Custom fabricate sheet metal flashing and trim to comply with recommendations in SMACNA's "Architectural Sheet Metal Manual" that apply to design, dimensions, metal, and other characteristics of item indicated. Shop fabricated items where practicable. Obtain field measurements for accurate fit before shop fabrication.
   1. Seams for Aluminium: Fabricate non-moving seams with flat-lock seams. Form seams and seal epoxy seam sealer. Rivet joints for additional strength.

B. Comply with details shown to fabricate sheet metal flashing and trim that fit substrates and result in waterproof and weather-resistant performance once installed. Verify shapes and dimensions of surfaces to be covered before fabricating sheet metal.

C. Form exposed sheet metal Work that is without excessive oil canning, buckling, and tool marks and that is true to line and levels indicated, with exposed edges folded back to form hems.

D. Separate metal from non-compatible metal or corrosive substrates by coating concealed surfaces at locations of contact with permanent separation as recommended by manufacturer.

E. Conceal fasteners and expansion provisions where possible. Exposed fasteners are not allowed on faces of sheet metal exposed to general view.

2.4 FINISHES

A. All aluminium visible surfaces exposed to the building interior and exterior: Aluminium surfaces shall be finished with Super Durable Powder Coating complying with AAMA 2604 and that are thermoset, free from VOCs, with minimum dry film thickness of 60 microns.
   1. Color and Finish: To match approved color sample.

B. Prepare, pretreat and apply coating (including protective coat) to exposed metal surface to comply with AAMA 2604.

C. Concealed Aluminum Surfaces: Any aluminum sections which are completely concealed and therefore do not require the facing color shall be anodized to a minimum standard of AAMA 611 class II.

D. Identification marks shall not appear on finished surfaces.

E. Adhesion: No removal of finish after 1.5mm cross hatching to base metal, impacting to the point of metal rupture, and subjecting to application and quick removal of cellophane tape.

2.5 REFERENCED STANDARDS

A. AAMA 611: Voluntary Specification for Anodized Architectural Alminium.

B. ASTM B 209: Specification for Aluminum and Aluminum-Alloy Sheet and Plate

C. ASTM B 209M: Specification for Aluminum and Aluminum-Alloy Sheet and Plate [Metric]

D. ASTM C 920: Specification for Elastomeric Joint Sealants

E. ASTM C 1311: Specification for Solvent Release Sealants

F. SSPC-Paint 12: Paint Specification No.12: Cold Applied Asphalt Mastic (Extra Thick Film)

G. The minimum standards for products specified in this section shall be including as under but not limited to the following. Except as otherwise specified herein, perform work in accordance with specifications, codes and standards cited therein, and their latest editions, revisions, applicable addenda and supplements. Where there is conflict between the referenced standards the most stringent of the conditions/requirements shall be applicable.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions under which sheet metal flashing and trim are to be installed and verify that Work may properly commence. Do not proceed with installation until unsatisfactory conditions have been corrected.

3.2 INSTALLATION-GENERAL

A. General: Anchor sheet metal flashing and trim and other components of the Work securely in place, with provisions for thermal and structural movement. Use fasteners, welding rods, protective coatings, separators, sealants, and other miscellaneous items as required to complete sheet metal flashing and trim system.
   1. Torch cutting of sheet metal flashing and trim is not permitted.

B. Metal Protection: Where dissimilar metals will contact each other or corrosive substrates, protect against galvanic action by painting contact surfaces with permanent separation as recommended by fabricator or manufacturers of dissimilar metals.
   1. Coat side of uncoated aluminium sheet metal flashing and trim with manufacturer's recommended protective coating where flashing and trim will contact wood, ferrous metal, or cementitious construction.
   2. Underlayment: Where installing metal flashing directly on cementitious substrates, install a course of felt underlayment and cover with a slip sheet or install a course of polyethylene underlayment.
   3. Bed flanges in thick coat of roofing cement where required for waterproof performance.

C. Install exposed sheet metal flashing and trim without excessive oil canning, buckling, and tool marks.

D. Install sheet metal flashing and trim true to line and levels indicated. Provide uniform, neat seams with minimum exposure of welds, and elastomeric sealant.

E. Install sheet metal flashing and trim to fit substrates and to result in watertight performance. Verify shapes and dimensions of surfaces to be covered before fabricating sheet metal.
   1. Space cleats not more than 300mm apart. Anchor each cleat with two fasteners. Bend tabs over fasteners.

F. Expansion Provisions: Provide for thermal expansion of exposed flashing and trim. Space movement joints at a maximum of 3m with no joints allowed within 600mm of corner or intersection. Where lapped or bayonet-type expansion provisions cannot be used or would not be sufficiently watertight, form expansion joints of intermeshing hooked flanges, not less than 25mm deep, filled with elastomeric butyl sealant concealed within joints.

G. Fasteners: Use aluminium or stainless-steel fasteners.

H. Seal joints with elastomeric sealant as required for watertight construction.
   1. Where sealant-filled joints are used, embed hooked flanges of joint members not less than 25mm into sealant.
   2. Prepare joints and apply sealants to comply with requirements in specification section "Joint Sealants".

I. Counterflashing: Coordinate installation of counterflashing with installation of base flashing. Insert counterflashing in reglets or receivers and fit tightly to base flashing. Extend counterflashing 100mm over base flashing. Lap counterflashing joints a minimum of 100mm and bed with elastomeric sealant.

3.3 CLEANING AND PROTECTION

A. Clean exposed metal surfaces, removing substances that might cause corrosion of metal or deterioration of finishes.

B. Provide final protection and maintain conditions that ensure sheet metal flashing and trim Work during construction is without damage or deterioration other than natural weathering at the time of Substantial Completion.

END OF SECTION 07 6200.01

SECTION 07 8400.01 - FIRE STOPPING

PART 1 - GENERAL

Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.1 SECTION INCLUDES

A. Provide fire stop systems consisting of a material, or combination of materials, installed to retain the integrity of fire-rated construction by maintaining an effective barrier against the spread of flame, smoke, or gases through penetrations in fire-rated barriers or linear openings in or between fire rated barriers, curtain wall perimeter. Fire stops shall be used in locations including, but not limited to, the following:
   1. Perimeter fire-resistive joint systems consisting of floor to wall joints between perimeter-edge of fire resistance rated floor assemblies and exterior wall/curtain wall assembly.
   2. Vertical joint interface between the exterior wall/curtain wall and fire resistance rated wall/partition assemblies.
   3. Cavity barrier at the rainscreen at blank wall/ spandrel cladding.
   4. Sealant joints in fire-resistance-rated construction.
   5. Joints in smoke barriers.

B. Design to follow the architectural intent indicated in the drawings.

C. This section to be read in conjunction with the related wall types/ exterior wall system drawings submitted as a part of the facade tender package.

1.2 RELATED SECTIONS:

A. This schedule is intended to be used as a helpful indication of the related sections within the Project Specifications. It is not necessarily comprehensive or complete and it is the Contractor's responsibility to ascertain all applicable sections required to understand the full scope of Works intended.

B. Work to be coordinated and used in conjunction with this specification includes but is not restricted to:

1.	Section 01 4500	-	Quality Control
2.	Section 03 3000	-	Cast-In-Place Concrete
3.	Section 04 2200	-	Concrete Unit Masonry
4.	Section 05 7500.01	-	Decorative Formed Metal
5.	Section 07 2100.01	-	Thermal Insulation
6.	Section 07 6113.01	-	Standing Seam Sheet Metal Roofing
7.	Section 07 6200.01	-	Sheet Metal Flashing and Trims
8.	Section 07 9200.01	-	Joint Sealants
9.	Section 08 4413.01	-	Glazed Aluminium Curtain Wall

1.3 PERFORMANCE REQUIREMENTS

A. General: Provide fire stopping systems that are produced and installed to resist the spread of fire, according to requirements indicated, and the passage of smoke and other gases in line with local authority / Civil Defence requirements.

B. Rated Joint Systems: Provide fire-rated joint system with assembly ratings, as determined per ASTM E 1966.

C. For fire stopping exposed to view, provide products with flame-spread values of less than 25 and smoke-developed values of less than 450, as determined per ASTM E 84.

D. Perimeter Fire-Resistive Joint Systems: For joints between edges of fire-resistance rated floor assemblies and exterior glazing systems, provide systems with latex based dispersion with movement capability up to 50% and with ratings indicated below as determined by ASTM E 2307.

E. Fire-Test-Response Characteristics:
   1. Perform joint firestopping system tests by a qualified testing agency acceptable to Civil Defense and authorities having jurisdiction.
   2. Test per testing standards referenced in "Joint Firestopping Systems" Article. Provide rated systems complying with the following requirements:
      a. Joint firestopping systems shall bear classification marking of a qualified testing agency.
         1) UL in its "Fire Resistance Directory."
         2) Intertek Group in its "Directory of Listed Building Products."

F. Provide firestopping systems of sufficient thickness, width and density to provide and maintain all required fire resistance rating, in accordance with Civil Defence and local authority's requirements.

1.4 REFERENCES

A. The minimum standards for products specified in this section shall be including as under but not limited to the following. Except as otherwise specified herein, perform work in accordance with specifications, codes and standards cited therein, and their latest editions, revisions, applicable addenda and supplements. Where there is conflict between the referenced standards the most stringent of the conditions/requirements shall be applicable.
   1. ASTM E84: Standard Test Method for Surface Burning Characteristics of Building Materials.
   2. ASTM E119: Standard Test Methods for Fire Tests of Building Construction and Materials.
   3. ASTM E814: Standard Test Method for Fire Tests of Through-Penetration Fire Stops.
   4. ASTM E1966: Standard Test Method for Fire Resistive Joint Systems.
   5. ASTM E 2307: Standard Test Method for Determining the Fire Resistance of Perimeter Fire Barrier Systems using the Intermediate Scale Multi Store Test Apparatus.
   6. ASTM E 2174: Standard Practice for On-Site Inspection of Installed Fire Stops.
   7. ASTM E 2393: Standard Practice for On-Site Inspection of Installed Fire Stop Joint Systems.
   8. UL 2079: Tests for Fire Resistance of Building Joint Systems.
   9. ASTM E1399: Test for Dynamic Movement Conditions.
   10. ASTM E1966: Standard Test Method for Fire-Resistive Joint Systems.

1.5 SUBMITTALS

A. Product data for each type of product specified.

B. Shop drawings detailing materials, installation methods, and relationships to adjoining construction for each through-penetration-fire stopping system, and each kind of construction condition penetrated and kind of penetrating item. Include fire stop design designation of qualified testing and inspecting agency evidencing compliance with requirements for each condition indicated.

C. Product certificates signed by manufacturers of fire stopping products certifying that their products comply with specific requirements.

D. Product test reports from, and based on tests performed by, a qualified testing and inspecting agency evidencing compliance of fire stopping with requirements based on comprehensive testing of current products.

E. Qualification data for firms and persons specified in "Quality Assurance" article to demonstrate their capabilities and experience. Include list of completed projects with project names, addresses, names of Engineers and Employers, and other information specified.

F. Samples: Submit representative samples of materials proposed for use under this section.

G. Certification by fire stopping manufacturer that products supplied, comply with local regulations controlling use of volatile organic components (VOCs) and are non-toxic to building occupants.

H. Material Safety Data Sheets (MSDS) Submit MSDS for each firestop product. Submit qualifications of firestop installer, including letter from firestop manufacturer of products proposed to be installed, wherein manufacturer approves / recognizes as trained / or certifies installer for installation of that manufacturer's products.

I. Manufacturer's Letters for installations, a recommendation shall be obtained from the manufacturer, in writing for the specific application.

J. Method statement for the installation, maintenance and repair of fire stopping assemblies.

1.6 QUALITY ASSURANCE

A. General: Provide firestopping systems that are produced and installed to resist the spread of fire, and the passage of smoke and other gases.

B. Submit documented proof that the material has a proven record of successful performance for more than 10 years, confirmed by at least Five (5) local projects of similar nature.

C. Installer qualifications: An experienced installer with minimum Five (5) years' project experience of similar nature, who is certified, licensed, and qualified by the fire stopping manufacturer as having the necessary experience, staff, and training to install manufacturer's products per specified requirements.
   1. A firm that has been approved by Civil Defence and local authorities having jurisdiction.
   2. A manufacturer's willingness to sell its fire stopping products to the Contractor or to an installer engaged by the Contractor does not in itself confer qualification on the buyer.

D. Single-Source Responsibility: Obtain through-penetration fire stop systems for each kind of penetration and construction condition indicated from a single manufacturer.

E. Firestopping materials shall conform to flame (F) and temperature (T) ratings as required by local building code and as tested by accepted test agencies per ASTM E 814 or UL 1479. The

F rating must be a minimum of one (1) hour but not less than the fire resistance rating of the assembly being penetrated. T rating, when required by code authority, shall be based on measurement of the temperature rise on the penetrating item(s). The fire test shall be conducted with a minimum positive pressure differential of 0.25mm of water column as per UBC standard 7-1.

F. Firestopping products shall be asbestos free and free of any PCBs (Polychlorinated biphenyls).

G. Do not use any product containing solvents or that requires hazardous waste disposal.

H. Do not use firestop products which after curing, dissolve in water.

I. Do not use firestop products that contain ceramic fibers.

J. Field-Construction Mock-up: Prior to installing fire- stopping, erect mock-ups for each different through-penetration firestop system indicated to verify selections made and to demonstrate qualities of materials and execution. Build mock-ups using materials indicated for final installations.
   1. Locate mock-up in the location and of the size indicated or, if not indicated, as directed by the Engineer.
   2. Build Mock-up to the satisfaction of the Engineer. If the required standard is not achieved, the Contractor shall replace (at the contractor's cost) until the required standard is achieved as judged by the Engineer.
   3. Demonstrate the proposed range of aesthetic effects and workmanship.
   4. Obtain written approval of mock-up from the Engineer before starting fabrication.
   5. Maintain mock-ups during construction in an undisturbed condition as a standard for judging the completed Work.
   6. Approved mock-ups in an undisturbed condition at the time of Substantial Completion may become part of the completed Work if approved by the Engineer.

K. Co-ordination Work: Co-ordinate construction of openings and penetrating items to ensure that designated through-penetration fire stop systems are installed per specified requirements.

L. For firestopping exposed to view, traffic, moisture, and physical damage, provides products that do not deteriorate when exposed to these conditions.

M. Notify the Employer's inspection agency at least seven (7) days in advance of through penetration firestopping system installation.

1.7 DELIVERY, STORAGE AND HANDLING

A. Deliver fire stopping products to project site in original, unopened containers or packages with intact and legible manufacturers' labels identifying product and manufacturer, date of manufacture, lot number, shelf life, if applicable; qualified testing and inspecting agency classification marking applicable to Project; curing time; and mixing instructions for multi component materials.

B. Store and handle fire stopping materials to prevent their deterioration or damage due to moisture, temperature changes, contaminants, or other causes.

C. Do not cover up those fire stopping installations that will become concealed behind other construction until Employer's inspection agency and local authorities, if required, have examined each installation.

1.8 WARRANTY

A. Special Warranty: Written warranty, signed by Contractor agreeing to repair or replace fire stopping system that do not comply with performance and other requirements specified in this Section within specified warranty period.
   1. Warranty Period: 5 years from date of Substantial Completion.

1.9 PROJECT CONDITIONS

A. Coordinate Work of this Section with interfacing and adjoining Work for proper sequencing of each installation. Ensure best possible weather resistance, durability of Work, and protection of materials and finishes.

B. Verify existing conditions and substrates before starting work.

C. Do not use materials that contain solvents, show sign of damage or are beyond their shelf life.

D. During installation, provide masking and drop cloths as needed to prevent firestopping products from contaminating any adjacent surfaces.

E. Conform to ventilation requirements if required by manufacturer's installation instructions or Material Safety Data Sheet (MSDS).

F. Weather Conditions: Do not proceed with installation of firestop products when temperatures are in excess or below the manufacturer's recommendations.

G. Schedule installation of firestop products after completion of penetrating item installation but prior to covering or concealing of openings.

H. Install and cure joint firestopping systems per manufacturer's written instructions using natural means of ventilation or, where this is inadequate, forced-air circulation.

1.10 SEQUENCING AND SCHEDULING

A. Pre-Installation Conference: Convene a pre-installation conference to establish procedures to maintain optimum working conditions and to coordinate this work with related and adjacent work.

B. Sequence: Perform work of this and other sections in proper sequence to prevent damage to the firestop systems and to ensure that their installation will occur prior to enclosing or concealing work.

C. Install all firestop systems after voids and joints are prepared sufficiently to accept the applicable firestop system.

D. Do not cover firestop systems until they have been properly inspected and accepted by the Employer.

1.11 COORDINATION

A. Coordinate construction of joints, opening and penetrating items to ensure that joint and penetrating firestopping systems can be installed according to specified firestopping system design.

B. Coordinate sizing of joints, sleeves, opening, core-drilled holes or cut openings to accommodate joint firestopping systems.

PART 2 - PRODUCTS

2.1 MANUFACTURRERS

A. Subject to compliance with the requirements of the specifications of this Section, the following manufacturers are acceptable.
   1. 3M
   2. Hilti
   3. Siderise
   4. Tremco
   5. Specified Technologies Inc. (STI)
   6. Fischer
   7. Promat
   8. Lamatherm
   9. Or Equivalent, Subjected to Engineer's acceptance.

   Conditions to qualify "equivalent": Qualification exercise consists of assessment of technical adaptability of the design and the material, quality procedures maintained, minimum 10 (Ten) years of its usage in similar project and credibility of its performance record and other project references. All cost related to this exercise shall be borne by the proposer. The proposer shall agree to use above listed pre-approved system manufactures, should the proposed system be not acceptable to the Engineer, after the completion of qualification exercise.

2.2 MATERIALS GENERAL

A. Compatibility: Provide firestopping composed of components that are compatible with each other, the substrates forming openings, and the items, if any, penetrating the firestopping under conditions of service and application, as demonstrated by firestopping manufacturer based on testing and field experience.

B. Fire stop Materials: Provide only Fire stop systems, which have been tested and listed as Fire stop systems to meet every condition of the work. Do not provide materials or systems not part of a tested and listed Fire stop system suitable for the condition.

C. Auxiliary Materials: Provide the miscellaneous accessory items necessary to complete each Fire stopping condition, including, but not limited to, fillers, mechanical fastenings, support devices, collars, sleeves, and other materials, as made or recommended by the Fire stop material manufacturer and tested and listed with the rated Fire stop system.

D. Fire stop Sealant: For small openings of 300mm diameter or less, penetrations subject to movement, in non-combustible pipes, non-combustible conduits, non-combustible ducts, control joints, slab edge and exterior wall voids, as a sealant for smoke barrier construction, fire and smoke dampers, head of wall details and fire doors in masonry or gypsum drywall partitions.
   1. Provide the following:
      a. Silicone Sealants (for non-fire resistive joints): One-part, low modulus moisture-activated alkoxy-cure silicone capable of withstanding high movement in compression and extension.
      b. Intumescent Sealant (for fire resistive joints): Graphite based, one-part intumescent paste unaffected by water or moisture when cured containing no

solvents, inorganic fibers, water soluble intumescent ingredients, or silicone compounds and capable of expanding a minimum 8 times.

c. Elastomeric Spray and Coatings (for non-fire resistive visible joints): Elastomeric coatings designed for architectural joints containing dynamic movement due to seismic, wind sway or thermal expansion conditions.

E. Fire stop Mortars/Compounds:
   1. For penetrations such as cable trays, electrical and communication bundles and non-combustible sleeves and pipes:
      a. Pre-packaged dry mixes of inorganic binders, Portland cement, fillers and lightweight aggregate mixed with water to form a non-shrinking, homogenous mortar shall be fire impact resistant, load bearing, non-moving.

F. Fire stopping for Combustible Penetrating Items:
   1. For use, in openings where plastic pipes insulated pipes or insulated cables are installed. Provide one of the following:
      a. Intumescent Graphite Based Sealants:
         One-part graphite based Intumescent sealant unaffected by water or moisture when cured and capable of expanding a minimum 8 times.
      b. Fire stop Devices
         Prefabricated devices unaffected by humidity, moisture and frost.
      c. Intumescent Wrap Strips:
         Solvent free Intumescent wrap strips, comprised of a graphite impregnated rubber matrix faced on both sides with polyethylene, unaffected by water, frost or ultraviolet light.

G. Fire stopping for Large Complex Multiple Service Penetrations:
   1. For large vertical openings, containing multiple complex penetrations, non-combustible pipes, non-combustible conduits, non-combustible ducts, control joints, slab edge and exterior wall voids, as a sealant for smoke barrier construction, fire and smoke dampers, head of wall details in masonry or gypsum drywall partitions.
      a. Composite Sheet:
         Lightweight firestop panel comprised of graphite-based intumescent material capable of expanding a minimum 15 times sandwiched between galvanized steel backer sheet and wire mesh covered with aluminum foil on one side for medium to large sized openings.

H. Firestopping for Cable Trunking Internals:
   1. For use inside cable trunking at the point of passage through compartment wall or floor.
      a. Pillows/Bags: Re-usable, dustless, intumescent heat expanding pillows/bags composed of plastic jacketed blocks of mineral fiber encapsulated on all six sides with graphite-based intumescent material.
      b. Formed Compressible Bricks: Re-usable, dustless, gas tight, Intumescent heat expanding bricks composed of water insoluble expansion agents and fire-retardant additives.

I. Firestopping for Diamond Cored Holes:
   1. Formed Compressible Plugs: Re-usable, dustless, gas tight, Intumescent heat expanding plugs composed of water insoluble expansion agents and fire-retardant additives.
   2. Intumescent Sealant: Graphite based, one-part Intumescent paste unaffected by water or moisture when cured containing no solvents, inorganic fibres, or silicone compounds capable of expanding a minimum 8 times.

J. Intumescent, Non-hardening Firestop Putty:

1. Non-hardening intumescent putty consisting of a 100% solids butyl rubber base for use in openings having a nominal diameter of 152mm or less.

2.3 JOINT FIRESTOPPING SYSTEMS

A. Joint Firestopping Systems: Systems that resist spread of fire, passage of smoke and other gases, and maintain original fire-resistance rating of assemblies in or between which joint firestopping systems are installed. Joint firestopping systems shall accommodate building movements without impairing their ability to resist the passage of fire and hot gases.

B. Joints in or between Fire-Resistance-Rated Construction: Provide joint firestopping systems with ratings determined per ASTM E 1966 or UL 2079.
   1. Fire-Resistance Rating: Equal to or exceeding the fire-resistance rating of the wall, floor, or roof in or between which it is installed.

C. Joints at Exterior Curtain-Wall/Floor Intersections: Provide joint firestopping systems with rating determined per ASTM E 2307.
   1. F-Rating: Equal to or exceeding the fire-resistance rating of the floor assembly.

D. Accessories: Provide components of fire-resistive joint systems, including primers and forming materials, that are needed to install elastomeric fill materials and to maintain ratings required. Use only components specified by joint firestopping system manufacturer and approved by the qualified testing agency for conditions indicated.

E. All fire and smoke stops shall be positively fixed in position in such a manner that they shall not become dislodged in the event of a fire. The fixing shall secure the stop in position for a period at least equal to that required for the compartment wall or floor against which the works abut.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions with installer present, for compliance, requirements for opening configurations, penetrating items, substrates, and other conditions affecting performance of firestopping.

B. Do not proceed with installation until unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Surface cleaning: Clean out openings and joints immediately prior to installing firestopping to comply with recommendations of firestopping manufacturer and the following requirements:
   1. Remove all foreign materials from surfaces of opening and joint substrates and from penetrating items that could interfere with adhesion of firestopping.
   2. Clean opening and joint substrates and penetrating items to produce clean, sound surface capable of developing optimum bond with firestopping. Remove loose particles remaining from cleaning operation.
   3. Priming: Prime substrate, where recommended by firestopping manufacturer using the manufacturers recommended products and methods. Confine primers to areas of bond; do not allow spillage and migration onto exposed surfaces.

4. Masking Tape: Use masking tape to prevent firestopping from contacting adjoining surfaces that will remain exposed upon completion of Work and that would otherwise be permanently stained or damaged by such contact or by cleaning methods used to remove smears from firestopping materials. Remove tape as soon as it impossible to do so without disturbing firestopping's seal with substrates.

3.3 INSTALLATION

A. General: Install fire-resistive joint systems to comply with manufacturer's written installation instructions and published drawings for products and applications indicated

B. Install forming/damming materials and other accessories of types required to support fill materials during their application and in the position needed to produce the cross-sectional shapes and depths required to achieve fire ratings indicated.
   1. After installing elastomeric fill materials and allowing them to fully cure, remove combustible forming materials and other accessories not indicated as permanent components of fire-resistive joint system.

C. Install joint fillers to provide support of sealants application and at position required to produce the cross-sectional shapes and depths of installed sealants relative to joint widths that allow optimum sealant movement capability and develop fire-resistance rating required.

D. Install elastomeric fill materials for fire-resistive joint systems by proven techniques to produce the following results:
   1. Elastomeric fill voids and cavities formed by joints and forming materials as required to achieve fire-resistance ratings indicated.
   2. Apply elastomeric fill materials so they contact and adhere to substrates formed by joints.

E. For elastomeric fill materials that will remain exposed after completing the Work, finish to produce smooth, uniform surfaces that are flush with adjoining finishes

F. Tool non-sag sealant immediately after sealant application and prior to the time skinning or curing begins. Form smooth, uniform beads of configuration indicated or required to produce fire-resistance rating, as well as to eliminate air pockets, and to ensure contact and adhesion of sealant with sides of joint. Remove excess sealant from surface adjacent to joint. Do not use tooling agents that discolour sealants or adjacent surfaces or are not approved by sealant manufacturer.

G. Obtain approval of design and works from Civil Defence / House of expertise for compliance.

3.4 IDENTIFICATION

A. Joint Identification: Identify joint firestopping systems with legible metal or plastic labels. Attach labels permanently to surfaces adjacent to and within 150 mm of joint edge so labels are visible to anyone seeking to remove or joint firestopping system. Use mechanical fasteners or self-adhering-type labels with adhesives capable of permanently bonding labels to surfaces on which labels are placed. Include the following information on labels:
   1. The words "Warning - Joint Firestopping - Do Not Disturb. Notify Building Management of Any Damage."
   2. Contractor's name, address, and phone number.
   3. Designation of applicable testing agency.
   4. Date of installation.
   5. Manufacturer's name.

6. Installer's name.

3.5 FIELD QUALITY CONTROL

A. Inspection: Engage a qualified independent inspection agency to examine penetration fire stopping with ASTM E 2174, “Standard Practice on Site Inspection of Installed Fire Stops and ASTM E-2393, “Standard Practice of On-Site Inspection of Installed Fire Stop Joint Systems. Inspection agency to examine firestopping and will determine, in general, that firestopping has been installed in compliance with requirements of tested and listed firestop system, and installation process conforms to FM 4991-Standard for Approval of Firestop Contractors.

B. The inspector shall advise the Contractor of any deficiencies noted within one (1) working day.

C. Do not proceed to enclose firestopping with other construction until inspection agency has verified that the firestop installation complies with the requirements.

D. Where deficiencies are found, repair or replace the firestopping so that it complies with requirements of tested and listed system design.

E. Proceed with enclosing joint firestopping systems with other construction only after inspection reports are issued and installations comply with requirements.

F. Provide technical representative from firestopping manufacturer to inspect the work being performed by installed and provide report to Engineer of the compliance with requirements.

3.6 JOINT FIRESTOPPING SYSTEM SCHEDULE

A. Where UL-classified systems are indicated, they refer to system numbers in UL's "Fire Resistance Directory" under product Category XHBN or Category XHDG.

B. Where Intertek Group-listed systems are indicated, they refer to design numbers in Intertek Group's "Directory of Listed Building Products" under product category Expansion/Seismic Joints or Firestop Systems.

3.7 CLEANING AND PROTECTION

A. Clean off excess fill materials and sealants adjacent to openings and joints as work progresses by methods and with cleaning materials approved by manufacturers of firestopping products and of products in which opening and joints occur.

B. Protect firestopping during and after curing period from contact with contaminating substances or from damage resulting from construction operations or other causes so that they are without deterioration or damage at time of substantial completion. If, despite such protection, damage or deterioration occurs, cut out and remove damaged or deteriorated firestoppping immediately and install new materials to produce firestopping complying with specified requirements.

END OF SECTION 07 8400.01

SECTION 07 9200.01 - JOINT SEALANTS

PART 1 - GENERAL

Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.1 SECTION INCLUDES

A. Furnish all labor, materials, tools appliances and equipment, and perform all operations and services necessary for the complete application of sealants for the following installations, including those specified by reference to this Section.

B. This Section includes sealants for the following applications:
   1. Exterior joints in the following vertical surfaces and non-traffic horizontal surfaces:
      a. Aluminium curtain wall glazing joints, and interfaces.
      b. Metal sheet cladding joints and interfaces.
      c. Exterior perimeter joints of the exterior openings/ elements such as aluminium doors, windows etc.
      d. Other building envelope related structural and weather joints as indicated in the drawings.
   2. Interior joints in the following vertical surfaces and horizontal non-traffic surfaces:
      a. Interior perimeter joints of glass façade, and exterior openings/ elements such as aluminium doors, windows, etc.
      b. Sheet metal cladding assembly joints.
      c. Framing joints and drills.
      d. Joints between insulated glass pane and IGU spacer.
      e. All perimeters' joints and interfaces, and areas to accomplish a sealed building.
      f. Other building envelope related structural and weather joints as indicated in the drawings.

C. Design to follow the architectural intent indicated in the drawings.

D. This section to be read in conjunction with the related wall types/ exterior wall system drawings submitted as a part of the tender package.

1.2 RELATED SECTION

A. This schedule is intended to be used as a helpful indication of the related sections within the Project Specifications. It is not necessarily comprehensive or complete and it is the Contractor's responsibility to ascertain all applicable sections required to understand the full scope of Works intended.

B. Work to be coordinated and used in conjunction with this specification includes but is not restricted to:

1.	Section 01 4500	-	Quality Control
2.	Section 03 3000	-	Cast-in-Place Concrete
3.	Section 04 2200	-	Concrete Unit Masonry

4.	Section 05 7313.01	-	Glazed Decorative Metal Railing
5.	Section 05 7500.01	-	Decorative Formed Metal
6.	Section 07 2100.01	-	Thermal Insulation
7.	Section 07 2500.01	-	Weather Barrier
8.	Section 07 6113.01	-	Standing Seam Sheet Metal Roofing
9.	Section 07 6200.01	-	Sheet Metal Flashing and Trim
10.	Section 08 4000	-	Aluminium Works
11.	Section 08 1116.01	-	Alum. Doors and Frames
12.	Section 08 4413.01	-	Glazed Alum. Curtain Wall
13.	Section 08 8000.01	-	Glazing
14.	Section 08 9100.01	-	Louvers

1.3 PERFORMANCE REQUIREMENTS

A. Provide elastomeric joint sealants that establish and maintain watertight and airtight continuous joint seals without staining or deteriorating joint substrates when testing against the following given below tests as a part of curtain wall or building envelope element.
   1. Resist air leakage when tested in accordance with ASTM E283 as an integral part of the curtain wall assembly.
   2. Resist water penetration when tested in accordance with ASTM E331 and AAMA 501.1 as an integral part of the curtain wall assembly.
   3. Resist sealant joint failure when tested in accordance with ASTM E330 as an integral part of the curtain wall assembly.

1.4 SUBMITTALS:

A. Product Data: For each joint-sealant product indicted.

B. Pre-Contract Samples: Manufacturer's color charts consisting of strips of cured sealants showing the full range of colors available for each product exposed to view.

C. Post Contract Samples: For each kind and color of joint sealant required, provide Samples with joint sealants in 13mm to 15mm wide joints formed between two (2) 300 mm long strips of actual material the sealant to be with. To be inspected for staining one week and four weeks after the sample is submitted.

D. Product Certificates: Signed by manufacturers of joint sealants certifying that products furnished comply with requirements and are suitable for the use indicated.

E. Compatibility and Adhesion Test Reports: From sealant manufacturer indicating the following:
   1. Materials forming joint substrates and joint sealant backings have been tested for compatibility and adhesion with joint sealants.
   2. Sealant will not stain metal, stone and glass panels or other substrates when tested in accordance with ASTM C 1248, or other acceptable testing criteria.
   3. Interpretation of test results and written recommendations for primers and substrate preparation needed for adhesion.

F. Preconstruction Field-Adhesion Test Reports: Indicate which sealants and joint preparation methods resulted in optimum adhesion to joint substrates based on testing specified in "Preconstruction Testing" Article.

G. Product Test Reports: From a qualified testing agency indicating sealants comply with

requirements, based on comprehensive testing of current product formulations.

H. Submit a detailed schedule of sealant usage for all locations. List each sealant material (including manufacturer and product name), joint fillers, and sealant color and product data for each location of use.

I. Method statement for tooling, maintenance and repair of joint sealant assemblies.

J. Qualification Data: For qualified Installer.

K. Printed statement for VOC content.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: Perform installation with experienced and trained installers supervised by trained personnel who shall have at least three (3) years successful experience in installation of similar size and scope.

B. Manufacturers Qualifications: A firm experienced a minimum five (5) years in manufacturing products or system similar to those indicated for this project with a record of successful in-services performance.

C. Manufacturer of sealants shall provide a qualified technical representative who will visit the site for the purpose of advising the Installer of the proper procedures and precautions in the application of the materials.

D. Source Limitations: Obtain each type of joint sealant through one source from a single manufacturer.

E. Preconstruction Compatibility and Adhesion Testing: Submit to joint-sealant manufacturers, for testing indicated below, samples of materials that will contact or affect joint sealants.
   1. Use ASTM C 1087 to determine whether priming and other specific joint preparation techniques are required to obtain rapid, optimum adhesion of joint sealants to joint substrates.
   2. Submit not fewer than ten pieces of each type of material, including joint substrates, shims, joint-sealant backings, secondary seals, and miscellaneous materials.
   3. Schedule sufficient time for testing and analyzing results to prevent delaying the Work.
   4. For materials failing tests, obtain joint-sealant manufacturer's written instructions for corrective measures including use of specially formulated primers.

F. Product Testing: Obtain test results from an independent qualified testing agency based on testing current sealant formulations within a 36-month period preceding the commencement of the Work.
   1. Testing Agency Qualifications: An independent testing agency qualified according to ASTM C 1021 to conduct the testing indicated, as documented according to ASTM E 548.
   2. Test elastomeric joint sealants for compliance with requirements specified by reference to ASTM C 920, and where applicable, to other standard test methods.
   3. Test other joint sealants for compliance with requirements indicated by referencing standard specifications and test methods.

G. Test Method: Test joint sealants according to Method A, Field-Applied Sealant Joint Hand Pull Tab, in Appendix X1 in ASTM C 1193. For joints with dissimilar substrates, verify adhesion to each substrate separately; extend cut along one side, verifying adhesion to opposite side. Repeat procedure for opposite side.

H. Preconstruction Field-Adhesion Testing: Before installing elastomeric sealants, field test their adhesion to Project joint substrates as follows:
   1. Locate test joints, where indicated on Project or, if not indicated, as directed by Engineer.
   2. Conduct field tests for each application indicated below:
      a. Each type of elastomeric sealant and joint substrate indicated.
      b. Each type of non-elastomeric sealant and joint substrate indicated.
   3. Notify Engineer seven days in advance of dates and times when test joints will be erected.
   4. Arrange for tests to take place with joint-sealant manufacturer's technical representative present.
      a. Test Method: Test joint sealants according to Method A, Field-Applied Sealant Joint Hand Pull Tab, in Appendix X1 in ASTM C 1193.
         1) For joints with dissimilar substrates, verify adhesion to each substrate separately; extend cut along one side, verifying adhesion to opposite side. Repeat procedure for opposite side.

I. Evaluation of Preconstruction Field-Adhesion-Test Results: Sealants not evidencing adhesive failure from testing, in absence of other indications of noncompliance with requirements, will be considered satisfactory. Do not use sealants that fail to adhere to joint substrates during testing.

J. Mockups: Before installing joint sealants, apply elastomeric sealants, as follows, to verify selections made under sample Submittals and to demonstrate aesthetic effects and to set quality standards for materials and execution:
   1. Locate mock-up in the location or, if not indicated, as directed by the Engineer.
   2. Build Mock-up to the satisfaction of the Engineer. If the required standard is not achieved, the Contractor shall replace (at the contractor’s cost) until the required standard is achieved as judged by the Engineer.
   3. Demonstrate the proposed range of aesthetic effects and workmanship.
   4. Obtain written approval of mock-up from the Engineer before starting fabrication.
   5. Maintain mock-ups during construction in an undisturbed condition as a standard for judging the completed Work.
   6. Approved mockups in an undisturbed condition at the time of Substantial Completion may become part of the completed Work if approved by the Engineer.

K. Conduct sealant field adhesion tests to Project joint substrates prior to installation.

1.6 DELIVERY, STORAGE AND HANDLING

A. Deliver materials to Project site in original unopened containers or bundles with labels indicating manufacturer, product name and designation, color, expiration date, pot life, curing time, and mixing instructions for multicomponent materials.

B. Store and handle materials in compliance with manufacturer’s written instructions to prevent their deterioration or damage due to moisture, high or low temperatures, contaminants, or other causes.

C. Mark product with shop drawing reference, unless already properly marked.

D. Sequence deliveries to avoid delays, but minimize on-site storage.

1.7 PROJECT CONDITIONS

A. Environmental Limitations: Do not proceed with installation of joint sealants under the following conditions:
   1. When ambient and substrate temperature conditions are outside limits permitted by joint sealant manufacturer.
   2. When joint substrates are wet or dusty.

B. Joint-Width Conditions: Do not proceed with installation of joint sealants where joint widths are less than those allowed by joint sealant manufacturer for applications indicated.

C. Joint-Substrate Conditions: Do not proceed with installation of joint sealants until contaminants capable of interfering with adhesion are removed from joint substrates.

D. Compatibility:
   1. Determine compatibility characteristics of sealants in contact with sealant backings by test method ASTM C1087.
   2. Provide joint sealants, joint fillers and accessory joint materials that are compatible with one another and with joint substrates under project conditions.
   3. Install joint sealants, joint fillers and related joint materials that are non-staining to visible joint surfaces and surrounding substrate surfaces

E. Scheduling:
   1. Schedule applications of waterproofing, water repellents and preservative finishes after sealant installation unless sealant manufacturer approves otherwise in writing.
   2. Ensure that installed sealant is allowed to cure sufficiently prior to subsequent applications.

1.8 WARRANTY

A. Special Installer's Warranty:
   1. Installer's standard form in which installer agrees to repair or replace joint sealants that do not comply with performance and other requirements specified herein within specified warranty period.
   2. Warranty Period: 10 (ten) years from date of Substantial Completion.

B. Special Manufacturer's Warranty:
   1. Manufacturer's standard form in which the joint sealant manufacturer agrees to furnish joint sealants to repair or replace those that do not comply with performance and other requirements specified herein within specified warranty period.
   2. Warranty Period: 10 (ten) years from date of Substantial Completion.

C. Special warranties specified herein exclude deterioration or failure of joint sealants from the following:
   1. Movement of the structure caused by structural settlement or errors attributable to design or construction resulting in stresses on the sealant exceeding sealant

manufacturer's written specifications for sealant elongation and compression.

2. Disintegration of joint substrates from natural causes exceeding design specifications.
3. Mechanical damage caused by individuals, tools, or other outside agents.
4. Change in sealant appearance caused by the accumulation of dirt or other atmospheric contaminants.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. Dow Corning Ltd.,
UK
B. GE,
Middle East
C. Sika Gulf,
Middle East
D. Tremco Ltd.,
Middle East
Or Equivalent, Subjected to Engineer's acceptance.

Conditions to qualify "equivalent": Qualification exercise consists of assessment of technical adaptability of the design and the material, quality procedures maintained, minimum 10 (Ten) years of its usage in similar project and credibility of its performance record and other project references. All cost related to this exercise shall be borne by the proposer. The proposer shall agree to use above listed pre- approved system manufactures, should the proposed system be not acceptable to the Engineer, after the completion of qualification exercise.

2.2 MATERIALS GENERAL

A. Compatibility: Provide joint sealants, backings, and other related materials that are compatible with one another and with joint substrates under conditions of service and application, as demonstrated by sealant manufacturer based on testing and field experience.

B. Comply with ASTM C 1193 – Standard guide for use of Joint Sealants.

C. Liquid Applied Joint Sealants: Comply with ASTM C 920 and other requirements indicated for each liquid applied joint sealant specified, including those referencing ASTM C 920 classifications for type, grade, class, and uses related to exposure and joint substrates.

D. Suitability for Immersion in Liquids: Where sealants are indicated for joints that will be continuously immersed in liquids, provide products that have undergone testing according to ASTM C 1247 or other similar testing methods acceptable to the authorities having jurisdiction.

E. Hardness specified is intended to indicate the general range necessary for overall performance. Consult manufacturer's technical representative to determine the actual hardness recommended for the conditions of installation and use. Unless otherwise recommended by the manufacturers technical representative provide sealant within the range of hardness (20 to 60) (Shore A, fully cured at 75°F).

F. Modulus of Elasticity: For joints subject to either thermal expansion or dynamic movement, provide elastomeric sealants which have the lowest modulus of elasticity which is consistent with exposure to abrasion or vandalism. For horizontal joints subject to traffic, provide sealants with high modulus of elasticity, as required to withstand indentation by stiletto heels. Comply with manufacturer's recommendations wherever no other requirements are indicated.

G. VOC Content of Interior Sealants: Sealants and sealant primers used inside the weatherproofing system shall comply with the following limits for VOC content when calculated according to 40 CFR 59, Subpart D (EPA Method 24):
   1. Architectural Sealants: 250 g/L.
   2. Sealant Primers for Nonporous Substrates: 250 g/L.
   3. Sealant Primers for Porous Substrates: 775 g/L.

H. Or comply with most stringent of the local code or acceptable national/international building standard requirements for VOC's limits.

I. Stain-Test-Response Characteristics: Where sealants are specified to be non- staining to porous substrates, provide products that have undergone testing according to ASTM C 1248 and have not stained porous joint substrates indicated for Project.

J. Suitability for Immersion in Liquids: Where sealants are indicated for joints that will be continuously immersed in liquids, provide products that have undergone testing according to ASTM C 1247 or other similar testing methods acceptable to the authorities having jurisdiction.

K. Suitability for Contact with Food and Potable water: Where sealants are indicated for joints that will come in repeated contact with food and potable water, provide products approved by NSF or similar acceptable to the authorities having jurisdiction.

L. Colors of Exposed Joint Sealants: To match adjacent surfaces and to Engineer's approval.

2.3 SEALANT MATERIALS

A. Joint Sealer Type AA (Acrylic acoustical): One component acrylic latex, permanently elastic, non-staining, non-shrinking, non-migrating and paintable.

B. Joint Sealer Type AP (Acrylic painters caulk): One component acrylic latex caulking/ polysulphide compound, conforming to ASTM C 834, paintable within 24 hours after application, with a minimum movement capability of ±12.5 percent.

C. Joint Sealer Type HL2 (Horizontal-self-Leveling, 2-component): Pouring grade self- leveling multi-component urethane sealant, conforming to FS TT-S-00227E (Interim Federal Specification), Type I, Class A, and ASTM C 920, with a minimum movement capability of ±25 percent.

D. Joint Sealer Type HT (Horizontal-Trowel): Trowel grade multi-component modified urethane/polysulphide sealant, conforming to FS TT-S-00227E (Interim Federal Specification), Type I, Class A, and ASTM C 920, with a minimum movement capability of ±25 percent.

E. Joint Sealer Type P1 (Silicone, general construction): One-part medium modulus, natural cure, synthetic sealant, having a useful life expectancy of at least 20 years, conforming to ASTM C 920, Type S, NS, Class 25, use NT, G, A, M, O with a minimum movement

capability of ±50 percent.

F. Joint Sealer Type BP2 (Polyurethane/polysulphide, Multi-component): Low modulus type, Multicomponent non-sagging gun-grade polyurethane sealant, conforming to FS TT-S-00227E (Interim Federal Specification), Type II, Class A, and ASTM C 920, Type M, Class 25, Grade NS, use NT, M, A and O with a minimum movement capability of ±50 percent.

G. Joint Sealer Type P2 (Polyurethane/polysulphide, Multi-component): Low modulus type, multi-component non-sagging gun-grade polyurethane sealant, conforming to FS TT-S-00227E (Interim Federal Specification), Type II, Class A, and ASTM C 920, Type M, Class 25, Grade NS, use NT, M, A and O with a minimum movement capability of ±50 percent.

H. Joint Sealer Type SC (Silicone, general construction): One-part medium modulus, natural cure, synthetic sealant, having a useful life expectancy of at least 20 years, conforming to ASTM C 920, Type S, NS, Class 25, use NT, G, A, M, O with a minimum movement capability of ±50 percent.

I. Joint Sealer Type SE (Silicone, Exterior construction): One-part low modulus, moisture curing, synthetic rubber sealant, having a useful life expectancy of at least 20 years, conforming to ASTM C 920, Type S, NS, Class 25, FS TT-S-001543A (Interim Federal Specification), Type, Class A with a minimum movement capability of +100 percent and -50 percent.

J. Joint Sealer Type SM (Silicone, Mildew-resistant): USDA approved one component acetoxy silicone rubber, mildew resistant, acceptable to local health officials, conforming to U.S. Food and Drug Administration regulation 21 CFR 177.2600, FS TT-S-001543A (Interim Federal Specification), Type Non-Sag, Class A, and FS TT- S-00230C (Interim Federal Specification), Type II, Class A and ASTM C 920, Type S, Class 25, Grade NS, use NT,G and A with a minimum movement capability of ±25 percent, and a Shore A hardness of 20.

2.4 ELASTOMERIC JOINT SEALANTS (SILICONE)

A. Elastomeric Sealant Standard: Ultra violet light resistant, nonbleeding, non-staining, non-sag, multi-component, polyurethane based elastomeric sealant. Comply with ASTM C 920 and other requirements indicated for each liquid-applied chemically curing sealant.

B. Silicone sealant: One part, ultra violet light resistant, nonbleeding, non-staining, non- sag, silicone sealant; capable of ±50% movement as measured in compliance with ASTM C 719; ASTM C920, Type S, Grade NS, Class 25, use 'O' related to joint substrate for vertical joints.

C. Glazing as per specification section “Glazing 08 8000”.

2.5 LATEX JOINT-SEALANT

A. Latex Sealant: One-part non-sag, mildew resistant, acrylic-emulsion sealant complying with ASTM C834, formulated to be paintable and recommended for exposed applications on interior locations involving joint movement of not more than plus or minus 5%.

2.6 JOINT-SEALANT BACKING

A. General: Provide sealant backings of material and type that are non-staining; are

compatible with joint substrates, sealants, primers, and other joint fillers; and are approved for applications indicated by sealant manufacturer based on field experience and laboratory testing.

B. Cylindrical Sealant Backings: ASTM C 1330, of size and density to control sealant depth and otherwise contribute to producing optimum sealant performance.

C. Elastomeric Tubing Sealant Backings: Neoprene, butyl, EPDM, or silicone tubing complying with ASTM D 1056, nonabsorbent to water and gas. Provide products with low compression set and of size and shape to provide a secondary seal, to control sealant depth, and otherwise contribute to optimum sealant performance.

D. Bond-Beaker Tape: Polyethylene tape or other plastic tape recommended by sealant manufacturer for preventing sealant from adhering to rigid, inflexible joint filler materials or joint surfaces at back of joint where such adhesion would result in sealant failure. Provide self-adhesive tape where applicable.

E. Sealant backer rod shall be compressible rod stock polyethylene foam, polyethylene- jacketed polyurethane foam, butyl rubber foam, neoprene foam or other flexible, permanent, durable non-absorptive closed cell material as recommended for compatibility with sealants by sealant manufacturer. Provide size and shape of rod which will control the joint depth for sealant placement, break bond of sealant at bottom of joint, form optimum shape of sealant bead on back side, and provide a highly compressible backer to minimize the possibility of sealant extrusion when joint is compressed.

2.7 MISCELLANEOUS MATERIALS

A. Primer: Material recommended by joint sealant manufacturer where required for adhesion of sealant to joint substrates indicated, as determined from preconstruction joint-sealant-substrate tests and field tests.

B. Cleaners for Nonporous Surfaces: Chemical cleaners acceptable to manufacturers of sealants and sealant backing materials, free of oily residues or other substances capable of staining or harming joint substrates and adjacent nonporous surfaces in any way, and formulated to promote optimum adhesion of sealants with joint substrates.

C. Masking Tape: Non-staining, non-absorbent tape product compatible with joint sealants and adjacent joint surfaces that is suitable for masking.

2.8 REFERENCED STANDARDS

A. The minimum standards for products specified in this section shall be including as under but not limited to the following. Except as otherwise specified herein, perform work in accordance with specifications, codes and standards cited therein, and their latest editions, revisions, applicable addenda and supplements. Where there is conflict between the referenced standards the most stringent of the conditions/requirements shall be applicable.

1. ASTM C 717: Terminology of Building Seals and Sealants
2. ASTM C 719: Standard Test Method for Adhesion and Cohesion of Elastomeric Joint Sealants under Cyclic Movement.
3. ASTM C 794: Standard Test Method for Adhesion-in-Peel of elastomeric Joint Sealants.
4. ASTM C 834: Standard Specifications for Latex Sealants
5. ASTM C 919: Standard Practice for Use of Sealant in Application

6. ASTM C 920: Standard Specifications for Elastomeric Joint Sealants.
7. ASTM C 1193: Standard Guide for use of Joint Sealants.
8. ASTM C 1248: Standard Test Method for Staining of Porous Substrate by Joint Sealants.
9. ASTM C 1330: Standard Specification for Cylindrical Sealant Backing of use with Cold Liquid Applied Sealants.
10. ASTM D 1056: Standard Specifications for Flexible Cellular Materials – Sponge or Expanded Rubber.
11. ASTM E 90: Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine joints indicated to receive joint sealants, with Installer present, for compliance with requirements for joint configuration, installation tolerances, and other conditions affecting joint-sealant performance.

B. Surface Condition: Prior to the work of this section, carefully inspect the installed work of other trade and verify that such work is complete to a stage where this installation may properly commence. In the event of any discrepancy do not proceed with the installation until all such discrepancies have been fully resolved.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

D. Verification of Conditions:
   1. Verify that field measurements, surfaces, substrates, structural support, utility connections, tolerances, levelness, plumbness, humidity, moisture content level, cleanliness and other conditions are as required by the manufacturer, and ready to receive Work.
   2. Consult with sealant manufacturers to determine whether priming is necessary.
   3. Provide joints properly dimensioned to receive the approved sealant system.
   4. Provide joint surfaces that are clean, dry, sound and free of voids, deformations, protrusions and contaminants that may inhibit application or performance of the joint sealant.
   5. Test substrate as needed to verify proper conditions.

E. Pre-installation Conference: Conduct conference at Project Site.

3.2 PREPARATION

A. Surface Cleaning of Joints: Clean out joints immediately before installing joint sealants to comply with joint sealant manufacturer's written instructions and the following requirements:
   1. Remove all foreign material from joint substrates that could interfere with adhesion of joint sealant, including dust, paints (except for permanent, protective coatings tested and approved for sealant adhesion and compatibility by sealant manufacturer), old joint sealants, oil, grease, waterproofing, water repellents, water, surface dirt, and frost.
      a. Clean porous joint substrate surfaces like concrete, masonry, unglazed surfaces of ceramic tiles etc. by brushing, grinding, blast cleaning,

mechanical abrading, or a combination of these methods to produce a clean, sound substrate capable of developing optimum bond with joint sealants. Remove loose particles remaining from above cleaning operations by vacuuming or blowing out joints with oil- free compressed air.

2. Remove laitance and form-release agents from concrete.
3. Clean nonporous surfaces like metal, glass, porcelain enamel, glazed surfaces of ceramic tile etc. with chemical cleaners or other means that do not stain, harm substrates, or leave residues capable of interfering with adhesion of joint sealants.

B. Joint Priming: Prime joint substrates, where recommended in writing by joint sealant manufacturer, based on preconstruction joint-sealant-substrate tests or prior experience. Apply primer to comply with joint sealant manufacturer's written instructions. Confine primers to areas of joint-sealant bond; do not allow spillage or migration onto adjoining surfaces.

C. Masking Tape: Use masking tape where required to prevent contact of sealant with adjoining surfaces that otherwise would be permanently stained or damaged by such contact or by cleaning methods required to remove sealant smears. Remove tape immediately after tooling without disturbing joint seal.

D. Substrate preparation and joint seal application in accordance with manufacturer's instructions.

3.3 ENVIRONMENTAL PROCEDUERS

A. Comply, at minimum, with sealant and sealant primer manufacturer recommendations for space ventilation during and after installation. Where feasible, the following ventilation conditions shall be maintained during the sealant/sealant primer curing period or for 72 (seventy-two) hours after installation.

B. To the extent practical, allow sealant and sealant primer installations to cure prior to the installation of materials that adsorb volatile organic content (VOC). Materials that adsorb VOCs include carpets, textiles, unprimed GWB (gypsum wall board), and acoustical ceiling panels.

3.4 INSTALLATION OF JOINT SEALANTS

A. General: Comply with joint sealant manufacturer's written installation instructions for products and applications indicated, unless more stringent requirements apply.

B. Sealant Installation Standard: Comply with recommendations of ASTM C 1193 for use of joint sealants as applicable to materials, applications, and conditions indicated.

C. Provide the selected sealant system, where shown on the Drawings, and in strict accord with the manufacturer's recommendations as reviewed by the Designer.

D. Install sealant immediately after joint preparation.

E. Mix and apply multi-component sealants in accord with manufacturer's printed instructions.

F. Install sealant to fill joints completely from the back, without void or entrapped air, using proven techniques, proper nozzles and sufficient force that result in sealants directly contacting and fully wetting joint surface.

G. Install sealant backings of type indicated or required to support sealants during application and at position required to produce cross-sectional shapes and depths of installed sealants relative to joint widths that allow optimum sealant movement capability.
   1. Do not leave gaps between ends of sealant backings.
   2. Do not stretch, twist, puncture, or tear sealant backings.
   3. Remove absorbent sealant backings that have become wet before sealant application and replace them with dry materials.

H. Install bond-breaker tape behind sealants where sealant backings are not used between sealants and back of joints.

I. Install sealants by proven techniques to comply with the following and at the same time backings are installed:
   1. Place sealants so they directly contact and fully wet joint substrates.
   2. Completely fill recesses provided for each joint configuration.
   3. Produce uniform, cross-sectional shapes and depths relative to joint widths that allow optimum sealant movement capability.

J. Tooling of Non-sag Sealants: Immediately after sealant application and before skinning or curing begins, tool sealants according to requirements specified below to form smooth, uniform beads of configuration indicated; to eliminate air pockets; and to ensure contact and adhesion of sealant with sides of joint.
   1. Remove excess sealants from surfaces adjacent to joint.
   2. Use tooling agents that are approved in writing by sealant manufacturer and that do not discolor sealants or adjacent surfaces.
   3. Provide concave joint configuration per Figure 8A in ASTM C 1193, unless otherwise indicated.
   4. Provide flush joint configuration, per Figure 8B in ASTM C 1193, where indicated.
   5. Provide recessed joint configuration, per Figure 8C in ASTM C 1193, of recess depth and at locations indicated.
   6. Use masking tape to protect adjacent surfaces of recessed tooled joints.

3.5 FIELD QUALITY CONTROL

A. Inspect joints for complete fill, for absence of voids, and for joint configuration. Any sealant found out of plumb or cracking or any work otherwise defective shall be taken out and replaced to approval, at no additional cost to the Project.

B. Manufacturer's Field Services: Manufacturer's field representative to inspect and approve of installation prior to issuing warranty.

C. Perform Field-Adhesion test.

D. Evaluation of Field-Adhesion Test Results: Sealants not evidencing adhesive failure from testing or non-compliance with other indicated requirements will be considered satisfactory. Remove sealants that fail to adhere to joint substrates during testing or to comply with other requirements. Retest failed applications until test results prove sealants comply with indicated requirements.

3.6 CLEANING

A. Clean off excess sealants or sealant smears adjacent to joints as the Work progresses by methods and with cleaning materials approved in writing by manufacturers of joint sealants and of products in which joints occur.

B. At completion of construction activity of other trades, touch up and restore damaged or defaced coated surfaces.

C. At end of each workday, remove rubbish and other discarded material from project site.

3.7 PROTECTION

A. Protect joint sealants during and after curing period from contact with contaminating substances and from damage resulting from construction operations or other causes so sealants are without deterioration or damage at time of Substantial Completion. If, despite such protection, damage or deterioration occurs, cut out and remove damaged or deteriorated joint sealants immediately so installations with repaired areas are indistinguishable from the original work.

END OF SECTION 07 9200.01

SECTION 08 1116.01 - ALUMINIUM DOORS AND FRAMES

PART 1 - GENERAL

Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.1 SECTION INCLUDES

A. Design, engineering, supply, fabrication and installation of the independent/ façade integrated aluminium and glass framed doors, accessories, complete with reinforcing, shims, anchors and attachment devices, etc. as specified and as shown on drawings and perform all services and operations necessary for the complete supply and installation including but not limited to the following:
   1. Aluminium and glass external doors, curtain wall integrated and independent.
      a. Thermal break double and single leaf swing door with handle and lock assembly as per Architects hardware schedule.
      b. Door design with full thermal break, operable leaves, hinged opening frame and threshold as indicated in the architect drawings; weatherproofing by thermoplastic gasket or double brush seal.
      c. Doors with Finger Protection and concealed pivot/hinge mechanism.
   2. Sealant and caulking for works of this section and adjacent construction.
   3. Aluminum-framed glass doors, including factory glazing, operating hardware and accessories designed for exterior applications.
   4. Glass and glazing for the doors specified above, with performance IGU.

B. Design to follow the architectural intent indicated in the drawings.

C. This section to be read in conjunction with the related wall types/ exterior wall system drawings submitted as a part of the facade tender package.

1.2 RELATED SECTIONS

1.	Section 01 4500	-	Quality Control
2.	Section 03 3000	-	Cast-in-Place Concrete
3.	Section 04 2200	-	Concrete Unit Masonry
4.	Section 07 2500.01	-	Weather Barrier
5.	Section 07 6200.01	-	Sheet Metal Flashing and Trim
6.	Section 07 9200.01	-	Joint Sealants
7.	Section 08 4000	-	Aluminium Works
8.	Section 08 4413.01	-	Glazed Aluminium Curtain Wall
9.	Section 08 8000.01	-	Glazing

1.3 REFERENCES

A. The minimum standards for products specified in this section shall be including as under but not limited to the following. Except as otherwise specified herein, perform work in accordance with specifications, codes and standards cited therein, and their latest editions, revisions, applicable addenda and supplements. Where there is conflict between the referenced standards the most stringent of the conditions/requirements shall be applicable.

B. ANSI Z97.1 - Safety Performance Specifications and Methods of Test for Safety Glazing Material Used in Buildings.

C. CPSC 16CFR-1201 - Safety Standard for Architectural Glazing Materials.

D. All applicable local codes, regulations, etc. of Authorities having Jurisdiction.

Note: Contractor shall note that he shall keep copies of all of the referenced standards on the site and shall provide the Engineer (Architect) copy of the requested referenced standard as and when requested.

1.4 PERFORMANCE REQUIREMENTS

A. General: Provide aluminium and glass framed door system; including anchorage, capable of withstanding, without failure, the effects of the following:
   1. Structural loads.
   2. Thermal Movements.
   3. Movements of supporting structure indicated on Drawings.
   4. Dimensional tolerances of building frame and other adjacent construction.
   5. Failure includes the following (not limited to):
      a. Deflection exceeding specified limits.
      b. Framing members transferring stresses, including those caused by thermal and structural movements, to glazing.
      c. Noise or vibration created by wind and thermal and structural movements.
      d. Loosening or weakening of fasteners, attachments, and other components.

B. Structural Loads:
   1. Wind load: Refer to the wind tunnel analysis or project wind report at its latest revision for the design loads.

   2. Deflection of framing members:
      a. Deflection Normal to Wall Plane: Limited to 1/175 of clear span.
      b. Deflection parallel to Glazing Plane: Limited to L/360 of clear span or 3.2mm, whichever is smaller.
      c. Operable Units: Provide a minimum 1.6mm clearance between framing members and operable units.
      d. Cantilever Deflection: Where framing members overhang an anchor point, limited to 2 times the length of cantilevered member, divided by 175.

   3. Deflection of Infills (Glass):
      a. Horizontal deflection for four-way supported glazing infills shall be as follows:
         1) For center of glass deflection: L/60 or 25mm whichever is minimum.
         2) For the supported glass edge L/175.

C. Thermal requirements: Framing systems and its associated structure like anchoring structures shall accommodate expansion and contraction movement due to surface temperature differential without causing buckling, stress on glass, failure of joint seals, excessive stress on structural elements, reduction of performance or other detrimental effects.

D. Base engineering calculation on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.
   1. Temperature Change (Range): 70 deg. C, ambient; 100 deg. C, material surfaces.
   2. Night time or lower range: +10 deg. C.

1.5 SUBMITTALS:

A. General: Prepare, review, approve, and submit specified submittals in accordance with "Conditions of the Contract" and Division 1 Submittals Sections. Product data, shop drawings, samples, and similar submittals are defined in "Conditions of the Contract".

B. Product Data: Manufacturer's data sheets on each type of door series (product) to be used, including:
   1. Preparation instructions and recommendations.
   2. Storage and handling requirements and recommendations.
   3. Installation methods.

C. Shop Drawings: Include outside net frame dimensioning, direction of swing (outswing or inswing), typical head, side jamb, sill and panel details and type of glazing material per vertical plan and elevations view drawings; including anchorage, accessories, finish colors and textures.

D. Selection Samples: For each finish product specified, two complete sets of color chips representing manufacturer's full range of available colors and patterns.

E. Verification Samples: For each finish product specified, two samples, minimum size 150 mm square, representing actual product, color, and patterns.

F. Manufacturer's Certificates: Certify products meet or exceed specified requirements.

G. Closeout Submittals: Provide manufacturer's warranty documents and maintenance instructions, that include recommendations for periodic checking and adjustment and periodic cleaning and maintenance of all components.

H. Other Action Submittals:
   1. Swing Door Hardware Schedule: Prepared by or under the supervision of supplier, detailing fabrication and assembly of door hardware, as well as procedures and diagrams. Coordinate final door hardware schedule with doors, frames, and related work to ensure proper size, thickness, hand, function, and finish of door hardware.

1.6 QUALITY ASSURANCE

A. Manufacturer Qualifications: Provide complete, engineered and high-quality swing door system by a single source manufacturer with at least Ten (10) years documented experience manufacturing required door systems in the region or the place of work.

B. Installer Qualifications: Installer with documented experienced of minimum Five (5) years in the installation of manufacturer's door systems or similar to the products specified.

C. Source Limitations: Obtain aluminum-framed glass door through one source from a single manufacturer.

D. Test Reports: Submit certified test reports showing compliance with specified performance characteristics and physical properties.

E. Qualification Data: Submit manufacturers and installer qualification data.

F. Mock-Up: Provide a mock-up for evaluation of surface preparation techniques and application workmanship.
   1. Finish areas designated by Engineer (Architect).

2. Do not proceed with remaining work until workmanship, color, finish and sheen are approved by Engineer (Architect).
3. Refinish mock-up area as required to produce acceptable work.

G. Do not modify intended aesthetic effects, as judged solely by Architect, except with Architect's approval. If modifications are proposed, submit comprehensive explanatory data to Architect for review.

H. Pre-installation Conference: Conduct conference at Project site to comply with requirements in Division 01 Section “Project Management and Coordination”.

1.7 DELIVERY, STORAGE AND HANDLING

A. As per manufacturer’s recommendations.

B. Store products in manufacturer's unopened packaging until ready for installation.

C. Protect stored product from damage. Store products flat in dry, well-ventilated area out of direct sunlight, under cover, protected from weather, moisture and excessive dryness and construction activities.

D. Deliver components in the manufacturer's original protective packaging.

E. Remove temporary protection after installation when acceptable to the Engineer. Do not leave coating residue on any surface.

1.8 SEQUENCING

A. Ensure that locating templates and other information required for installation of products of this section are furnished to affected trades in time to prevent interruption of construction progress.

B. Ensure that products of this section are supplied to affected trades in time to prevent interruption of construction progress.

1.9 PROJECT CONDITION

A. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's absolute limits.

B. Field Measurements:
1. Verify actual locations of structural supports for aluminium framed doors by field measurements before installation and indicate measurements on Shop Drawings.
2. Coordinate fabrication schedule with construction progress to avoid delay of the work.

1.10 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace aluminum framed swing doors that fails in materials or workmanship within specified warranty period.
1. Failures include, but are not limited to, the following:

a. Failure to meet performance requirements.
b. Structural failures including excessive deflection, water leakage, air infiltration, or condensation.
c. Faulty operation of movable sash and hardware.
d. Deterioration of metals, other materials, and metal finishes beyond normal weathering.
e. Failure of insulating glass.

2. Warranty Period:
a. Doors and Window: Ten (10) years from date of substantial completion.
b. Glazing: Ten (10) years from date of substantial completion.
c. Metal Finish: Ten (10) years from date of substantial completion.
d. Anodized Coating Finish: Twenty (20) years from date of substantial completion.
e. Super Durable Powder Coating Finish: Twenty-five (25) years (for product) from date of substantial completion.

A. The warranty shall not include damage caused by vandalism, or natural conditions exceeding the performance requirements.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. Section Manufacturers
1. Technal, Middle East/Germany.
2. Wicona, Middle East/Germany.
3. Schuco, Middle East/Germany.
4. Gutmann, Middle East/Germany.
5. Hueck, Middle East/Germany.
Or Equivalent, subjected to Engineer's acceptance.

B. Fabricators/Assemblers/System manufacturers:
1. Alico, Middle East.
2. Technical Glass, Middle East.
3. Al Abbar, Middle East.
4. Alumco, Middle East.
5. Permasteelisa Gartner, Middle East.
6. Glassline Industries, Middle East.
7. Reem Emirates Aluminium LLC, Middle East.
Or Equivalent, subjected to Engineer's acceptance.

C. Hardware Manufacturers/Supplier (Supplied or approved by system supplier):
1. Dorma, Middle East.
2. ASSA ABLOY Entrance Systems, Middle East.
3. Geze, Middle East.
4. Blasi Automatic Doors, Middle East
Or Equivalent, subjected to Engineer's acceptance.

Condition to qualify "equivalent": Qualification exercise consists of assessment of technical adaptability of the façade system design and the material, quality procedures maintained, minimum 10 (ten) years of its usage in similar project and credibility of its performance record and other project references. All cost related to this exercise shall be borne by the proposer. The proposer shall agree to use above

listed pre-approved system manufactures, should the proposed system be not acceptable to the Engineer, after the completion of qualification exercise.

2.2 HINGES, GENERAL

A. Quantity: Provide the following, unless otherwise indicated:
   1. Two Hinges: For doors with heights up to 1250 mm.
   2. Three Hinges: For doors with heights 1550 to 2300 mm.
   3. Four Hinges: For doors with heights 2300 to 3050 mm.
   4. For doors with heights more than 3050 mm, provide 4 hinges, plus 1 hinge for every 750 mm of door height greater than 3048 mm.

B. Hinge Weight: Unless otherwise indicated, provide the following:
   1. Entrance Doors: Heavy-weight hinges.
   2. Doors with Closers: Antifriction-bearing hinges.
   3. Interior Doors: Standard-weight hinges.

C. Hinge Base Metal: Unless otherwise indicated, provide the following:
   1. Exterior Hinges: Stainless steel, with stainless-steel pin.
   2. Interior Hinges: Stainless steel, with stainless-steel pin.
   3. Hinges for Fire-Rated Assemblies: Steel, with steel pin, Stainless steel, with stainless-steel pin.

D. Hinge Options: Where indicated in door hardware sets or on Drawings:
   1. Maximum Security Pin: Fix pin in hinge barrel after it is inserted.
   2. Nonremovable Pins: Provide set screw in hinge barrel that, when tightened into a groove in hinge pin, prevents removal of pin while door is closed; for outswinging exterior doors and outswinging corridor doors with locks.

E. Fasteners: Comply with the following:
   1. Machine Screws: For metal doors and frames; install into drilled and tapped holes.
   2. Screws: Phillips flat-head; machine screws drilled and tapped holes for metal doors. Finish screw heads to match surface of hinges.
   3. Zinc plated steel concealed fasteners; Hardened aluminum alloys or AISI 300 series stainless steel exposed fasteners, countersunk, finish to match aluminum color.

2.3 HINGES: Supply and fit hinges that comply with the noted BHMA Standards as set out in HMMA Certified Product Directory.

A. Butts and Hinges: BHMA A156.1.

B. Template Hinge Dimensions: BHMA A156.7.

C. Spring Hinges
   1. Self-Closing Hinges: BHMA A156.17.

D. Closers: Pivots and Pivot Hinges
   1. Closers: BHMA A156.4.
   2. Self-Closing Hinges and Pivots: BHMA A156.17

E. Continuous Hinges
   1. Standard: BHMA A156.26.

2. General: Minimum 3 mm thick, hinge leaves with minimum overall width of 100 mm; fabricated to full height of door and frame and to template screw locations; with components finished after milling and drilling are complete.
   a. Fire Pins: Steel pins to hold fire doors in place if required by tested listing.
3. Continuous, Barrel-Type Hinges: Hinge with knuckles formed around a pin that extends entire length of hinge. Base metal for all hinges to be stainless steel.

2.4 LOCKS AND LATCHES, GENERAL

A. Accessibility Requirements: Where indicated to comply with accessibility requirements, comply with "Americans with Disabilities Act (ADA), Accessibility Guidelines for Buildings and Facilities (ADAAG)." [ANSI A117.1.] and in compliance with all prevailing local regulations.

B. Latches and Locks for Means of Egress Doors: Comply with NFPA 101. Latches shall not require more than 65 N to release the latch. Locks shall not require use of a key, tool, or special knowledge for operation.

C. Electrified Locking Devices: BHMA A156.25.

D. Lock Trim:
   1. Levers.
   2. Knobs.
   3. Escutcheons (Roses).
   4. Dummy Trim: Match knob or lever lock trim and escutcheons.
   5. Lockset Designs: Provide design indicated on Drawings or, if sets are provided by another manufacturer, provide designs that match those designated.

E. Strikes: Manufacturer's standard strike with strike box for each latch bolt or lock bolt, with curved lip extended to protect frame, finished to match door hardware set, and as follows:
   1. Strikes for Bored and Preassembled Locks and Latches: BHMA A156.2.
   2. Strikes for Mortise Locks and Latches: BHMA A156.13.
   3. Strikes for Auxiliary Deadlocks: BHMA A156.5.
   4. Aluminum Frame Strike Box: Manufacturer's special strike box fabricated for aluminum framing.

2.5 MECHANICAL LOCKS AND LATCHES

A. Lock Functions: Function numbers and descriptions indicated in door hardware sets comply with the following:
   1. Bored Locks: BHMA A156.2.
   2. Mortise Locks: BHMA A156.13.
   3. Interconnected Locks: BHMA A156.12.

2.6 ELECTROMAGNETIC LOCKS

A. General: BHMA A156.23; electrically powered, of strength and configuration indicated; with electromagnet attached to frame and armature plate attached to door.
   1. Type: Full exterior or full interior, as required by application indicated.
   2. Strength Ranking 4500 N.
   3. Inductive Kickback Peak Voltage: Not more than 50 V.
   4. Residual Magnetism: Not more than 10 N to separate door from magnet.

B. Delayed-Egress Locks: BHMA A156.24.

1. Means of Egress Doors: Lock releases within 15 seconds after applying a force not more than 65 N for not more than 3 seconds, as required by NFPA 101.
2. Security Grade: Activated from secure side of door by initiating device.
3. Movement Grade: Activated by door movement as initiating device.

2.7 SELF-CONTAINED ELECTRONIC LOCKS

A. General: Internal, battery-powered, self-contained electronic locks; consisting of complete lockset, motor-driven lock mechanism, and actuating device; enclosed in zinc-dichromate-plated, wrought-steel case. Provide key override, low-battery detection and warning, LED status indicators, and ability to program at the lock; type and function indicated.
   1. Type: Mortise.
   2. Actuating Device: Digital keypad, Magnetic-stripe card reader or biometric finger point reader.
      a. Card: Manufacturer's standard.
   3. Faceplate Material: Stainless steel.
   4. Trim: Lever or Match trim specified for mechanical locks.

B. Accessories: Card encoder and software.

2.8 EXIT LOCKS AND EXIT ALARMS

A. Exit Locks: BHMA A156.29, Grade 1, surface mounted, battery powered, housed in metal case; with red-and-white lettering reading "EMERGENCY EXIT PUSH TO OPEN-ALARM WILL SOUND. "Include the following features:
   1. Low-battery alert.
   2. Outside key control.
   3. Audible alarm that sounds when unauthorized use of door occurs.
   4. Silent alarm with remote signal capability for connection to remote indicating panel.

2.9 EXIT DEVICES

A. Exit Devices: BHMA A156.3, Grade 1 or as is indicated.

B. Accessibility Requirements: Where handles, pulls, latches, locks, and other operating devices are indicated to comply with accessibility requirements, comply with "Americans with Disabilities Act (ADA), Accessibility Guidelines for Buildings and Facilities (ADAAG), ANSI A117.1. and in strict compliance with all applicable local regulations.

C. Exit Devices for Means of Egress Doors: Comply with NFPA 101. Locks shall not require use of a key, tool, or special knowledge for operation.

D. Panic Exit Devices: Listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for panic protection, based on testing according to UL 305.

E. Fire Exit Devices: Devices complying with NFPA 80 that are listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for fire and panic protection, based on testing according to UL 305 and NFPA 252.

F. Outside Trim: Lever or Knob; material and finish to match locksets, unless otherwise indicated.
   1. Match design for locksets and latch sets, unless otherwise indicated.

2.10 LOCK CYLINDERS

A. Standard Lock Cylinders: BHMA A156.5, Grade 1.

B. High-Security Lock Cylinders: BHMA A156.30, Grade 1.
   1. Key Control Level: Category A.

C. Cylinders: Manufacturer's standard tumbler type, constructed from brass or bronze, stainless steel, or nickel silver, and complying with the following:
   1. Number of Pins: Six or seven as indicated on hardware schedule.
   2. Mortise Type: Threaded cylinders with rings and straight- or clover-type cam.
   3. Bored-Lock Type: Cylinders with tailpieces to suit locks.
      a. High-Security Grade: BHMA A156.5, Grade 1A, listed and labeled as complying with pick- and drill-resistant testing requirements in UL 437 (Suffix A).

D. Permanent Cores: Manufacturer's standard; finish face to match lockset; complying with the following:
   1. Interchangeable Cores: Core insert, removable by use of a special key; usable with other manufacturers' cylinders.

E. Construction Keying: Comply with the following:
   1. Construction Master Keys: Provide cylinders with feature that permits voiding of construction keys without cylinder removal. Provide 10 construction master keys.
   2. Construction Cores: Provide construction cores that are replaceable by permanent cores. Provide 10 construction master keys.
      a. Replace construction cores with permanent cores as indicated in keying schedule and as directed by Engineer (Architect).
      b. Furnish permanent cores to Owner (Client) for installation.

F. Manufacturer: Same manufacturer as for locks and latches.

2.11 KEYING

A. Keying System: Factory registered, complying with guidelines in BHMA A156.28, Appendix A. Incorporate decisions made in keying conference, and as follows:
   1. Master Key System: Cylinders are operated by a change key and a master key.
   2. Grand Master Key System: Cylinders are operated by a change key, a master key, and a grand master key.
   3. Great-Grand Master Key System: Cylinders are operated by a change key, a master key, a grand master key, and a great-grand master key.

B. Keys: Nickel silver.
   1. Stamping: Permanently inscribe each key with a visual key control number and include the following notation:
      a. Notation: "DO NOT DUPLICATE" – To be coordinated with Facility Management.
   2. Quantity: In addition to one extra key blank for each lock, provide the following in coordination with the Facility Management:
      a. Cylinder Change Keys: Three.
      b. Master Keys: Five.
      c. Grand Master Keys: Five.
      d. Great-Grand Master Keys: Five.

C. Keying System shall meet the requirements of the Owner (Client).

2.12 CLOSERS

A. Accessibility Requirements: Where handles, pulls, latches, locks, and other operating devices are indicated to comply with accessibility requirements, comply with "Americans with Disabilities Act (ADA), [ANSI A117.1.] and in strict compliance with local regulations.

B. Door Closers for Means of Egress Doors: Comply with NFPA 101.

C. Hold-Open Closers/Detectors: Coordinate and interface integral smoke detector and closer device with fire alarm system.

D. Flush Floor Plates: Provide finish cover plates for floor closers unless thresholds are indicated. Match door hardware finish, unless otherwise indicated.

E. Recessed Floor Plates: Provide recessed floor plates with insert of floor finish material for floor closers unless thresholds are indicated. Provide extended closer spindle to accommodate thickness of floor finish.

F. Surface Closers: BHMA A156.4, Grade 1 is indicated. Provide type of arm required for closer to be located on non-public side of door, unless otherwise indicated.

G. Concealed Closers: BHMA A156.4, Grade 1.

H. Coordinators: BHMA A156.3.

2.13 STOPS AND HOLDERS

A. Stops and Bumpers: BHMA A156.16, Grade 1.
   1. Provide floor stops for doors unless wall or other type stops are scheduled or indicated. Do not mount floor stops where they will impede traffic. Where floor or wall stops are not appropriate, provide overhead holders.

B. Mechanical Door Holders: BHMA A156.16, Grade 1.

C. Combination Floor and Wall Stops and Holders: BHMA A156.8, Grade 1.

D. Electromagnetic Door Holders: BHMA A156.15.
   1. Coordinate with fire detectors and interface with fire alarm system and fire door assemblies.

E. Silencers for Metal Door Frames: BHMA A156.16, Grade 1; neoprene or rubber, minimum diameter 13 mm; fabricated for drilled-in application to frame.

2.14 DOOR GASKETING SYSTEMS

A. Standard: BHMA A156.22.

B. General: Provide continuous weather-strip gaskets on exterior doors and provide smoke, light, or sound gaskets on interior doors where indicated or scheduled. Provide non-corrosive fasteners for exterior applications and elsewhere as indicated.
   1. Perimeter Gaskets: Apply to head and jamb, forming seal between door and frame.
   2. Meeting Stile Gaskets: Fasten to meeting stiles, forming seal when doors are closed.
   3. Door Bottoms: Apply to bottom of door, forming seal with threshold when door is closed.

C. Replaceable Seal Strips: Provide only those units where resilient or flexible seal strips are easily replaceable and readily available from stocks maintained by manufacturer.

D. Gasket Materials: ASTM D 2000 and AAMA 701/702.

2.15 THRESHOLDS

A. Standard: BHMA A156.21.

B. Accessibility Requirements: Where thresholds are indicated to comply with accessibility requirements, comply with "Americans with Disabilities Act (ADA), Accessibility Guidelines for Buildings and Facilities (ADAAG) and in strict compliance include with local regulations.
   1. Bevel raised thresholds with a slope of not more than 1:2. Provide thresholds not more than 13 mm high and 19 mm high for exterior doors.

C. Thresholds for Means of Egress Doors: Comply with NFPA 101. Maximum 13 mm high.

2.16 SWING DOORS AND FRAMES

A. Materials:
   1. Extrusions: ASTM B 221 (ASTM B 221M), 6063-T5 and 6063-T6 Aluminum Alloys.
   2. Solid Aluminum Sheet:
      a. For Painted Finish: ASTM B 209 (ASTM B 209M), 3003-H14 Aluminum Alloy, 0.080” (1.5 mm) minimum thickness.
      b. For Anodized Finish: ASTM B 209 (ASTM B 209M), 5005-H14 Aluminum Alloy, 0.050” (1.5 mm) minimum thickness.

B. Aluminum Swing Door:
   1. Frame and Panels: Complete single or double outswing or inswing door system including head, side jambs, thresholds and aluminum panels to sizes indicated on the Drawings.
      a. Panel: Extruded aluminum stile and rail panel with glazing as indicated in the drawings.
         1) Thickness: As indicated in the drawing.
         2) Stile and Rail: As indicated in the drawing.
         3) Bottom rail: As indicated in the drawing.
         4) Finish/Color: To match approved sample.
      b. Frame and Sill: Extruded aluminum.
      c. Weatherstripping: Weather seal inserted in frame and sill to provide perimeter seal, as well as between door panels.
   2. Configuration:
      a. Complete swing door system including head, side jambs, sills and aluminum door panels.
      b. Sizing and Configuration Options: Provide frames and door panels to sizes and swings indicated on the Drawings.
   3. Glass: All glass to comply with safety glazing requirements of ANSI Z97.1 and CPSC 16CFR 1201.
      a. Glazing: Silicone bedding on exterior surfaces and glazing seal on the interior of the panel.
      b. Refer to Section “Glazing- 80 8000” for glass and glazing.
   4. Locking Hardware and Handles: Manufacturer's standard multi-point locking system.
      a. Handle:
         1) Finish: Brushed Satin Stainless (Inox) Steel for non-marine environment and Mirror polished for the marine zones.

2) Vertical full height bar handle on both the sides at exterior entrance door or as indicated in the drawings.

b. Provide panel handle height centered at 36 inches (914 mm) from bottom of panel if not indicated in the drawing.

5. Hardware: Manufacturer's standard hardware integrated with engineered side jambs and threshold frame system.
   a. Hinges: Double leaf bearing hinges.
   b. Concealed panic hardware.
   c. Threshold:
      1) Standard Self-draining Weather Resistant Sill.
      2) Flush sill.
      3) Ramp sill.
      4) ADA Compliant ramp sill.
   d. Hardware finish to match the door frame or profile finish.
6. Aluminum Finish: Provide same finishes on inside and outside.
   a. Refer Section 2.18 “ALUMINIUM FINISHES” for frame or profile finish.
7. Threshold Finish:
   a. Same as door frame finishes.

2.17 FABRICATION

A. Sizes and Configurations: Fabricate to sizes and configurations indicated on the Drawings. Verify opening sizes by field measurement prior to completing fabrication.

B. Verify opening direction.

C. Fabricate and assemble units with joints only at intersection of aluminum members with hairline joints; rigidly secure, and sealed in accordance with manufacturer’s recommendations.

D. Manufacturer's non-integral nail fin for use in conjunction with standard flashing, weather resistant barriers and compatible sealants.

2.18 ALUMINUM FINISHES

A. All aluminium visible surfaces exposed to the building interior and exterior: Aluminium surfaces shall be finished with Super Durable Powder Coating complying with AAMA 2604 and that are thermoset, free from VOCs, with minimum dry film thickness of 60 microns.
   1. Color and Finish: To match approved color sample.

B. Prepare, pretreat and apply coating (including protective coat) to exposed metal surface to comply with AAMA 2604.

C. Concealed Aluminum Surfaces: Any aluminum sections which are completely concealed and therefore do not require the facing color shall be anodized to a minimum standard of AMMA 611 Class II.

D. Identification marks shall not appear on finished surfaces.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Do not begin installation until substrates have been properly prepared.

B. Verify that dimensions of rough opening will fit the net frame dimensions of door system; verify that rough openings are level, plumb, and square, with no unevenness in the floor.

C. If substrate preparation is the responsibility of another installer, notify Engineer (Architect) of unsatisfactory preparation before proceeding.

D. Verify dimensions, tolerances, and method of attachment with other Work

3.2 PREPARATION

A. Clean surfaces thoroughly prior to installation.

B. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

C. Remove dust and other loose material from openings.

3.3 INSTALLATION

A. Install in accordance with manufacturer's instructions, and within specified tolerances.

B. Properly flash and waterproof around the perimeter of the opening and frame; when fixed independently on the concrete opening. Adequate overhangs to prevent the effects of sheeting water from above are recommended.

C. Provide adequate anchorage devices. Securely fit frame in place, level, straight, plumb and square. Install frame in proper elevation, plane and location, and in proper alignment with other work.

D. Protect aluminum members in contact with masonry, steel, concrete, or dissimilar materials using nylon pads or Acrylic based one component protective coating.

E. Shim and brace aluminum system before anchoring to structure.

F. Ensure doors are adjusted for proper operation.

3.4 ADJUSTMENT, CLEANING AND PROTECTION

A. Adjusting: Adjust operating items as recommended by manufacturer.

B. Protect installed products until completion of project.

C. Protect installed product from construction activities, particularly thresholds and floor channels.

D. Touch-up, repair or replace damaged products before Substantial Completion.

E. Cleaning: The General Contractor shall clean installed products in accordance with manufacturer’s instructions prior

F. To Owner's acceptance, and remove construction debris from project site. Legally dispose of debris.

3.5 SCHEDULES

A. Refer Engineer's (Architect's) Door schedule.

END OF SECTION 08 1116.01

SECTION 08 4413.01 - GLAZED ALUMINIUM CURTAIN WALLS

PART 1 - GENERAL

Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.1 SECTION INCLUDES

A. Design, engineering, supply, fabrication, installation and test of Aluminum curtain wall systems, accessories, complete with reinforcing, shims, anchors and attachment devices, etc. as specified and as shown on drawings and perform all services and operations necessary for the complete supply and installation of all glazed curtain walls including but not limited to the following:

1. Unitized curtain wall system for Tower-H32 façade; based on Female – Female configuration integrated with solid metal sheet cladding C-feature at slab junction, weather louvers, façade lighting and BMU restraint sockets. Including, UCW integrated automated Guillotine windows.
2. Aluminium stick curtain wall system with/without reinforcement for Building-H27 and H30; Tower H-32 winter garden, below sawtooth façade area, entrance facade, and event/ entertainment box façade integrated/ making an interface with:
   a. Solid alum. metal sheet cladding.
   b. Alum. operable elements, main entrance curved sliding and hinged doors.
   c. Vertical and horizontal glass fins as architectural/ shading feature.
   d. Extruded alum. vertical aero-foil architectural/ shading feature.
3. Add-on steel curtain wall at entrance skylight and sawtooth façade; steel framing members wrapped with solid alum. sheet.
4. Point fixed glazing with curved IGU at event box cylindrical lift.
5. Roof trellis or shading canopy with extruded alum. profiles, fixed/ connected within the canopy steel structure.
6. Integrated with accessible type façade lighting as indicated/ detailed in the drawings; with concealed fixtures and cable arrangement.
7. Double glazed insulated vision glass; spandrel glass with ventilated back pan, mechanically held in place using toggles.
8. Aluminum trims, snap in sealant stops, flashings and similar items in conjunction with aluminum curtain wall systems and assemblies.
9. Painting and coating in conjunction with the above aluminum and steel items.
10. Internal and perimeter sealants, weeps, vents and gasket systems. Anchors, embeds, shims, fasteners, inserts, expansion devices, accessories, support brackets and attachments.
11. Perimeter fire resistive joint system as required to satisfy the latest edition international code requirements.
12. Smoke seals, thermal breaks and insulations and water proofing assemblies.
13. Sealant and caulking for works of this section and adjacent construction.
14. Provide façade glass with compatible Nano coating for reducing the façade cleaning frequency.
15. Façade connections to the main structure shall be designed not to transfer any load from main to the secondary façade supporting structure.

B. This part in conjunction with the architectural drawings define the scope of the works and establishes requirements for the materials /assembly's performance requirements and aesthetic effects as indicated by dimensions, arrangements, alignment, and profiles of components and assemblies as they relate to sightlines, to one another, and to adjoining construction.

C. Delegated Design: Notwithstanding the drawings and other information provided, the Contractor shall be fully responsible for the design including comprehensive engineering analysis by an experienced façade professional engineer, detailing of the works, described herein, and shall provide a complete, and secure installation, which shall meet the specified performance requirements and the design criteria indicated in this specification.

D. The contractor shall be responsible for obtaining all relevant data including the full extent and detailing of the works specified in this part, and allow for all necessary monetary and time provision to ensure the proper and complete installation of the system; and shall be fully responsible for all engineering calculations and detail designs of the works.

E. This specification shall be taken as a single contract for the works specified in this part and shall not be sub-divided into multi-trade packages.

F. Design to follow the architectural intent indicated in the drawings.

G. This section to be read in conjunction with the related wall types/ exterior wall system drawings submitted as a part of the facade tender package. Including, and not limited to Structural, Thermal, Sustainability and Acoustic Report at its latest revision.

H. RELATED SECTIONS

1.	Section 01 4500	-	Quality Control
2.	Section 03 3000	-	Cast-in-Place Concrete
3.	Section 04 2200	-	Concrete Unit Masonry
4.	Section 05 7313.01	-	Glazed Decorative Metal Railing
5.	Section 05 7500.01	-	Decorative Formed Metal
6.	Section 07 2100.01	-	Thermal Insulation
7.	Section 07 2500.01	-	Weather Barrier
8.	Section 07 6113.01	-	Standing Seam Sheet Metal Roofing
9.	Section 07 6200.01	-	Sheet Metal Flashing and Trim
10.	Section 07 8400.01	-	Fire Stopping
11.	Section 07 9200.01	-	Joint Sealants
12.	Section 08 1116.01	-	Alum. Doors and Frames
13.	Section 08 4000	-	Aluminium Works
14.	Section 08 8000.01	-	Glazing
15.	Section 08 9100.01	-	Louvers
16.	Section 11 8123.01	-	Façade Cleaning and Maintenance Equipments

1.2 REFERENCES

A. The minimum standards for products specified in this section shall be including as under but not limited to the following. Except as otherwise specified herein, perform work in accordance with specifications, codes and standards cited therein, and their latest editions, revisions, applicable addenda and supplements. Where there is conflict between the referenced standards the most stringent of the conditions/requirements shall be applicable.

B. Aluminum Association (AA):
   1. DAF-45 Designation System for Aluminum Finishes.

C. American Architectural Manufacturers Association (AAMA):
   1. Aluminum Curtain Wall Design Guide Manual.
   2. 501.2: Field Check of Metal Curtain Walls for Water Leakage.

3. AAMA 612: Voluntary Specification, Performance Requirements, and Test Procedures for Combined Coatings of Anodic Oxide and Transparent Organic Coatings on Architectural Aluminum.
4. 2604: Voluntary Specification for High Performance Organic Coatings on Architectural Extrusions and Panels.
5. 606.1: Specifications and Inspection Methods for Integral Color Anodic Finishes for Architectural Aluminum.
6. 607.1: Specifications and Inspection Methods for Clear Anodic Finishes for Architectural Aluminum.
7. 608.1: Specification and Inspection Methods for Electrolytic ally Deposited Color Anodic Finishes for Architectural Aluminum.
8. 701.2: Specifications for Pile Weather-stripping.
9. Manual#10: Care and Handling of Architectural Aluminum from Shop to Site.

D. American National Standards Institute (ANSI):
   1. Z97.1: Specifications and Methods of Test for Safety Glazing Material Used in Buildings.

E. American Society for Testing and Materials (ASTM):
   1. A36: Structural Steel.
   2. A123: Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
   3. A525: General Requirements for Steel Sheet, Zinc-Coated (Galvanized) by the Hot-Dip Process.
   4. A526: Sheet Steel, Zinc Coated (Galvanized) by the Hot-Dip Process, Commercial Quality.
   5. B209: Aluminum and Aluminum-Alloy Sheet and Plate.
   6. B221: Aluminum-Alloy Extruded Bars, Rods, Wire, Shapes, and Tubes.
   7. B308: Aluminum-Alloy 6061-T6 Standard Structural Shapes, Rolled or Extruded.
   8. ASTM B429/ B429M: Standard Specification for Aluminum- Alloy Extruded Structural Pipe and Tube.
   9. C509: Cellular Elastomeric Preformed Gasket and Sealing Material.
   10. C716: Installing Lock-Strip Gaskets and Infill Glazing Materials.
   11. E 783: Standard Test Method for Field Measurement of Air Leakage through Installed Exterior Windows and Doors.
   12. 501.1: Standard Test Method for Water penetration of Windows, Curtain Walls and Doors Using Dynamic Pressure.
   13. C864: Dense Elastomeric Compression Seal Gaskets, Setting Blocks and Spacers.
   14. C920: Elastomeric Joint Sealants.
   15. E283: Rate of Air Leakage through Exterior Windows, Curtain Walls, and Doors.
   16. E330: Structural Performance of Exterior Windows, Curtain Walls, and Doors by Uniform Static Air Pressure Difference.
   17. E331: Test Method for Water Penetration of Exterior Windows, Curtain Walls, and Doors by Uniform Static Air Pressure Difference.
   18. E773: Test Method for Seal Durability of Sealed Insulating Glass Units.
   19. E774: Sealed Insulating Glass Units.

F. Consumer Product Safety Commission (CPSC):

1. 16 CFR 1201: Safety Standard for Architectural Glazing Materials.

G. Federal Specifications (FS):
   1. TT-P-645A: Primer, Paint, Zinc Chromate, Alkyd Type.
   2. AWS D1.2: American Welding Society, Structural Welding Code- Aluminum.

H. GANA Glass Association of North America Glazing Manual.

I. Flat Glass Marketing Association (FGMA):
   1. Glazing Manual.

J. Steel Structures Painting Council (SSPC):
   1. SP2: Hand Tool Cleaning.
   2. SP3: Power Tool Cleaning.
   3. Paint 12: Cold-Applied Asphalt Mastic (Extra Thick Film).

K. American Society of Civil Engineers:
   1. ASCE 7-10 Wind Loads: Guide to the Wind Load Provisions.

L. Centre for Window and Cladding Technology (CWCT)
   1. Standard for systemized building envelopes.

M. All applicable local codes, regulations, etc. of Authorities having Jurisdiction.

Note: Contractor shall note that he shall keep copies of all of the referenced standards on the site and shall provide the Engineer copy of the requested referenced standard as and when requested.

1.3 PERFORMANCE REQUIREMENTS

A. This section of specifications is the performance type. The design and performance specified are minimum requirements. Methods which may be suggested by these specifications are not intended to limit methods of achieving the required design and performance criteria. All the drawings must be reviewed, approved, stamped and certified for the design of this project by the system supplier company (not manufacturer) and a Third-Party professional Engineer.

B. General: Provide aluminium framed doors and windows system, stick curtain wall system, metal sheet decorative cladding system, façade lighting and rope access fixtures; including anchorage, capable of withstanding, without failure, the effects of the following:
   1. Structural loads.
   2. Thermal Movements.
   3. Movements of supporting structure indicated on Drawings including, but not limited to, story drift, twist, column shortening, long-term creep, and deflection from uniformly distributed and concentrated live loads.
   4. Dimensional tolerances of building frame and other adjacent construction.
   5. Failure includes the following (not limited to):
      a. Deflection exceeding specified limits.
      b. Thermal stresses transferred to building structure.
      c. Framing members transferring stresses, including those caused by thermal and structural movements, to glazing.
      d. Glazing-to-glazing contact.
      e. Noise or vibration created by wind and thermal and structural movements.
      f. Loosening or weakening of fasteners, attachments, and other components.
      g. Spontaneous breakage of the tempered glass as a result of NiS inclusion.
      h. Sealant failure.

i. Air leakage, water leakage, exceeding specified limits.

C. Air Infiltration: Air movement through the work, building interior to building exterior and building exterior to building interior shall not exceed.
   1. For Fixed Element: 1.5 cum/hr. / sq.m when tested according to ASTM E 283 at a minimum static-air-pressure differential 600 Pa.
   2. For Operable Elements When closed: 2.0 cum/hr./m of crack length when tested according to ASTM E 283 at a minimum static-air-pressure differential 300 Pa.

D. Water Infiltration:
   1. Work to quickly drain any water and condensate moisture which may enter within the cladding components to the building exterior.
   2. Work to prevent entry of water and moister to the building interior side of the work.
   3. Provide curtain wall system with glazing and panel fittings that does not show evidence of water leakage when tested according to ASTM E 331 at minimum pressure differential of 20 percent of inward acting wind-load design pressure, but not less than 600 Pa.
   4. Provide curtain wall system with glazing and panel fittings that does not show evidence of water penetration through fixed glazing and framing areas when tested according to AAMA 501.1 at dynamic pressure equal to 20 percent of wind-load design pressure, but not less than 600 Pa.
   5. Water leakage is defined as follows:
      a. Uncontrolled water infiltrating system or appearing on system's normally exposed interior surfaces from sources other than condensation according to AAMA 501.1. Water controlled by flashing and gutters that is drained back to the exterior and not causing damage to the adjacent materials or finishes is not water leakage.

E. Structural Loads:
   1. Wind load: Refer to the available wind tunnel analysis or project wind report at its latest revision for the design loads.

   2. Deflection of framing members:
      a. Deflection Normal to Wall Plane: Limited to 1/175 of clear span up to a length of 4.1m. Above 4.1m, the deflection is limited to 1/240+6.35. However, such deflection shall not influence the integrity of adjacent trade like bulkhead, wall connections etc.
      b. Deflection parallel to Glazing Plane: Limited to L/360 of clear span or 3.2mm, whichever is smaller.
      c. Operable Units: Provide a minimum 1.6mm clearance between framing members and operable units.
      d. Cantilever Deflection: Where framing members overhang an anchor point, limited to 2 times the length of cantilevered member, divided by 175.
   3. Deflection of Infills:
      a. Horizontal deflection for four-way supported glazing infills shall be as follows:
         1) For center of glass deflection: L/60 or 25mm whichever is minimum.
         2) For the supported glass edge L/175.
         3) Maximum center deflection (vertical) for four side supported horizontal glazing shall be L/100.
         4) For center of aluminium cladding sheet panels or aluminium sandwich panel: Maximum deflection (ver./hor.); L/120 or 15mm, whichever is smaller; including for the panel stiffeners.
   4. Seismic Performance: Glazed aluminium curtain wall system shall withstand the effects of earthquake motions in accordance to ASCE 7-16 as acceptable to the local authorities.
   5. Seismic load – As per ASCE 7-16 as acceptable to the local authorities.

6. Structural Test Performance: Provide glazed curtain wall glazing systems tested according to ASTM E330 as follows:
    a. Test Pressure: At 150 percent of positive and negative wind-load design pressures, systems, including anchorage, do not evidence material failures, structural distress, and permanent deformation of main framing members exceeding 0.2 percent of span.
    b. Test Duration: As required by design wind velocity but not less than 10 seconds.
7. Dead loads:
    a. Maximum full dead load deflections, parallel (in-plane) to wall plane, of framing members shall not reduce glass bite or glass coverage, to less than 75% of the design dimension, and shall not reduce edge clearance to less than 25% of design dimension or 3 mm whichever is greater.
    b. Twisting (rotation) of the horizontals due to the weight of the glass shall not exceed 1 degree, measured between ends and center of each span.

F. Pressure Equalization (Exterior Walls):
1. Provide pressure equalization to glazing pockets, to exterior components of frames.
2. Provide baffles to prevent air flow within components and spaces as may be induced by stack effect and by wind forces. Locate baffles:
    a. Vertically oriented at every building corner and at every mullion (Spandrel areas).
    b. Horizontally sills and heads.
    c. Provide baffles to prevent excessive rain entry in to the components which could lead to water pounding and the prevention of pressure equalization.
    d. Provide baffles to prevent excessive sand entry in to the components. The entry of sand lead to blockages built-up which could prevent expansion possibilities and obstruct pressure equalization of the system.

G. Thermal requirements: Framing systems and its associated structure like anchoring structures shall accommodate expansion and contraction movement due to surface temperature differential without causing buckling, stress on glass, failure of joint seals, excessive stress on structural elements, reduction of performance or other detrimental effects.

H. Base engineering calculation on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.
1. Temperature Change (Range): 70 deg. C, ambient; 100 deg. C, material surfaces.
2. Night time or lower range: +10 deg. C.

I. Thermal Criteria (Exterior Walls)
1. The design should accommodate thermal expansion and contraction tolerances, within the components of façade system and its associated structure like anchoring structures, due to difference in temperature.
2. Thermal performance requirement for the building envelope (Glass & Glass +Frame) is to be referred from the specification Section “Glazing 08 8000”.

J. Glazing: As specified in Part "Glazing 08 8000”, of this Specification.
1. Glazing systems to allow for characteristics of framing and effects of connections and sealant.
2. Work metal, glass, glazing to ensure no more than 8 glass units per 1000 break during the building’s life.
3. Glazing gasket, sealant on building exterior side of glass in exterior walls shall be non-conductive. Gaskets used in exterior walls shall be fused or sealed at corners and shall be of high-performance EPDM quality.
4. Glass retaining devices shall apply uniform continuous pressure to glass to prevent:
    a. Distortion, oil canning and pillowing of glass.
    b. Pressure points on the glass which could contribute to stress breakage.

5. Metal in direct contact with glass not permitted.
6. Insulating glass units must not have their seal in contact with water.

K. Glazing-to-glazing joints accommodate thermal and mechanical movements of glazing and system, prevent glazing-to-glazing contact, and maintain required edge clearances.

L. Acoustic Requirement: Provide building envelope and cladding system which fulfills all requirements of the local authority and comply with the projects Acoustic Report (for the selected Noise Scenario by the Engineer) at its latest revision.
1. Provide complete installation free of rattles wind whistles and other noise due to thermal and structural movements and due to differential atmospheric pressure.

M. UV and Heat Resistance: All material provided which are exposed to atmospheric heat and which are exposed to sun-light including materials exposed through glass shall be formulated not to lose properties due to exposure to heat and due to exposure to ultra-violet radiation.

N. Joints:
1. The joint between the transom and mullions shall be sealed in accordance with the system supplier's standard EPDM solution to be used as sealing strips or equivalent option (not sealant).
2. Transom and mullions shall be assembled by using miter corner design, wherever possible. Visible screws or connectors will not be acceptable.

O. Interface:
1. Furnish inserts and anchoring devices which need to be preset and built into structure to appropriate trade.
2. Supply on timely basis to avoid delay in Work.
3. Instruct other trades of proper location and position.
4. Furnish setting drawings, diagrams, templates and installation instructions.

P. External Feature:
1. In case of structural support penetrating the building envelop, thermal bridging has to be minimized, either with the design or by using low conductive material.
2. The connection and joint design will have to consider the building movement and thermal expansion possibilities.
3. The sizes and dimensions shown in the drawing is the minimum requirement. However, the dimension of the members shall be decided as per the structural requirements.
4. The module, shape, section and distribution as per the drawing.

Q. Building Frame Movement: Design, fabricate and install glazed aluminum curtain walls to withstand building movements including thermal movements, loading deflections, shrinkage, creep and similar movements without glass breakage, anchor failures, or structural damage.

1.4 DESIGN REQUIREMENTS

A. Have this work designed by a Professional Engineer licensed to design structures and registered in the Place of the Work.

B. System manufacturer/supplier shall verify and confirm the façade design prepared by the façade manufacturer/fabricator.

C. Design shall be verified and confirmed by a qualified third-party façade specialist to ensure design compliance to performance requirements.

D. Third party specialist Engineer to be appointed for the review and approval of design calculation and shop drawings.

E. System manufacturer shall provide curtain wall systems, including necessary modifications to meet specified requirements and maintaining visual design concepts.

F. Provide concealed fastening.

G. Exposed steel to be in S/S grade or hot dip galvanized to the specified coating thickness.

H. Design members and their connections to withstand, within specified deflection limitations, their own weight, the weight of the glass and external feature, loads imposed by the motion of operable elements, maintenance loads and the maximum design loads and combination of loads due to rain, seismic loads, the pressure and suction of wind.

I. Follow profiles perimeter sizes and arrangements of components and assemblies indicated to obtain the required appearance.

J. All structural silicone adhesion applications and exposed weather-seal gaskets shall be shop installed under environmentally controlled conditions.

K. Make provisions to drain to the exterior face of the assembly, any water entering at joints and any condensation occurring within the building envelope assembly.

L. Design, fabricate and install the assembly to be watertight to the interior under the interior and exterior design conditions in combination with movements occurring due to loads imposed and all allowable tolerances as specified.

M. The Work shall accommodate, by means of expansion and contraction provisions, any movements in the building envelope assemblies themselves and between the assemblies and the building structure, caused by structural movements, both deflection and racking; and/or thermal expansion and contraction, without distortion, damage, misalignment of joints, breakage of air/vapor barriers, water and air penetration through the assembly, or glass breakage.

N. The Work shall have a method of attachment to the structure which shall take into account site peculiarities so that there shall be no possibility of site and air vibrations or normal temperature movements of the building to loosen, weaken and/or fracture the connection between building envelope assembly components and the structure or between the components themselves.

O. Reinforce building envelope assembly components, as required, so that the members can safely sustain design loads.

P. Unless otherwise indicated, the requirements for an air barrier and a vapor barrier are intended to be provided at the same plane in the building envelope design. The definition of the air/vapor barrier for the purpose of these Specifications is "a continuous membrane including joints of membrane between components and to adjacent construction which prevents or retards penetration of moisture laden air and the diffusion of water vapor through it".

Q. Design, engineer, assemble and secure assemblies in manner which will keep stresses on sealants below the sealant manufacturer's recommended maximum.

R. All glass units of each type shall be the same and shall have a consistent, uniform color and appearance. The external glass layer shall have a constant thickness in directly connected façade areas (elevation).

S. Installed glass components shall be free of oil caning and pillowing.

T. Exposed metal components shall be of uniform color, sheen and appearance, within components and from component to component and shall match approved samples and mock-ups.

U. Installed components panels shall be of uniform color, and visually flat under all lighting conditions.

V. Exposed metal sheets, plates and extrusions shall be visually flat under all lighting conditions.

W. Engineer and install glass units so as each glass unit can be removed without removing adjacent glass units. Glazing shall be replaceable from the exterior of the Curtain Wall from BMU unit access or equivalent.

X. Accurately size mullions at intersecting joints to obtain minimal joints just wide enough to permit thermal expansion and contractions so no damages and distortions to the work occurs due to thermally induced movements.

Y. Provide a continuous fire and smoke seal at junctions between edges of suspended concrete slabs and building envelope to prevent passage of fire and smoke from floor to floor and from floor to roof.

Z. Ensure components and assemblies drain to building exterior.

AA. Ensure components are designed, fabricated and installed to prevent build-up of sand within the facade and cladding cavities and glazing pockets which could cause blockages to air pressure equalization and to drainage of water.

BB. Work to resist forces imposed by window washing stages, maintenance unit cradles, any other building services.

CC. Design vertical mullions to include means of preventing the passage of smoke, noise, airflow and constant electrical connection within the mullions over the height of the building using closure pieces or seals or insulating barriers (disconnectors).

DD. All tempered glass shall be heat soaked. Use a safety factor for tempered glass to the statistical probability of failure. Provide statistical analysis of test data that the probability of failure is not greater than 1 in 54,000 sq.ft. of tempered glass. Provide certifications, that the oven is QM controlled, and these stating the manner. Written warranties against nickel sulphide inclusions or statistical heat soaking in lieu of heat soaking will not be accepted.

EE. Use a safety factor of 5:1 and a design strength of 138 kPa (20 psi) maximum for structural silicone sealant design, including glass unit edge seal.

FF. The vertical aluminum/steel profile shall be drained and ventilated to the outside before profile expansion joints.

GG. The expansion joint of the vertical profile shall be sealed to prevent any leakage of water from the profile drainage groove to the inside section.

HH. The transom-mullion joint shall be overlapped and watertight, airtight developed by a system-controlled solution.

II. All the gasket joints shall be welded compatible to the gasket material. Glue connections are only acceptable for site installed continuous turning gutter gasket. The connection shall have an overlapping lip and a controlled bonding surface.

JJ. Drawings and details are diagrammatic and are intended to show design concept, configuration, components and arrangements; they are not intended to identify nor solve completely the problems of thermal and structural performance and movements, air pressure equalization, air and vapor barriers, assembly framing, fixings and anchorages, moisture disposal, water penetration and problems at the glass line associated with glazing installation, movements, pressure fracture or thermal shock and weather seal.

KK. Tolerances shall not be cumulative.

LL. The external cladding material shall be in accordance to the local authority / civil defence department regulations / requirements.

1.5 SUBMITTALS:

A. General: During the tender bid submission, provide documents to confirm the design intent compliance. Submit in accordance with the Section “Submittal”.

B. Submittals for tender submission shall include but not limited to the following:
1. Initial design drawings.
2. Initial structural calculations.
3. Initial thermal calculations.
4. Criteria compliance statement.
5. Proposed system performance information.
6. Product details of major façade elements.

C. Products Data:
1. Submit system supplier’s printed literature indicating product specification and installation instructions and all relevant testing reports for each product required by this section.
2. Glass performance parameter calculated by manufacturer using Window software and verified by a certified third party.
3. Include erection and glazing details and maintenance and cleaning instructions.
4. Include information for factory finishes, accessories and other required components.
5. Include colour chart for finish indicating manufacturer’s standard colours available for selection.
6. Furnish data clearly describing re-glazing techniques.
7. Furnish maintenance and cleaning instructions.
8. Thermal assessment to express thermal performance of proposed system.
9. Statement to confirm the compliance to acoustic requirement.
10. Submit system supplier’s instructions for manufacturers to be an approved, certified manufacturer.

D. Calculations:
1. Submit calculations, under seal and signature of the sub-contractor’s professional Engineer, showing following:
a. Design load overview according to specifications and wind tunnel analysis report, as available.
b. Detailed engineering of all structural system members of CW.

c. Moment of inertia of all designed compounds (typical and non-typical areas).
d. Detailed engineering of anchorage hardware, sub structure, brackets, clip angles, washers, and anchor bolts, welds, torque pressures.
e. Materials proposed and their allowable sheet and bending stresses.
f. Deflections and expansion and contractions in accordance with all thermal and structural impacts.
g. Comprehensive glass calculation (dead load, live load, thermal load, bonding surfaces IGU).

2. Prepare calculations in a clear comprehensive manner to permit easy review together with drafts of the unit and location at the project and the applicable specific wind load according to the project wind tunnel analysis report. Incomplete calculations will be rejected un-reviewed.

E. Shop Drawings:
1. Submit shop drawings covering fabrication, installation of specified system.
2. Submit drawings indicating elevations, detailed design, dimensions, member profiles, and joint locations, arrangement of units, member connections, and thickness of various components.
3. Each shop drawing shall bear seal and signature of the sub-contractors third party Professional Engineer.
4. Show the following:
   a. Elevations.
   b. Details of special shapes.
   c. Reinforcing.
   d. Anchorage system.
   e. Interfacing with building construction.
   f. Furthermore:
      Detailed design, dimensions, members, profiles, joint location, and sizes, arrangement of units, horizontal to vertical connections thickness details of special shapes, reinforcing, drainage details, and flow diagrams, pressure equalization system and air barrier system interfacing with building structure, connections and joints and seals to adjacent building envelop systems, provisions for thermally induced expansion, and contraction, provisions for absorption of building structure movement and creep, thermal break details and locations, acoustic provision, safety provisions.
5. Indicate glazing details, method, locations of various types and thickness of glass, emergency breakout locations, and internal sealant requirements.
6. Location of exposed fasteners and joints for Engineer’s acceptance.
7. Clearly show where and how manufacturer’s system deviates from Contract Drawings and these specifications.
8. Panels, related reinforcement and securing methods.
9. Installation space for getting each unit in position.
10. All treatments of material to reach the architectural intention and the expected long thermal quality (coatings, tapings, bonding, e.g.)
11. All used material in general as numbered material component list without any exception.

F. Samples

G. Pre-Contract Samples
1. Glazed Curtain Walling/ Window Systems:
   a. Technical Literature.
   b. Minimum 300mm x 300mm sample of framing material inclusive of proposed finish.
   c. Minimum 300mm x 300mm sample of proposed glass build-up including coatings and frit patterns.

d. Minimum 300mm length or 300mm x 300mm sample of all other visible elements inclusive of proposed finish.
e. Sample of all ironmongery.

2. Panel Cladding System:
   a. Technical Literature.
   b. Minimum 300mm x 300mm sample of cladding panel inclusive of proposed finish.
3. External Through Colour Rendering:
   a. Technical Literature.
   b. Minimum 300mm x 300mm sample with applied finishes.
4. Doors:
   a. Doorsets:
      1) Technical Literature.
      2) Minimum 300mm x 300mm samples of door and framing inclusive of proposed finish.

H. Post-Contract Samples
1. Glazed Curtain Walling/ Window Systems:
   a. Minimum 1000mm length of framing inclusive of proposed finish.
   b. Minimum 2000mm x 1500mm sample of proposed glass build-up including coatings and frit patterns.
   c. Minimum 1000mm length or 1000mm x 1000mm sample of all other visible elements inclusive of proposed finish.
   d. Provide samples for the following where applicable:
      1) All items of ironmongery.
      2) Structural silicone.
      3) Fire stop.
      4) Insulation.
      5) Gasket.
2. Panel Cladding System:
   a. Minimum 1000mm x 1000mm sample of cladding panel inclusive of proposed finish.
   b. Minimum 1000mm length of framing sample inclusive of proposed finish.
   c. Sample of all visible fixings and fittings.
3. External Through Colour Rendering:
   a. Minimum 600mm x 600mm sample with applied finishes.
   b. Sample of all colour-matching fixings.
4. Doors:
   a. Doorsets:
      1) Minimum 600mm x 600mm section of door leaf and frame inclusive of proposed finish.
      2) Samples of ironmongery.
      3) Samples of fire and smoke seals.
      4) Samples of acoustic seals.
      5) If veneered, provide samples to indicate maximum range of variance.

I. Stresses in Glazing: Ensure that no glass or glazing combination develops stresses that may lead to the damage/ failure of glass, glazing materials, components, and/or framing system.
   a. Conduct thermal stress analysis performed by glass manufacturer supplier of all glass. Identify locations of each glass unit for the building, its stress factors and statement whether glass unit is to be annealed, heat strengthened or tempered. Bid price shall include for providing glass consistent with minimum results as determined by the glass stress analysis.
   b. Take into account shading stresses that might occur from adjacent component, including solar shading devices, signage applications, and the like.

J. Certifications:
   1. Submit statements certified by the Professional Engineer stating:
      a. Work of this section is engineered, supplied and installed as specified herein.
   2. Submit statement certified by the sealant manufacturers attesting that:
      a. Sealants selected are suitable for purpose intended.
      b. Sealants will not deteriorate due to exposure to ultra violet radiation / sun light.
      c. Joint designs are correct.
      d. Sealants are compatible with other materials with which they come in contact.
      e. Sealants are suitable for temperature and humidity conditions at time of application.
      f. Surfaces are properly prepared and primed to accept sealants.

K. Manufacturer's Instructions: Submit Data books, manufacturers printed installation instructions, Operating Instructions Manual for work of this section and as specified.

L. Preconstruction Test Reports from a qualified independent testing agency indicating and interpreting test results relative to compliance with performance requirements of glazed curtain wall system.

M. Product Test Reports from a qualified independent testing agency evidencing compliance of glazed curtain wall system with requirements based on comprehensive testing of manufacturer's current system.

N. Project Closeout Data: Submit Data Books, Operating Instructions Manual and Preventive Maintenance Manual for work of this section and as specified.

O. Submit test reports of performance testing being carried out by an independent testing agency, including controlled dismantling report.

P. Structural Sealant Data
   1. Submit product information on the sealants to be used including the following from sealant manufacturer:
      a. Recommendations and installation instructions, including cleaning and priming procedures.
      b. Test reports on adhesion to substrate and glass production samples tested in accordance with ASTM C 794.
      c. Compatibility statement that all materials in contact with the sealants and gaskets are compatible with the sealants in accordance with procedures of ASTM C 1087.
      d. Statement and test data indicating that the stress on the sealants and gaskets when exposed to the maximum load does not exceed 138 kPa and a safety factor of 5:1.
      e. Verification that sealants and gaskets are suitable for purposes intended, and is suitable for temperature, humidity and weather conditions at time of application.

Q. Qualification Data: For qualified testing agency, Manufacturer, and Installer.

R. Welding certificates according to the following:
   1. AWS D1.2/D1.2M, "Structural Welding Code-Aluminum".
   2. AWS D1.3/D1.3M, "Structural Welding Code-Sheet Steel".
   3. AWS D1.6/D1.6M, "Structural Welding Code-Stainless Steel".

S. Preconstruction Test Reports: For preconstruction mock-up submittals of glazed aluminum curtain walls, accessories, and components, from manufacturer, to include the following:

1. Testing Program: Developed specifically for Project.
2. Test Report: Prepare by a qualified preconstruction testing agency for each mock-up test.
3. Record Drawings: As-built drawings of preconstruction laboratory mock-ups showing changes made during preconstruction laboratory mock-up testing.

T. Field quality-control reports.

U. Warranties: Sample of special warranties.

V. Energy Performance Certificate: for glazed aluminium curtain wall, accessories, and components, from manufacturer.

W. Submit copies of instructions covering re-glazing, adjustments and other relevant maintenance data for inclusion in Operating and Maintenance Instructions Manual. Submit detailed cleaning procedures for glazing and prefinished aluminum. Include requirements for removal of excess fluid run-offs from silicon sealants, sealant primers and adhesive.

1.6 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as minimum.

B. Testing Agency Qualifications: To qualify for approval, an independent testing agency must demonstrate to the Engineer satisfaction, based on evaluation of agency-submitted criteria conforming to ASTM E699, that it has the experience and capability to satisfactorily conduct the testing indicated without delaying the Work.

C. Testing agency shall have a valid UKAS/ ENSA or equivalent accreditation.

D. Professional Engineer Qualifications: A professional Engineer who is legally qualified to practice in the jurisdiction where Project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for installations of Glazed Aluminum Curtain Wall that are similar to those indicated for this Project in material, design and extent.

E. Installer Qualifications: A qualified firm specializing in performing the work of this Section with minimum ten years documented experience and that is approved, authorized, or licensed by the product manufacturer to install his product and that is eligible to receive manufacturer's warranty. Include project names and addresses, names and addresses of Engineers and Employers, and other information specified:
   1. Engineering Responsibility: Engage a qualified professional Engineer to prepare or supervise the preparation of data for the Glazed Aluminum Curtain Wall, including drawings, testing program development, test-result interpretation and comprehensive engineering analysis that shows systems' compliance with specified requirements.
   2. Able to demonstrate thorough understanding of the principles of rain screen, stack effect, air seals, dynamic forces behavior of cast-in-place concrete structures, water and weather controls as affecting building cladding systems.

F. Source Limitations: Obtain Glazed Curtain Wall System from one source and by a single system supplier confirming the availability of components during the warranty period.

G. Insulating glass units shall be manufactured by company specializing in the fabrication of insulating glass units with minimum 10 (ten) years' experience, ample facilities to produce unit required and which have installations of units of similar composition to that required on the

Project that have been in place without failure for 5 years. Glass of the same type shall be by one manufacturer.

H. Tempering glass units shall be performed by a company specializing in the production of tempered glass with 10 (ten) years' experience, modern tempering equipment and facilities capable of treating post temperable glass units (low-E coating applied to glass prior to tempering) of maximum sizes required on the project, while maintaining tight roller wave tolerances specified and which have installations of post temperable units of similar composition to that required on the Project that are in place showing acceptable limits of distortion, by the Engineer/Employer.

I. Glazing installer shall be an experienced installer who has completed glazing work similar in material, design, and extent to that indicated for this Project; whose work has resulted in glass installations with a record of successful in-service performance.

J. Product Options: Information on Drawings and in Specifications establishes requirements for systems' aesthetic effects and performance characteristics. Aesthetic effects are indicated by dimensions, arrangements, alignment, and profiles of components and assemblies as they relate to sight lines and relationships to one another and to adjoining construction. Performance characteristics are indicated by criteria subject to verification by one or more methods including preconstruction testing, field testing, or in-service performance.

K. Structural sealant work, both insulating glass unit fabrication and curtain wall, shall be performed by specialists, experienced and qualified in structural silicone sealant application of similar nature, and approved in writing by the manufacturer of the sealant materials.

L. Comply with most stringent of the local code or acceptable national/international building standard requirements for VOC's limits.

M. Welding Standards: Comply with applicable provisions of AWS D1.2, "Structural Welding Code-Aluminum."
   1. Engage welders who have satisfactorily passed AWS qualification tests for welding processes involved and who are currently certified for these processes.

N. Review of the system design and drawing is required to be done by approved 3rd. party and system supplier.

O. Certified testing supervisor to be engaged for project installation and supervision till the project completion.

P. Engage an approved 3rd. party for onsite and factory inspection of the panel fabrication and installation. This should also be cross verified by the system supplier.

Q. Provide an on-site visual mock-up of the exterior wall assembly complete with all materials and products used to provide the finished installation, corner mullions, finishes, sealants and glazing, insulation, air/vapor barrier, weep/drainage system, as or shown on drawings or directed by Engineer, for review of quality of workmanship and appearance before proceeding with the remainder of the installation.
   1. Curtain wall mock-up shall be approximately 2 storey high and 3~4 module wide, erected in location as shown on drawings or determined by Engineer/Employer. Adjust mock-up as required to gain acceptance. Accepted work may form a part of the final installation.

R. Wet applied seals for the purpose of preventing the ingress of water shall not be accepted with the exception of the structural silicone seals or wet seals used for sealing the works against adjacent interfacing systems.

S. The Detailed Design and construction of the works shall be such that all rigid or fixed joints shall remain rigid and accommodate all specified thermal, building structure or other movements and any applicable loads without compromising its watertightness. All movement joints shall also be finally designed and constructed to accommodate such loads or movements without compromising the glazing's watertightness of glazing, glazed walls and associated cladding.

1.7 OFF-SITE MOCKUP

A. Shop Drawings of Glazed Curtain Wall Mock-Up: Submit shop drawings of curtain wall mock-up showing details of the test specimen and locations of instrumentation requirements under full design loading:
   1. Submit engineering calculations to show that maximum deflection does not exceed specified performance.
   2. Submit a detailed outline of each test procedure specified.

B. Provide an off-site test mock-up including an exterior wall assembly installation of curtain wall, insulation, air/vapor barrier, sealants, and all associated accessories; areas as indicted on drawings. Unless a larger test assembly is required for proper evaluation, mock-up shall be a minimum of 2 stories high, 4 modules wide. The façade shall also include its external feature which is making a part of the curtain wall system.

C. Erect test assemblies securely connected to test frame to simulate actual job conditions. Connections to test frame shall be exactly as per connections to the building.

D. Mock-up shall be delivered to and erected at inspected and testing company facilities, as directed, to the requirements of the testing company and to the approval of the Engineer. Cooperate with the inspection and testing company with regards to scheduling, delivery of materials and removal of test panel after completion of testing.

E. Testing Service: Engage a qualified independent testing agency to the approval of the Engineer which is equipped and experienced in conducting the required tests. The Contractor’s own test facilities won’t be acceptable. Testing agent shall have UKAS or EIAC (Emirates International Accreditation Center) or equivalent certificate. Façade contractor shall provide minimum of two options for the testing agency.

F. All costs and traveling expenses for 6 representatives of the Employer and Engineer shall be borne by the Contractor in addition to the Contractor’s own representatives. This visit shall be of three days duration.

G. Cost shall include business class return airfare, five-star hotel accommodation, food, taxi fare and subsistence.

H. The system shall not be considered as suitable for the project, if the performance mock-up is failed in two attempts and such failure is due to system design deficiency.

I. Contractor to confirm the supervisor and the leads of performance mock-up preparation team shall be from the team assigned for site installation of the façade works.

J. Mock-Up
   1. To be provided where agreed, to the acceptance of the client and the architect.

K. Quality Benchmarks
   1. External Wall Systems: First completed structural bay on Site accepted by the architect.

2. Doors: First fully installed of each doorset on Site accepted by architect.

1.8 OFF-SITE TESTING

A. Refer to Article 1.4 Performance Requirements.

B. Static Air Infiltration and Exfiltration Test.
   1. Perform testing in accordance with ASTM E283, except as otherwise specified herein.
   2. From these tests the following results shall be attained.
      a. Air infiltration and exfiltration through the completed curtain wall cladding systems shall not exceed requirements specified herein.

C. Static Water Penetration Test.
   1. Perform testing in accordance with ASTM E331, except as otherwise specified herein.
   2. From these tests the following results shall be attained:
      a. No water penetration inboard of the air barrier plane of the assembly shall occur at design conditions.

D. Dynamic Wind and Rain Test.
   1. Perform testing in accordance with Architectural Aluminum Manufacturers Association, AAMA 501.1 or as otherwise approved.
   2. From these tests the following results shall be attained:
      a. There shall be no infiltration of water to the interior face of the assembly (any water inboard of the air barrier plane).

E. Static Structural Performance Test.
   1. Perform testing in accordance with ASTM E330, except as otherwise specified herein.
   2. From these tests the following results shall be attained:
      a. The inward deflection on any part of the assembly shall not be sufficient to cause such member to touch any part of the main structure or impose loads onto any non-load bearing exterior wall members.
      b. All components shall return to their original positions under zero load, proving that the elastic limit of the materials has not been exceeded. Test pressure to be held for 60 seconds for 100% load, and 10 seconds for 150% load.
      c. In all cases the deflection of members shall not exceed specified tolerances. No failures or permanent deformation in excess of 1/1000 shall occur.
      d. Anchorage shall not show permanent set or signs of slackening off. The deflection of these members shall not be sufficient to damage non-load bearing members of the inner wall assembly.
      e. No noise shall occur due to movement of components.

F. Tested curtain wall system with the similar parameter as that of the project such as wind, span, modulation, test criteria's etc., is acceptable, if not objected by the client and the Engineer.

1.9 OFF SITE TESTING SEQUENCE AND PRESSURES

A. Perform tests in the following sequence and at the pressures stated:
   1. Test for air infiltration not exceeding 1.5 cum/hr/sq.m at 600 Pa static pressure difference to ASTM E283. Pressure shall be applied as per CWCT recommendations.
   2. Test for water penetration under design static pressure of 600 Pa.
   3. Test for water penetration under design dynamic pressure of 600 Pa.

4. Test for structural performance under positive and negative loading at one half and full design load with allowable deflections per Specifications and in accordance with wind pressures as defined by wind study report.
5. Repeat test for air filtration.
6. Repeat test for water penetration under static pressure.
7. Test for structural performance under positive and negative loading at full design load and at 1 ½ time full design load (to demonstrate a 1.5 safety factor) with no failures nor permanent deformation in excess of 1/1000.
8. Controlled Dismantling: Contractor shall discuss with the Engineer for the sequence and program of controlled dismantling.

1.10 ACOUSTICAL FIELD TESTING (As per Employers Discretion)

A. If no prior sound transmission data is available for the work of this section to demonstrate compliance with the Acoustic Engineer's Specifications or as stipulated in Acoustic Report, then acoustic testing may be required. Testing shall be completed in accordance with the following:
   1. ASTM E90 (laboratory measurement of airborne sound transmission)
   2. ASTM E966 (field measurement of airborne sound transmission through a facade)

1.11 PRE-INSTALLATION CONFERENCE

A. Conduct pre-installation conference in accordance with general requirements.

B. Conference Purpose and Agenda:
   1. Arrange with Engineer and representative of window and sealant manufacturer to visit Project site before beginning glazing operation to analyze site condition, and inspect surfaces and joints to be sealed in order that recommendations may be made should adverse conditions exist.
   2. Discuss following items:
      a. Weather conditions under which work will be done.
      b. Anticipated frequency and extend of joint movement.
      c. Joint design.
      d. Glazing procedures.

1.12 DELIVERY, STORAGE AND HANDLING

A. As per manufacturer's recommendations.

B. Deliver components in the manufacturer's original protective packaging.

C. Handle and store material in such a manner that no damage will be done to the materials or to the work of other Sections.

D. Provide removable protective covering over all aluminum to be exposed in the completed work.

E. Do not use adhesive papers or sprayed coatings which will become bonded when exposed to sunlight or weather.

F. Remove temporary protection after installation when acceptable to the Engineer. Do not leave coating residue on any surface.

G. Deliver glass units with manufacturer's labels intact on interior side of glass. Ensure labels indicate glass thickness, unit location, glass strength and orientation of units in vertical position.

H. Protect glass edges and corners to prevent chipping, cracking, and other similar damages.

I. Store components in a clean dry location away from uncured masonry or concrete. Cover components with waterproof paper, tarpaulin or polyethylene sheeting in a manner to permit circulation of air.

1.13 PROJECT CONDITION

A. Field Measurements:
   1. Verify actual locations of structural supports for glazed aluminum curtain walls by field measurements before installation and indicate measurements on Shop Drawings.
   2. Coordinate fabrication schedule with construction progress to avoid delay of the work.

1.14 CORROSION PROTECTION

A. Allow for protection against all corrosion arising from exposure to seawater, non-saline water, soil, high humidity, low or high temperatures, chemical acids and alkalis, abrasion and impact, fungi and bacteria.

B. Ensure that protective measures are taken to avoid any corrosion or any deleterious effects caused by manufacturing, finishing, transportation, storage and installation of materials.

C. Ensure full resistance to any corrosion for components that are secured or bolted to each other, paying particular attention to the surface damage caused by such bolting or securing.

D. Contractor upon Engineers approval, to install non-corroding products and materials where appropriate including aluminium, stainless steel, other non-corroding metals, and plastics (fire resistant). In the case where no alternative to steel or other oxidizing metal is possible (or where the specifications specifically indicate the use of steel/stainless steel), the steel or other material must be effectively protected against corrosion by galvanization and/or protective paint coatings, for example with an anti-corrosion guarantee of 10 (ten) years.

E. In all other cases, the Contractor will be asked to modify, if necessary, the proposed product (even if a catalogue or manufactured item) and to replace the corroding-type materials by non-corroding materials.

1.15 MAINTENANCE AND REPAIR

A. Contractor shall engage suppliers of materials, products, and equipment’s having an office and maintenance service based in the country of project location.

1.16 WORK REASONABLY INFERRED

A. Where Contractor shall execute the Work in accordance with the intent and meaning of the Drawings and Specification and shall supply all accessories and other items essential for the proper Performance and Completion of the Work and shall execute all Work, including the supply of necessary materials, which can be reasonably inferred, whether or not specifically shown or described in the Drawings and Specification.

1.17 WARRANTY

A. General Warranty: The Contractor is to provide a written agreement to indemnify the Employer against any defects in the design, fabrication, workmanship, and quality of materials, water tightness or performance of the Works included in this Part and to repair or replace defective design, fabrication, workmanship or materials of the curtain walling system during the warranty period. Defective materials and workmanship include but not limited to:
   1. Abnormal deterioration, ageing and weathering of the walling system.
   2. Leakage of water or air exceeding specified limits.
   3. Structural failure of components resulting from exposure to pressures and forces within specified limits.
   4. Failure of operating parts to function normally.
   5. Glass breakage due to defective design, manufacture or installation or exposure to pressures and forces within specified limits.
   6. Spontaneous breakage of the tempered glass.
   7. Deterioration or discoloration of finishes in excess of normal weathering and ageing.
   8. Failure of the curtain walling system to meet any other specified performance requirements.
   9. Noise or vibration caused by thermal movement.
   10. Adhesive or cohesive sealant failures.
   11. Glass and glazing damage from the weld spatters.
   12. Water leakage.
   13. Glazing breakage.

B. The warranty shall not include damage caused by vandalism, or natural conditions exceeding the performance requirements.

C. Warranty Period (in refer to contract conditions):
   1. Doors and Window: Ten (10) years from date of substantial completion.
   2. Glazing: Ten (10) years from date of substantial completion.
   3. Metal Finish: Ten (10) years from date of substantial completion.
   4. Anodized Coating Finish: Twenty (20) years from date of substantial completion.
   5. Super Durable Powder Coating Finish: Twenty-five (25) years (for product) from date of substantial completion.

D. Finish repairs to be covered by the same warranty period requirements of the section 1.17 C.

E. The Operation and Maintenance (O&M) operation manual, need to be provided by façade contractor (and approved by Employer's Engineer) and has to be in compliance with the façade warranty period.

F. Provide a written warranty of 10 (ten) years for gaskets to be free from defects and to be replaced when failed.

G. Special Finish Warranty: Where special finishes (a Finish required by the design, but not a manufacturer's standard Finish) are specified, the Contractor shall obtain a warranty from the manufacturer covering failure of the applied finish, and agreeing to repair or replace items that show evidence of Finish deterioration. Deterioration of Finish includes but is not limited to, color fade, chalking, cracking, peeling, and loss of film integrity. Special Warranty to be confirmed by Employer/Engineer.

1.18 PROPOSAL SUBMISSION

A. Contractor shall indicate in his Bid, which of the Subcontractors bidding the work of this Section is included in the Contractor's Bid Price.

B. Proposals from the Subcontractors bidding the work of this section shall be accompanied with two sets of drawings and data details which show how manufacturer intends to comply with requirements of this specification section for the exterior elements and all other works as follows:
   1. Show typical head, sill, jamb and corner details and typical intermediate mullions and external and internal features attached to the façade.
   2. Junctions with anchorage to the building structure.
   3. Air barriers and vapor barriers.
   4. Dielectric separator details.
   5. Thermal separator details.
   6. Mechanical retention of the glazing.
   7. Provisions for thermally induced expansion and contraction of members.
   8. Provisions for absorption of the building structure movements.
   9. Water drainage system.
   10. Pressure equalization system and air ventilation to the glass rebate.
   11. Initial structural calculations.
   12. Initial thermal calculations.
   13. Criteria compliance statement.
   14. Proposed system performance information.
   15. Product details of major façade elements.
   16. Method of erection.
   17. Method statement for glass manufacturer, assembly, structural glazing etc.

C. Contractor shall provide details of the certified system used, including the name, sizes and origin (country) of profiles, arrangement of components and assemblies for the approval by the Engineer.

D. In the proposed Bid, Contractor to provide a letter of authorization from the principal system supplier, that the fabricator is approved manufacturer and installer of the company. In addition, Contractor to provide list of works executed by the Fabricator using the manufacturer's profile as part of the experience certificate.

PART 2 - PRODUCTS

2.1 SUBSTITUTIONS

A. Manufacturers and products specified hereinafter are for the purpose of establishing minimum quality standards. Manufacturers which are equal to or better than those specified, and which conform to the design requirements and color selections, may be acceptable subject to Engineer's approval. This request shall be accompanied by:
   1. A certificate of compliance from the material manufacturer stating that the proposed products meet or exceed the requirements of this specification.
   2. Documented proof that the proposed material has a proven record of successful performance for more than 10 (ten) years, confirmed by at least five local projects that may be examined.

2.2 ACCEPTABLE MANUFACTURERS

A. Section Manufacturers
   1. Technal, Middle East/Germany
   2. Wicona, Middle East/Germany
   3. Gutmann, Middle East/Germany
   4. Schuco, Middle East/Germany

5. Hueck, Middle East/Germany
   Or Equivalent, subjected to Engineer's acceptance.

B. Fabricators/Assemblers/System manufacturers:
   1. Alico, Middle East
   2. Technical Glass, Middle East
   3. Al Abbar, Middle East
   4. Permasteelisa Gartner, Middle East
   5. Glassline Industries, Middle East
   6. Reem Emirates Aluminium LLC, Middle East
   Or Equivalent, subjected to Engineer's acceptance.

   Condition to qualify "equivalent": Qualification exercise consists of assessment of technical adaptability of the façade system design and the material, quality procedures maintained, minimum 10 (ten) years of its usage in similar project and credibility of its performance record and other project references. All cost related to this exercise shall be borne by the proposer. The proposer shall agree to use above listed pre-approved system manufactures, should the proposed system be not acceptable to the Engineer, after the completion of qualification exercise.

2.3 MATERIALS

A. Aluminum: Conform to the requirements published in the Aluminum Association's "Aluminum Standards and Data", referenced ASTM standards and the following. All aluminum extrusions shall be manufactured to dimensional tolerances so as to eliminate any edge projection or misalignment at joints. Unless otherwise specified, provide alloy and temper as required to suit performance requirements and finishes indicated. Provide concealed extruded bars, rods, shapes and tubes in alloys as recommended by the fabricator to join or reinforce assembly of exposed aluminum components.
   1. Extruded Bars, Rods, Shapes, and Tubes: ASTM B 221 (ASTM B 221M). Alloy 6063, T54, T5, T6 tempered.
   2. Extruded Structural Pipe and Tubes: ASTM B 429.
   3. Welding Rods and Bare Electrodes: AWS A5.10.

B. Sheet and plate aluminum: ASTM B209M, extruded anodizing quality, AA5052-H14 or 3003-H 14 or other acceptable temper and alloy suitable for purpose (exposed/non-exposed) and finish required, of one type throughout, nominal thickness of 3.2 mm, special hardness for flat panel application, tension leveled quality of flatness, free from perceptible distortions, waves, twists, bucking or other deficiencies in appearance and performance, spray coat finished after forming and welding is complete.

C. Steel Reinforcement: ASTM A 36 (ASTM A 36) for structural shapes, plates, and bars; ASTM A 1008/A 1008 M for cold-rolled sheet and strip; or ASTM A 1011 (ASTM A 1011 M) for hot-rolled sheet and strip.

D. Anchors and Fasteners:
   1. Material: Stainless Steel.
   2. Anchor and Fastener Metal Alloy Types, Designations and Standards: Alloys as selected by fabricator to prevent corrosion resistance with the components fastened.
   3. Where fasteners are subject to loosening or turn out from thermal and structural movements, wind loads, or vibration, use self-locking devices.

E. Spacers, Setting Blocks, Gaskets, and Bond Breakers: Manufacturer's standard permanent, non-migrating types compatible with sealants and suitable for joint movement and system performance requirements. Gaskets shall be black resilient, elastomeric type.

F. Air/vapor barrier gasket seals: Neoprene, ethylene propylene diene monomer (EPDM), thermoplastic, silicone, or other accepted material, designed to remain flexible and heat resistant.

G. Concrete Protective Coating: Acrylic based one component protective coating, solvent free with excellent resistance against carbonation and ingress of chloride ions, sulphates and oxygen. Suitable for use in hot climate condition. Class A, according to ASTM E84-16.

H. Primer for ferrous metals: ASTM D4146.

I. Touch up primer for galvanized steel: Zinc rich anti-corrosion coating containing 92-95% zinc in dried film providing electro-chemical protection on ferrous metals.

J. Thermal Breaks: Include thermal breaks in all members. Thermal break material shall be polyamide or EPDM. Provide thermal breaks tested as per AAMA TIR A8 and AAMA 505 for dry shrinkage and composite performance.

K. Shims: Non-staining, non-ferrous, type as recommended by system manufacturer.

L. Expansion Anchor Devices: Lead-shield or toothed-steel, drilled-in, expansion bolt anchors.

M. Fire stopping sealant: One part/component silicon manufactured by Dow Corning Ltd. Or equivalent silicon as approved.

N. Mineral fiber safing insulation: Complete with galvanized steel sheets, silicone, insulation and assemblies. Fire stopping shall be minimum 33% wider than the space to be filled, compressed in right direction. Civil defence pre-approved firesafing materials are recommended.

O. Glazing Gaskets:
   1. Compression type design, replaceable, molded or extruded ethylene propylene diene monomer (EPDM), designed to remain flexible and heat resistant.
   2. Comply with ASTM C509 or C864.
   3. Profile and hardness as necessary to maintain uniform pressure for watertight seal.
   4. Manufacturer's standard black colour unless otherwise specified, all gaskets should be marked with the supplier and the section reference.

2.4 SEALANTS

A. Concealed Sealing Materials: All sealing materials concealed within the glazed aluminum curtain walls shall be silicone compatible with and adherent to each material it will be in contact with, as recommended by the manufacturer to fulfill performance requirements.

B. Sealants and gasket seals shall be non-bleeding, non-staining and capable of supporting their own weight. Sealants and gasket seals exposed to exterior shall also be UV resistant and ozone resistant. Sealants used in glazing systems shall also be capable of supporting specified loads associated with the glazing systems.

C. Primers, cleaning materials and joint backing: Recommended by the sealant manufacturer to suit the various substrates and conditions of the Work, compatible with each other and with sealant.

D. Sealants shall be from the same manufacturer for all work of this Section.

E. Sealant at exposed locations: ASTM C920, type which minimize silicone fluid leaching, one component, neutral cure, medium modulus, 'Silruf NB-SCS9000' by GE Silicones or Spectrum 3 or 'Spectrum 4' by Tremco Ltd., or 756 SMS' by Dow Corning Corporation or equivalent.

F. Structural silicone and heal bead for shop and field application (where shop application is not possible): ASTM C920, one part/component, '795' by Dow Corning Corporation, 'Utraglaze 4000' by GE Silicones, or 'Spectrum 2' by Tremco Ltd. or equivalent.

G. Structural silicone sealant for shop glazing: Two part/component '983' by Dow Corning Corporation, 'Ultraglaze 4400' by GE Silicones, or 'Proglaze II' by Tremco Ltd. or equivalent.

H. Sealants used in structural joints shall have adequate strength to retain insulating units to the metal framing under the design conditions.

I. Sealants used in contact with edges of laminated glass shall not cause damage or disfigurement or laminate interlayer of the laminated glass.

J. Sealant where concealed: Sealant as specified for exposed locations.

K. Air seal sealants: For back pan seals; one part/component elastomer sealant.

L. Sealant colors: To be selected by the Engineer, not necessarily from the sealant manufacturer's standard color range.

M. Sealant tape (concealed locations): Extruded, ribbon-shaped, non-drying, non-skinning, non-oxidizing, reinforced, polyisobutylene tape of sufficient width and thickness, 6-7mm minimum, to permit a continuous seal, "Tremco 440 Tape' by Termco Ltd., or equivalent from other approved manufacture.

2.5 COMPONENTS

A. Insulating glass units: ASTM E2190, and IGMA TM-3000, certified by the Insulating Glass Certification Council (IGCC) or equivalent European standards BS EN 1279, consisting of 2 panes separated by a 12/16/20 mm wide factory double time sealed, dehydrated and air-filled cavity space utilizing black (recommended for aesthetics) stainless steel/warm edge spacer, without metal edge banding. Seal color to be selected by Engineer. Air pressure within the glass cavity space shall suit atmospheric conditions at the location of the installation to prevent distortion of the installed units. Edges of glass shall be straight cut, free from nicks and other imperfections conducive to breakage. Units shall be manufactured in IGMA or equivalent European certified facility certified facility. Aris all edges of glass adjacent to structural silicone glazing.
   1. Glass Type: As indicated on drawings and as per specification section "Glazing".
   2. Glass: Fully tempered shall be heat soak tested to remove Nickel Sulfide inclusions, which may otherwise cause spontaneous breakage in situ. All edgework and holes must be completed before tempering. Written warranties against nickel sulphide inclusions in lieu of heat soaking will not be accepted.
   3. Insulated glass spacers shall be straight and evenly set into glass units. Reflective and low emissivity coatings shall be edge deleted over entire depth of spacer and edge seal at locations where silicone sealant is in contact with glass.
   4. Use 2-part silicone sealant for secondary seal of the insulating glass units at structural silicone glazing. Remove excess silicone seal from glass edges of outer lite.
   5. The glass manufacturer shall confirm that the recommended glass thickness and stress levels are compatible with a safe and service design.

6. All glass will be manufactured and processed in a factory where the Quality Control procedures are accredited to ISO 9000 and are independently maintained.

B. Setting Blocks: Silicone rubber, compatible with silicone sealants, 85 plus or minus 5 Durometer A hardness, by Tremco Ltd. or equivalent from other approved manufacturer. Silicone compatible EPDM may be used instead of silicone where tested and approved by the manufacturers of the silicone sealant and the setting blocks.

C. Spacers: Continuous, bond breaker type and compatible with silicone sealant, 'Norton Thermabond V-2100 or V-2200' by Saint-Gobain Performance Plastics Corporation, approved equivalent by Tremco Ltd., or equivalent from other approved manufacturer.

D. Gaskets: Continuous silicone rubber, compatible with silicone sealant, or silicone compatible EPDM, 60 plus or minus 5 Durometer A hardness.

E. Prefinished Aluminum Panel System
   1. Prefinished Aluminum Panel system shall consist of pre-finished solid sheet aluminum compiling ASTM B209M, extruded anodizing quality, AA5052-H14 or 3003-H 14 or other acceptable temper and alloy suitable for purpose and finish required, of one type throughout, nominal thickness of 3 or 3.2mm, special hardness for flat panel application, tension leveled quality of flatness, free from perceptible distortions, waves, twists, bucking or other deficiencies in appearance and performance, spray coat finished after forming and welding is complete.
   2. Prefinished sheet shall be non-sequential, non-progressive, flat profiled aluminum panel system mounted with concealed fasteners, and clips.
   3. Provide continuous insulation and air/vapor barrier behind wall cladding assemblies.
   4. Provide matching flashings, copings, trims and closures, and sealant work.
   5. Provide all primary and secondary steel framing secured to structure.

F. Air/vapor barrier
   1. Self-adhesive membrane air/vapor barrier: High temperature resistant, self-adhesive type with release liner on adhesive side, EPDM membrane, minimum 1mm thick or as recommended by the manufacturer. Primer, mastic, stainless steel fastening bars and fasteners, as recommended by membrane manufacturer.
   2. Sheet metal for metal air/vapor barrier and air seal tie-ins to adjacent construction: Structural quality sheet steel, ASTM A653/A653M Grade 33, tension leveled and galvanized by hot dip process, minimum coating weight of 380 g/m2, nominal core thickness 0.91mm; where possible provide preformed sheet steel liner panels with factory applied caulking at side lap joints.
   3. Waterproofing sheet membrane at ledges, projecting features, copings and parapets: High temperature resistant, sheet rubber, 1.5mm thick EPDM, neoprene, butyl, or other approved manufacturer.
   4. Adhesive, tapes and sealant for membrane: Recommended by the membrane manufacturer. Ensure compatibility of flashing membrane with roofing and waterproofing membranes, and air/vapor barriers.
   5. Support for flexible membrane across air spaces: 0.55mm sheet steel to ASTM A653/653, Z275 hot dip galvanized.

G. Cladding Panel for Add-On Steel Frames:
   1. Prefinished solid aluminum panels as indicated on drawings.
   2. Insulation: As per specification sections "Building Insulation".

H. Brackets and Reinforcements: Provide manufacturer's standard high-strength aluminum brackets and reinforcements. Provide non-staining, nonferrous shims for aligning system components.

I. Metal Panels and Copings:
   1. Supports: Metal panel, and coping supports shall be custom fabricated so that the panels and copings are an integral part of the framing for the glazed aluminum curtain wall system; are secure, yet accommodate expansion and contraction; and that individual panels and copings may be installed or removed with a minimum amount of disturbance to adjacent components.
   2. Component thicknesses must be as follows:
      a. Alum. Copings: 3mm minimum thickness.
      b. Alum. Shadowbox Panels: 2mm minimum thickness.
      c. Galvanized Back Pans: 1.5~2mm minimum thickness, for non-visible application.
   3. Finishing: After forming metal panels and copings, but before finishing, remove abrasions, scratches, die markings and dents.

J. Fire Resistive Joint System:
   1. Provide Perimeter Fire-Resistive Joint Systems consisting of a material, or combination of materials, (fill materials backed by slag-wool-/rock-woolfiber board insulation and other components), installed to retain the integrity of fire-rated construction by maintaining an effective barrier against the spread of flame, smoke, or gases through the joints between edges of floor and exterior curtain walls. Place firestop material under permanent 33% compression. Use impaling clips or metal trims to hold insulation in place. Local Authority / Civil Defence pre-approved perimeter system as applicable.
   2. Perimeter Fire-Resistive Joint Systems: For joints between edges of fireresistance-rated floor assemblies and exterior curtain walls, provide systems of type and with ratings indicated below as determined by UL 2079.
      a. UL-Listed, Perimeter Fire-Containment Systems: Integrity ratings equaling or exceeding fire-resistance ratings of floor or floor/ceiling assembly forming one side of joint.
   3. Compatibility: Provide fire-resistive joint systems that are compatible with joint substrates, under conditions of service and application, as demonstrated by fire-resistive joint system manufacturer based on testing and field experience.
   4. Accessories: Provide components of fire-resistive joint systems, including primers and forming materials. Use only components specified by fireresistive joint system manufacturer and approved by the qualified testing and inspecting agency for systems indicated.
   5. Fire resistive joint system as a whole assembly shall comply with Local Authority / Civil Defence requirements as applicable.

K. Concealed Flashing: Manufacturer's standard corrosion-resistant, non-staining, non-bleeding flashing, compatible with adjacent materials, and of type recommended by manufacturer.

2.6 FABRICATION-CURTAIN WALL

A. General: Fabricate glazed aluminum curtain wall system according to Shop Drawings. Fabricate components that, when assembled, will have accurately fitted joints with ends coped or mitered to produce hairline joints free of burrs and distortion. After fabrication, clearly mark components to identify their locations in Project according to Shop Drawings.

B. Take field measurements and levels required to verify or supplement those shown on the Drawings for the proper layout and installation of the work. Coordinate dimensional tolerances in adjacent building elements and confirm prior to the commencement of the work. Execute fitting and assembly in the shop with the various parts or assemblies ready for erection at the building site.

C. Thermally Improved Construction: Fabricate all exposed units with an integral, concealed, low conductance thermal barrier, located between exterior materials and members exposed on interior side, in manner that eliminates direct metal to metal contact.
   1. Provide thermal break construction system that has been in use for not less than three years and has been tested to demonstrate resistance to thermal conductance and condensation and to show adequate strength and security of retention for all fitting components.
   2. Provide thermal barriers tested according to AAMA 505; determine the allowable design shear flow per the appendix in AAMA 505.
   3. Provide hardware with low conductivity or nonmetallic material for hardware bridging thermal breaks at frame or vent sash.

D. Forming: Form shapes with sharp profiles, straight and free of defects or deformations, before finishing.

E. Neoprene, polyvinylchloride or other approved material and having a minimum tensile strength of 13.8 MPa and Durometer A Hardness of 60, plus/minus 5; softening temperature not less than 110°C.

F. Air/vapor barrier gasket seals: Neoprene, ethylene propylene diene monomer (EPDM), thermoplastic, silicon, or other accepted material, designed to remain flexible and heat resistant.

G. Accurately fit, remove burrs and rigidly frame together joints and corners. Match joints toward the exterior weathertight and joints towards the interior air and vapor tight. Metal in contact shall have hairline joints. Location of exposed joints shall be subject to the acceptance of the Engineer.

H. Reinforce frames and assemblies by concealed means as necessary to meet the specified design requirements and the installation required. Reinforcement shall be hot-rolled mild steel and shall be securely anchored to horizontal and vertical members by accepted positive mechanical means.

I. Prepare components to receive concealed fasteners and anchor and connection devices.

J. Fabricate components to drain water passing joints, condensation occurring in glazing channels, condensation occurring within framing members, and moisture migrating within the system to the exterior.

K. Welding: Complete the welding of exposed surfaces prior to finishing:
   1. All welding shall be in accordance with the recommendations of the AWS and shall be performed with electrodes and/or by methods recommended by suppliers of the metal being welded. Fabricate welded aluminum assemblies so that fraying surfaces are free rinsing and will not trap coating solutions.
   2. Welds behind finished surfaces shall be so performed as to eliminate distortion, and discoloration, on the finished side. Plug, puddle, and spot welding are not permitted. Provide low heat filled welds using a chill bar on finished side to eliminate dimpling, distortion, and/or discoloration on the finished side. If weld heads appear on the finished surface, the weld head shall be ground, and polished to match and blend with the finish on adjacent parent metal. Weld spatter and welding oxides on finished surfaces shall be removed immediately.
   3. At joints where welding cannot be performed, used concealed stainless-steel fasteners to join assembly.

L. Shop Painting of Carbon Steel: Un-galvanized steel items shall be thoroughly cleaned of all loose scale, filings dirt, and other foreign matter, in accordance with SSPC-SP3 “Power Tool Clean”’ and painted with coating as specified for carbon steel surfaces.

M. Glazing Pockets: Provide minimum clearances for thickness and type of glass indicated according to GANA "Glazing Manual."

N. Metal Protection: Where aluminum will contact dissimilar metals, protect against galvanic action by painting contact surfaces with primer or by applying sealant or tape recommended by manufacturer for this purpose. Where aluminum will contact concrete or masonry, protect against corrosion by painting contact surfaces with acrylic based protective paint/coating.

O. Shop Assembly: As far as practical, all fitting and assembly work shall be done in a fabrication shop.
   1. Framing members attaching curtain wall components to building supports shall provide for 3-way adjustment to accommodate fabrication and construction tolerances and allow for thermal and building movements.
   2. Provide vents, weep holes and internal water passages in the glazing framing recesses as recommended by the respective glass and framing recesses as recommended by the respective glass and framing manufacturers to conduct infiltration water to the exterior, and to avoid condensation at glass or spandrel unit air spaces. Provide weep baffles secured to inside of frame behind vents and weep holes.

P. Air/Vapor Barriers
   1. Provide and maintain the continuity of the rain screen air/vapor barrier and insulation integrity of the building envelope.
   2. Provide continuous and homogenous air/vapor barriers for this work, for installation on the interior side of the thermal insulation, and provide for connection of air/vapor seal at junction with adjacent air/vapor barrier work provided under other sections.
   3. Brake-form sheet metal barriers to permit assembly using self-tapping screws, and other mechanical means of secure fastening. No exposed fastening permitted. Provide reinforcing and stiffeners as required to comply with specified deflection allowances.
   4. Make provision in barrier design to accommodate movement resulting from thermal change and from structural deflection, and to accommodate positive and negative wind pressures.
   5. Form edges of sheet metal air/vapor barriers to permit peripheral and joint sealing at concealed locations, where indicated and required.
   6. Cut, fit and form metal air/vapor barriers as required to accommodate conflicting framing, anchors, connections, mechanical and electrical appurtenances and other obstructions.
   7. Fabricate supplementary framing from rolled steel sections by welding. After fabrication remove all grease, scale and other substances deleterious to bond and apply one coat of priming paint.

Q. Install glazing according to Shop Drawings. Comply with requirements of specification section "Glazing," unless otherwise indicated.
   1. Prepare surfaces that will contact sealant according to sealant manufacturer's written instructions to ensure compatibility and adhesion. Preparation includes, but is not limited to, cleaning and priming surfaces.
   2. Remove excess sealant from component surfaces before sealant has cured.

2.7 ALUMINUM FINISHES

A. All aluminium visible surfaces exposed to the building interior and exterior: Aluminium surfaces shall be finished with Super Durable Powder Coating complying with AAMA 2604 and that are thermoset, free from VOCs, with minimum dry film thickness of 60 microns.
   1. Colour and Finish: To match approved sample.

B. Prepare, pretreat and apply coating (including protective coat) to exposed metal surface to comply with AAMA 2604.

C. Concealed Aluminum Surfaces: Any aluminum sections which are completely concealed and therefore do not require the facing color shall be anodized to a minimum standard of AAMA 611, class II.

D. Identification marks shall not appear on finished surfaces.

E. Adhesion: No removal of finish after 1.5mm cross hatching to base metal, impacting to the point of metal rupture, and subjecting to application and quick removal of cellophane tape.

2.8 STEEL FINISHES

A. General: Requirements and specification for finishes on carbon steel as referred to this Section shall comply to the following:
   1. All surfaces of steel members incorporated in Works shall be finished as required in this Section and based on environmental classification for exposed and no exposed surfaces.

B. Finishes (Mild Steel and Galvanized Steel)
   1. Application shall be in strict conformance with the manufacturer's recommendations. Substrate preparation requirements as per manufacturer's instructions shall be fully compiled.
   2. Finish Systems:
      a. Paint System (For Mild Steel)

- Description: Epoxy – Polyurethane System
- Type: Epoxy primer/Polyurethane topcoat
- Surface(s): Mild Steel
- Preparation: Blast cleaning to Sa 2.5 for Mild Steel
- Initial Coats: Stripe coating and, one full coat of two pack epoxy primer @100 microns as a primer (Solids % by volume: >48, Mixing ratio 4:1)
- Intermediate: One full coat of two pack epoxy primer @125 microns as an intermediate coat. (Solids % by volume: >48, Mixing ratio 4:1)
- Finishing Coats: Two coats of polyurethane topcoat each @50 microns (solid % by volume: >48, Mixing ratio 4:1)

      b. Paint Systems (G.I Surfaces)

- Description: Epoxy – Polyurethane System

- Type: Epoxy primer/Polyurethane topcoat
- Surface(s): Galvanized Steel
- Preparation: Blast cleaning to Sa 2.5
- Initial Coats: Stripe coating and, one full coat of two pack epoxy primer @125 microns as a primer (Solids % by volume: >48, Mixing ratio 4:1)
- Finishing Coats: Two coats of polyurethane topcoat each @50 microns (solid % by volume: >48, Mixing ratio 4:1)

3. Time between coatings shall be as per manufacturer's instruction.
4. Use spray coating for painting. The minimum thickness shall be maintained on all corners, edged and ends of pieces. Do not paint outdoors in rainy and dusty weathers.

2.9 REFERENCE STANDARDS

A. ASTM A 36/A 36M: Standard Specification for Carbon Structural Steel.

B. ASTM A 123/A 123M: Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.

C. ASTM A 153/A 153M: Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.

D. ASTM A 1008/A 1008M: Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy and High-Strength Low-Alloy with Improved Formability, Solution Hardened and Bake.

E. ASTM A 1011/A 1011M: Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy and High-Strength Low-Alloy with Improved Formability.

F. ASTM B 209/B 209M: Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate.

G. ASTM B 221/B 221M: Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes.

H. ASTM B 429: Standard Specification for Aluminum-Alloy Extruded Structural Pipe and Tube.

I. ASTM E 90: Standard Test Method for Laboratory Measurement of Airborne, Sound Transmission Loss of Building Partitions and Elements.

J. ASTM E 699: Standard Criteria for Evaluation of Agencies Involved in Testing, Quality Assurance, and Evaluating Building Components in Accordance with Test Methods Promulgated by ASTM Committee E-6.

K. ASTM E 783: Standard Test Method for Field Measurement of Air Leakage through Installed Exterior Windows and Doors.

L. ASTM E 1105:	Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform or Cyclic Static Air Pressure Difference.
M. ASTM E219:	Standard Specification for Insulating Glass Unit Performance and Evaluation.
N. AAMA CW-RS-1-96:	The Rain Screen Principle and Pressure Equalized Design.
O. AAMA 501.1:	Voluntary Specification Standard Test Method for Exterior Windows, Curtain Walls and Doors for Water Penetration Using Dynamic Pressure.
P. AAMA 501.5-07:	Test method for Thermal Cycling of Exterior Walls.
Q. AAMA 2603:	Voluntary Specification, Performance Requirements and Test Procedures for Pigmented Organic Coatings on Aluminum Extrusions and Panels.
R. AAMA 2605:	Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing High Performance Organic Coatings on Aluminum Extrusions and Panels.
S. AWS D1.2:	American Welding Society, Structural Welding Code-Aluminum.
T. GANA:	Glass Association of North America Glazing Manual.
U. SSPC-Paint 12:	Society for Protective Coatings Paint 12: Cold-Applied Asphalt Mastic (Extra Thick Film).

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of glazed curtain wall system. Do not proceed with installation until unsatisfactory conditions have been corrected or accommodations acceptable.

B. Verify dimensions, tolerances, and method of attachment with other Work

3.2 PREPARATION

A. Supply anchorage devices and inserts to the appropriate trades where required for building in or casting-in-place and instruct as to proper location and position.

B. Ensure that masonry and concrete surfaces to receive adhesive and sealants are dry, firm, sound, smooth, suitable for bond, and free from loose material, projections, ice, frost, slick, grease, oil and other matter detrimental to bond.

C. Maintain surface of substrates and ambient temperatures constantly within range recommended by material manufacturer during application and curing of sealants and adhesives, and during installation of glazing.

D. Remove dust and other loose material from openings.

3.3 INSTALLATION

A. General: Comply with manufacturer's written instructions for protecting, handling, and installing glazed curtain wall system and ventilation systems. Do not install damaged components. Fit frame joints to produce hairline joints free of burrs and distortion. Rigidly secure non-movement joints. Seal joints watertight, unless otherwise indicated. Provide means to drain water to the exterior to produce a permanently weatherproof system.

B. Metal Protection: Where aluminum will contact dissimilar metals, protect against galvanic action by painting contact surfaces with primer or by applying sealant or tape recommended by manufacturer for this purpose. Where aluminum will contact concrete or masonry, protect against corrosion by applying the aluminium with powder coating or anodized finishes.

C. Air/Vapor Barrier and Insulation
1. Provide thermal insulation and air/vapor barriers to ensure a uniform, continuous thermal and vapor barrier effect. Where adjacent insulation and vapor barriers are to be provided under other Sections, coordinate the work such that thermal and vapor barrier continuity is achieved.
2. Coordinate work of this Section with, and provide connection for compartmentalization of air spaces provided under other Sections.
3. Gun applied continuous bead of sealant to joints and air/vapor barrier junctions with adjacent construction. Liberally butter screw fastenings with sealant.
4. Provide air tight seals at penetrations and joints in air/vapor barriers.
5. In addition, at back sloped metal air/vapor barrier locations, apply minimum 100 mm wide strip of silicone membrane air/vapor barrier centred over joints in metal air/vapor barrier, lap ends for continuity.
6. Isolate metal air/vapor barriers with thermal breaks and spacers.
7. Provide sheet steel and/or flexible, continuous membrane and gasket air/vapor barrier seals between work of this Section and adjacent construction, and at deflection and expansion connections, where required. Prime substrates, apply membranes and gaskets to concrete and masonry with adhesive and retain with continuous aluminum or galvanized steel plates or bars and non-corrosive mechanical fasteners. Vulcanize or overlap joints to ensure a continuous seal.
8. Install self-adhesive membrane air/vapor barrier installation and mechanical fastening of insulation at exterior sheathing locations, around structural steel and at concrete and masonry substrate locations.
9. Apply insulation to the exterior side of air/vapor barriers.
10. Adhere stick clips for insulation to metal air/vapor barriers at 300 mm o.c. both ways. Support adhesive-applied clips in place until adhesive has set. As an alternative, gun weld applies pins to metal substrates in lieu of stick clips, provided clips do not easily break off and weld burn-through does not occur.
11. Cut insulation as required and fit snugly to penetrations, obstructions, openings and corners. Butt insulation boards tightly. Cut out back of board insulation as required to accommodate substrate irregularities and build up over cut out areas on the other side as required to ensure thermal barrier uniformity unless otherwise indicated or approved.
12. Press insulation boards firmly and tightly to barrier or substrate impaling them on clips without bending clips. Butt insulation boards tightly at joints. Install retainers to clips.
13. Fill irregular shaped voids within assemblies with fibrous packing insulation to maintain continuity of thermal barrier.

D. Provide structural framing and supports required to support work of the section.

E. Install operable window, as applicable and related hardware, at location indicated and ensure weathertight, rattle-free closure when unit is in the closed and locked position.

F. Install components to drain water passing joints, condensation occurring in glazing channels, condensation occurring within framing members, and moisture migrating within the system to the exterior.

G. Install framing components plumb and true in alignment with established lines and grades.

H. Anchorage: After system components are positioned, fix connections to building structure as indicated on Shop Drawings.
   1. Provide separators and isolators to prevent metal corrosion and electrolytic deterioration and to prevent impeding movement of moving joints.

I. Welding: Weld components to comply with referenced standard and Shop Drawings, unless otherwise indicated. Weld in concealed locations to minimize distortion or discoloration of finish. Protect glazing surfaces from welding.
   1. Install glazing according to Shop Drawings. Comply with requirements of specification Section 08 8000 "Glazing," unless otherwise indicated.

J. Fire Safing: At juncture of curtain wall and edges of floor and roof structure, provide fire stop and safing materials between floors and curtain wall. Install fire stopping to comply with Local Authority / Civil Defence regulations and requirements of tested assemblies as applicable. Rating of fire stopping shall be equivalent to the rating of floor slab construction as tested.
   1. Also, refer specification section “Fire Stopping” and “Building Insulation”.

3.4 INSTALLATION-GLAZING

A. General: Install glazing gaskets and sealants in accordance with manufacturer's instructions without exception, including surface preparations.

B. Structural silicone glazing adhesive sealant and gaskets shall be installed in curtain wall fabricator’s shop.

C. Clean rabbets, stops and glass edges of dust, dirt, moisture, oil and other foreign matter detrimental to glazing material adhesion. Ensure drainage holes are not blocked.

D. Mask surfaces subject to staining, and whenever necessary to ensure neat appearance of the glazing bead. Remove masking as work progresses.

E. Accurately size glass to fit openings allowing clearances following acceptable trade practices. Cut glass cleanly and carefully; replaces glass with nicked or otherwise damaged edges, scratches or abrasion of faces, or other damage.

F. Use spacers, setting blocks and shims of proper size to support and hold glass in position independent of the glazing gaskets. Place 2 setting blocks under each unit at adequate points. Place spacers located directly opposite each other on both sides of the glass, at maximum 600 mm centres and maximum 300 mm from corners and uniformly spaced. Arrange spacers, setting blocks and shims so as to avoid blocking water transfer inside window frames and to stay in place during transportation and installation.

G. Install glazing gaskets in continuous lengths between corners, not stretched, and seal, vulcanize, or weld joints at corners to prevent entry of water and air.

H. Set glass properly centred with uniform bite and face and edge clearance, free from twist, warp or other distortion likely to develop stress.

I. Handle, prepare and install glass in accordance with manufacturer's directions.

J. Install glazing materials to obtain complete adhesion over the full bite area of the unit and to be free from gaps, air bubbles and embedded foreign matter.

K. Ensure a weathertight continuous seal between glazed element and frame flush with sightline and rattle free setting for glass.

3.5 INSTALLATION-SEALANT

A. Seal joints between frame assemblies and adjacent construction, and within glazed assemblies where required to maintain weather-tightness and integrity of air/vapor barrier. Seal junctions in sheet metal air/vapor barriers and between air/vapor barriers and adjacent construction. Cover fasteners penetrating the air/vapor barriers with sealant.

B. Preparation
   1. Ensure that ambient and surface temperatures are within range recommended by sealant manufacturer and joint conditions are suitable for the materials to be installed.
   2. Ensure that surfaces to be sealed are sound, dry, and free from dirt, water loose scale, corrosion, or other contaminants which may adversely affect the performance of the sealing material. Remove oil or grease films.
   3. Perform cleaning to the extent required to achieve acceptable joint surfaces.
   4. Protect cleaned and primed surfaces from further contamination by oil, dust, rain, condensation and other materials detrimental to sealant bonding strength. Re-clean and re-prime contaminated surfaces.
   5. Install filler strips as backup for sealant to provide optimum joint profile, but not less than 6mm depth of sealant bead.
   6. Mask areas adjacent to the joints to prevent contamination of adjacent surfaces. Remove masking promptly after the joint has been completed.
   7. If recommended by the manufacturer of the sealing material, prime joints to prevent attaining, or to assist the bond.
   8. Apply primer with a brush which will permit all joint surfaces to be primed. Perform priming immediately before installation of sealant.

C. Cleaning and Priming for Structural Silicone Glazing.
   1. Do not clean and prime surfaces for structural glazing that cannot be glazed within 2 hours.
   2. Use clean, soft, absorbent, lint-free cloth for cleaning and priming. Each piece of cloth shall be used only once.
   3. Do not use brush for cleaning operation.
   4. Pour cleaning solvent from container onto cloth to avoid contamination; do not dip cloth into solvent.
   5. Dry wipe immediately before cleaning solvent evaporates and dries on the substrate.
   6. Prime all surfaces to receive glazing materials unless recommended specifically to the contrary, in writing, by the sealant manufacturer.
   7. Apply a thin film (one pass) of primer; remove excess primer by wiping with a clean dry cloth according to manufacturer's specification.

D. Installation
   1. Install materials in accordance with the manufacturer's instructions.
   2. Do not exceed shelf life, and pot life of the materials and installation times, as stated by the manufacturers.

3. Be familiar with the work life of the sealant to be used.
4. Before any sealing is commenced, test the materials for indications of staining or poor adhesion.
5. Sealants shall be of gun grade consistency to suit the joint condition. Use pressure operated guns with nozzles of the proper sizes to suit the joints and the sealant material.
6. Use sufficient pressure to fill all voids and joints full. Sealants shall bond to both sides of joint and shall not adhere to the filler material.
7. Ensure that the correct sealant depth is maintained. Superficial painting with a skin bead will not be accepted.
8. Sealant installations shall be full bead free from air pockets and embedded impurities and having smooth surfaces, free from ridges, wrinkles and sags.
9. After joints have been completely filled tool them neatly to a slight concave surface.
10. Install exposed silicone sealants at glazing so that top surfaces of the beads are formed to drain water away from the glass.
11. Trim and clean excess sealants from glass and framing surfaces immediately after installation.
12. Immediately clean adjacent surfaces which have been soiled and leave work in a neat, clean condition. Remove excess materials and droppings using recommended cleaners and solvents.

E. Fire Safing and Curtain Wall Insulation:
   1. Install firesafing and curtain wall insulation specified in relevant sections of the specification set.
      a. Refer specification section “Fire Stopping” and “Building Insulation” for fire safing and curtain wall insulation respectively.

3.6 INSTALLATION – FIRE RESISTIVE JOINT SYSTEMS

A. General: Install fire-resistive joint systems to comply with fire-resistive joint system manufacturer’s written installation instructions for products and applications indicated.

B. Install forming/packing/backing materials and other accessories of type required to support fill materials during their application and in position needed to produce cross-sectional shapes and depths required to achieve fire ratings indicated.

C. Install fill materials for fire-resistive joint systems by proven techniques to produce the following results:
   1. Fill voids and cavities formed by openings and forming/packing/backing materials as required to achieve fire-resistance ratings indicated.
   2. Apply fill material so they contact and adhere to substrates formed by joints.

D. Obtain approval of design and works from Local Authority / Civil Defence / House of expertise for compliance as applicable.

3.7 TOLERANCES

A. Erection Tolerances: Install glazed curtain wall system to comply with the following maximum tolerances:
   1. Plumb: plus/minus 2 mm in 3 m; plus/minus 3 mm in 12 m, 5 mm overall.
   2. Level: plus/minus 2 mm in 6 m; plus/minus 3 mm in 12 m, 5mm overall.
   3. Alignment/ Line: Where surfaces abut in line, limit offset from true alignment to plus/minus 1.5 mm; where a reveal or protruding element separates aligned surfaces by less than 50mm, limit offset to plus/minus 3 mm.

4. Location/ Plane: Limit variation from plane or location shown on Shop Drawings to plus/minus 2 mm in 3 m; and plus/minus 3 mm over total length.

3.8 CLEANING

A. Test and adjust hardware installed under this Section and replace or repair all faulty items. Adjust weather stripping so as to leave each opening unit in its most weathertight position. Test operable elements and ensure easy smooth operation.

B. Repair and touch-up minor surface damage. Remove and replace damaged panels as directed.

C. Clean and make good surfaces soiled or otherwise damaged in connection with this work. Replace finishes or components that cannot be satisfactory cleaned.

D. Handover a proper cleaned hardware.

3.9 FIELD QUALITY CONTROL

A. Testing Agency: Shall be responsible during the installation of the Glazed Aluminum Curtain Walls for the supervision, inspections and tests carried out in accordance with, but not limited to, the requirements detailed in this Specification Section, and those adopted from the accepted Method Statement prepared by the Inspection and Testing Agency. Inspection and Test Reports shall be submitted periodically to the Contractor, Installer and Engineer during installation of the Works, and in accordance with the stipulations of the accepted Method Statement.

B. Testing Services: Testing and inspecting of representative areas to determine compliance of installed systems with specified requirements shall take place in successive stages as required. Do not proceed with installation of the next area until test results for previously completed areas show compliance with requirements.

C. Air Infiltration: Test areas of installed system for compliance with system performance requirements according to ASTM E 783.

D. Water Spray Test: After completing the installation of 2-story area of glazed curtain wall system, test system for water penetration according to AAMA 501.2 in a 2-bay area directed.

E. Repair or remove Work that does not meet requirements or that is damaged by testing; replace to conform to specified requirements.

3.10 MANUFACTURER'S FIELD SERVICES

A. Curtain wall and glass product manufacturers to provide field surveillance of installation of their products.

B. Monitor and report installation procedures and unacceptable conditions.

C. Repair and touch-up minor surface damage. Remove and replace damaged panels as directed.

3.11 PROTECTION

A. Provide final protection and maintain conditions, in a manner acceptable to Manufacturer and Installer that ensure glazed curtain wall system is without damage or deterioration at the time of Substantial Completion.

END OF SECTION 08 4413.01

SECTION 08 8000.01 - GLAZING

PART 1 - GENERAL

Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.1 SECTION INCLUDES

A. This Section includes glazing for the following products and applications, including those specified in other Sections, where glazing requirements are specified by reference to this Section;
   1. Unitized curtain wall system, stick curtain wall system, Add-On steel curtain wall and skylight system.
   2. Fixed and operable glazing, including glass balustrade.
   3. Provide compatible Nano/Protective coating for reducing the façade cleaning frequency.

B. Includes all labor, material, equipment and services necessary to complete related trades with glazing components as shown on the drawings and specified herein.

C. Design to follow the architectural intent indicated in the drawings.

D. This section to be read in conjunction with the related wall types/ exterior wall system drawings submitted as a part of the tender package.

E. Glazing shall be executed by an approved specialist sub-contractor.

1.2 RELATED SECTIONS

1.	Section 01 4500	-	Quality Control
2.	Section 05 7313.01	-	Glazed Decorative Metal Railing
3.	Section 07 2500.01	-	Weather Barrier
4.	Section 07 9200.01	-	Joint Sealants
5.	Section 08 1116.01	-	Alum. Doors and Frames
6.	Section 08 4413.01	-	Glazed Aluminium Curtain Wall
7.	Section 11 8123.01	-	Façade Cleaning and Maintenance Equipments

1.3 PERFORMANCE REQUIREMENTS

A. General: Provide glazing systems capable of withstanding thermal movement, building movements, wind and impact loads (where applicable) without failure, including loss or glass breakage attributable to the following: defective manufacture, fabrication, and installation; failure of sealants or gaskets to remain watertight and airtight; deterioration of glazing materials; or other defects in construction.

B. Glass Design: Glass thicknesses indicated are minimum and are for detailing only. Confirm glass thicknesses by analyzing Project loads and in-service conditions. Provide glass lites for various size openings in nominal thicknesses indicated, but not less than thicknesses and in strengths (annealed or heat treated) required to meet or exceed the following criteria:

1. Glass Thicknesses: Select minimum glass thicknesses to comply with ASTM E 1300, according to the following requirements:

   a. Specified Design Wind Loads: Refer to the available wind tunnel analysis or project wind report at its latest revision for the design loads.
   b. Probability of Breakage for Vertical Glazing: 5 lites per 1000 for lites set vertically or not more than 15 degrees off vertical and under wind action.
   c. Probability of Breakage for Sloped Glazing: 1 lite per 1000 for lites set more, than 15 degrees off vertical and under wind action.
   d. Probability of Breakage for Tempered Glass: 5 lites per 1000 due to nickel sulfide inclusions. Tempered glass shall be Heat Soak tested.
   e. Maximum Lateral Deflection: For the following types of glass supported on all 4 edges, provide thickness required that limits center deflection at design wind pressure to L/60 times the short side length or 25mm, whichever is smaller.
      1) Monolithic glass, heat treated to resist wind loads.
      2) Insulating Glass.
      3) Laminated Glass.
   f. Pillow and overall bow effect: The glass thickness configuration of IGU must be done in such a way that any distortion due to pillow effect is minimized. Since pillow effect and distortion are of aesthetical importance, a theoretical explanation of distortion shall not be accepted.
      1) Control of distortion due to overall bow shall be ensured by:
         a) Introducing efficient quality control during glass heat treatment, IGU processing, structural bonding, curing etc.
         b) The IGU processing shall be carried in vertical position, with long side of the pane horizontally supported.
         c) The ambient climate condition of the location of IGU processing and site shall be studied in detailed to avoid climate load difference impact on reflection due to distortion.
         d) Avoid symmetric pane thickness configuration of the IGU construction. Double glazed units shall utilize panes of unequal thickness, unless otherwise specified. The outer panes shall be thicker/stiffer to minimize the effects of pillowing.
         e) Structural bonding, as and if applicable shall be applied between the glass and frame by supporting the glass panel in the middle to avoid permanent glass bow due to its own weight.
         f) Ensure IGU panel is kept for enough curing time, as recommended by the sealant manufacturer, before its transportation.
         g) Allowed overall bow on the glass, in its installed position shall be maximum 1/1250 at the center when measured diagonally.
         h) Provide a detailed method statement to confirm the measurements implemented for controlling overall bow/ distortion in glass pane.

C. Thermal Movements: Provide glazing that allows for thermal movements resulting from the maximum change (range) in ambient and surface temperatures acting on glass framing members and glazing components. Base engineering calculation on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.
   1. Temperature Change (Range): 70 deg. C, ambient; 100 deg. C, material surfaces.
   2. Night time or lower range +10 deg. C.

D. Comply with the specification document “Glass Types Overview” at its latest revision:
   1. Tower H32: QC00912012-SSH-DRW-GEN-FAC-EV-H32-00003 and QC00912012-SSH-DRW-GEN-FAC-EV-H32-00004.
   2. Building H30: QC00912012-SSH-DRW-GEN-FAC-EV-H30-00003 and QC00912012-SSH-DRW-GEN-FAC-EV-H30-00004.
   3. Building H27: QC00912012-SSH-DRW-GEN-FAC-EV-H27-00003 and QC00912012-SSH-DRW-GEN-FAC-EV-H27-00004.

E. Minimum thermal performance criteria for the glass; the objective is to achieve an overall (including frames) U-Value <= 1.8 W/m2k with the maximum targeted shading coefficient (SC) of 0.25 (SHGC 0.22), for vertical curtain wall/ glazing and U-Value <= ~~1.6~~ 2.0~2.2 W/m2k and shading coefficient (SC) of 0.21 (SHGC 0.18) with no glare, for horizontal glazed surfaces such as Skylights, as considered in the thermal modeling by the Engineer.

F. The spandrel glass panel back pan aluminum sheet to match as closely as possible to the glazing color unless otherwise indicated/ mentioned in the drawing while maintaining/ adhering to the energy requirements.

G. Acoustic Requirement: Glass thickness, type, configuration etc. has to be decided to comply with acoustic requirement as per project Acoustic Report (for the selected Noise Scenario by the Engineer) at its latest revision. To meet the acoustic design requirement laminated glass shall be used.

H. In case of selection of glass combination where the external glass thickness varies largely between adjacent panels, the inside layer of outside laminated glass shall be low iron glass to reduce the apparent variance in color due to larger variance in thickness.

I. Glazing shall not be used as a diaphragm to brace glazed elements unless design / build trade vendor demonstrates the ability of the glass and sealants to resist forces without putting undue stress on the glass.

1.4 SUBMITTALS:

A. Product Data: For each glass product and glazing material indicated.

B. Samples: 300mm x 300mm Samples for glass types and of 300mm long Samples for sealants. Install sealant Samples between two strips of material representative in color of the adjoining framing system.

C. Product Certificates: Signed by manufacturers of glass and glazing products certifying that products furnished comply with requirements.

D. Preconstruction Adhesion and Compatibility Test Report: From glazing sealant manufacturer indicating glazing sealants were tested for adhesion to glass and glazing channel substrates and for compatibility with glass and other glazing materials.

E. Glazing Schedule: Use same designations indicated on drawings for glazed openings in preparing a schedule listing glass types and thicknesses for each size opening and location.

F. Qualification data: For firms and persons specified in “Quality Assurance” to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of Engineer and Employer, and other information specified.

G. Product Test Reports: From a qualified testing agency indicating the products comply with requirements, based on comprehensive testing of current products.
   1. Tinted float glass.
   2. Coated float glass.
   3. Insulating glass.
   4. Glazing sealants (UV resistant).
   5. Glazing gaskets.
   6. Spacer Stainless Steel
   7. Spacer bonding / sealant (UV resistant).
   8. Laminated glass.

1.5 QUALITY ASSURANCE

A. Manufacturers and Fabricator's Qualification: Manufacturer/Fabricator shall be ISO 9001:2000 certified or one certified as employing an equivalent or higher system of quality management; having a minimum of 10 (Ten) years documented experience in the manufacture/fabrication of product/s specified in this Part; capable of providing field service representation, as required by Engineer, during installation.

B. Glazier Qualifications: An experienced and ISO certified glazier who has completed glazing similar in material, design, and extent to that indicated for Project and whose work has resulted in construction with a record of successful in-service performance.

C. Insulating glass units shall be manufactured by company specializing in the fabrication of insulating glass units with minimum 10 (Ten) years' experience, ample facilities to produce unit required and which have installations of units of similar composition to that required on the Project that have been in place without failure for 5 (five) years. Glass of the same type shall be by one manufacturer.

D. Glass tempering shall be performed by a company specializing in the production of tempered glass with 10 (Ten) years' experience, modern tempering equipment and facilities capable of treating post temperable glass units (low-E coating applied to glass prior to tempering) of maximum sizes required on the project, while maintaining tight roller wave tolerances specified and which have installations of post temperable units of similar composition to that required on the Project that are in place showing acceptable limits of distortion.

E. The use of toughened glass shall be justified by risk assessment and/or calculations, with the general aim of minimizing its use. In case of application of toughened glass, the glass panels required to be 100% heat soak tested. Oven shall be certified by a third party.

F. All toughened glass shall be heat soak tested to BS EN 14179: Part 1 paying particular attention to temperature and duration of treatment. Demonstrate that the surface temperature of all glass was maintained at 290° C±10° for 2 hours holding time. Keep records of heat soaking for each batch for QA/QC inspection purposes before dispatch to site.

G. Edge processing of thermally tempered single panes that are built into a glass system shall be carried out in line with EN 12150.
   1. Maximum overall bow: 2mm per 1000mm measured along the glass edge.
   2. Maximum local bow: The maximum deviation for flatness from peak to trough shall not exceed 0.3mm per 300mm or 0.15mm at the edge.

3. Rollerwave: Glass shall be sized to provide for the consistent and horizontally aligned orientation of ripples throughout the Works. The maximum deviation for flatness from peak to trough shall not exceed 0.08mm.

H. Verification of visual quality/defects or manufacturing imperfections such as pin holes, scratches, chipping etc. of toughened glass shall be carried out at a distance of 1000mm from the viewed surface of the glass at a viewing angle which corresponds to the normal usage of the room and carried out under diffuse daylight conditions (e.g., overcast sky), without direct sunlight or artificial lighting.

I. Exposed edge working shall be flat ground with small ground aris and have a frosted appearance.

J. Small shells and / or chips, exceeding a maximum diameter of 2 mm, shall be ground out prior to toughening.

K. All glass units of each type shall be the same and shall have a consistent, uniform color and appearance. The external glass layer shall have a constant thickness in directly connected façade areas (elevation). A patch effect, on adjacent façade elements, due to different glass thickness, shall not be accepted. In such cases, contractor has to sort the solution like, integration of low iron glass panes in the glass build-up.

L. Glass Product Testing: Obtain glass test results for product test reports in "Submittals" Article from a qualified testing agency based on testing glass products.
   1. Glass Testing Agency Qualifications: An independent testing agency with the experience and capability to conduct the testing indicated, as documented according to ASTM E 548.

M. Elastomeric Glazing Sealant Product Testing: Obtain sealant test results for product test reports in "Submittals" Article from a qualified testing agency based on testing current sealant formulations within a 36-month period.
   1. Sealant Testing Agency Qualifications: An independent testing agency qualified according to ASTM C 1021 to conduct the testing indicated, as documented according to ASTM E 548.
   2. Test elastomeric glazing sealants for compliance with requirements specified by reference to ASTM C 920, and where applicable, to other standard test methods.

N. Preconstruction Adhesion and Compatibility Testing: Submit to elastomeric glazing sealant manufacturers, for testing indicated below, samples of each glass type, tape sealant, gasket, glazing accessory, and glass-framing member that will contact or affect elastomeric glazing sealants.
   1. Use manufacturer's standard test methods to determine whether priming and other specific preparation techniques are required to obtain rapid, optimum adhesion of glazing sealants to glass, tape sealants, gaskets, and glazing channel substrates.
   2. Submit not fewer than eight pieces of each type and finish of glass-framing members and each type, class, kind, condition, and form of glass (monolithic, laminated, and insulating units) as well as one sample of each glazing accessory (gaskets, tape sealants, setting blocks, and spacers).
   3. Schedule sufficient time for testing and analyzing results to prevent delaying the Work.
   4. For materials failing tests, obtain sealant manufacturer's written instructions for corrective measures, including the use of specially formulated primers.

O. Safety Glazing: Weather specified or not on Drawings or other parts of the Contract Document, provide safety glazing where required or where risk of impact or falling glass is

likely, as determined according to the following:

1. BS 6262 Part 4, “Glazing for Building Safety Related to Human Impact”
2. ANSI Z97.1, “Safety Glazing Materials Used in Buildings-Safety Performance Specifications and Methods of Test”

P. Safety Glass: Category II materials complying with testing requirements specified in 16 CFR 1201 (Safety Standard for Architectural Glazing Material) and ANSI Z97.1 (American National Standard for Safety Glazing Materials Used in Buildings -Safety Performance Specifications and Methods of Test).

Q. Source Limitations for Glass: Obtain each type of glass from single source from single manufacturer for each glass type.

R. Source Limitations for Glazing Accessories: Obtain from single source from single manufacturer for each product and installation method.

S. Glazing Publications: Comply with published recommendations of glass product manufacturers and organizations, unless more stringent requirements are indicated. Refer to these publications for glazing terms not otherwise defined in this Section or in referenced standards.
   1. GANA Publications GANA Laminated Division’s “Laminated Glass Design Guide” and GANA “Glazing Manual”.
   2. IGMA Publication for Insulating Glass. SIGMA TM-3000 “Glazing Guidelines for Sealed Insulating Glass Units.
   3. IGMA Publication for Sloped Glazing IGMA TB-3001. “Sloped Glazing Guidelines.

T. Insulating-Glass Certification Program: Permanently marked either on spacers or on at least one component lite of units with appropriate certification label of the following inspecting and testing agency:
   1. Insulating Glass Certification Council.
   2. Associated Laboratories, Inc.
   3. National Accreditation and Management Institute.

U. Comply with most stringent of the local code or acceptable national/international building standard requirements for VOC’s limits.

V. Mockups: Before glazing, build mockups for each glass product indicated to verify selections made under sample Submittals and to demonstrate aesthetic effects and qualities of materials and execution. Build mockups to comply with the following requirements, using materials indicated for the completed Work:
   1. Build mockups in the location and of the size indicated or, if not indicated, as directed by the Engineer.
   2. Notify seven days in advance of dates and times when mockups will be constructed.
   3. Built mock-up of each type of glass to match glazing systems required for the Project, including typical lite size, framing systems, and glazing methods.
   4. Obtain approval of mockups before starting fabrication.
   5. Maintain mockups during construction in an undisturbed condition as a standard for judging the completed Work.
   6. Approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion subject to approval.
   7. If Engineer determines mockups do not comply with the requirements, reconstruct mockup for the approval of Engineer.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Protect glazing materials according to manufacturer's written instructions and as needed to prevent damage to glass and glazing materials from condensation, temperature changes, direct exposure to sun, or other causes.

B. For insulating-glass units that will be exposed to substantial altitude changes, comply with insulating-glass manufacturer's written recommendations for venting and sealing to avoid hermetic seal ruptures and pillow effect.

1.7 PROJECT CONDITIONS

A. Environmental Limitations: Do not proceed with glazing when ambient and substrate temperature conditions are outside limits permitted by glazing material manufacturers and when glazing channel substrates are wet from rain, frost, condensation, or other causes.

B. Insulating and laminating glass shall be processed in controlled environment as per glazing elements manufacturer's recommendation.

1.8 WARRANTY

A. General: Warranties shall be in addition to, and no limitation of other rights, the Employer may have under the Contract Documents.

B. Manufacturer's Special Project Warranty on Coated Glass Products: Provide written warranty signed by manufacturer of coated glass agreeing to furnish, within specified warranty period indicated below, replacements for those coated glass units which develop manufacturing defects. Manufacturing defects are defined as peeling, cracking or deterioration in metallic coating due to normal conditions.
   1. Warranty Period: Manufacturer's standard but not less than 10 (Ten) years from date of Substantial Completion.

C. Manufacture's Special Project Warranty on Laminated Glass: Provide written warranty signed by Laminated Glass Manufacturer agreeing to replace within the warranty period indicated below, laminated glass units that develop manufacturing defects. Manufacturing defects are defined as edge separation, delamination, materially obstructing vision through glass, and blemishes exceeding those allowed by referenced laminated glass standards.
   1. Warranty Period: Manufacturers standard but not less than 10 (Ten) years from date of Substantial Completion.

D. Manufacture's Special Project Warranty on Insulating Glass: Provide written warranty signed by manufacturer of insulating glass agreeing to finish within specified warranty period indicated below, replacements for those insulating glass units developing manufacturing defects. Manufacturing defects are defined as failure of hermetic seal of air space (beyond that due to glass breakage) as evidenced by intrusion of dirt of moisture, internal condensation or fogging, deterioration of protected internal glass coatings, if any and other visual indications of seal failure or performance.
   1. Warranty Period: Manufacturer's standard but not less than 10 (Ten) years form date of Substantial Completion.

E. Manufacturer's Special Warranty on Heat-Treated Glass: Manufacturer's standard form,

made out to the Employer and signed by the heat-treated glass manufacturer, agreeing to replace all fully-tempered glass units used in exterior assemblies that have broken spontaneously due to the inclusion of nickel sulphides, within the specified warranty period.

1. Warranty Period: Manufacturer's standard but not less than 10 (Ten) years form date of Substantial Completion.

F. Contractor's Warranty on Coated Glass, Laminated Glass, Insulating Glass, and Heat-Treated Glass: Provide Written warranty, made out to the Employer and signed by the Contractor, agreeing to install replacements, at his own expense, for glass units that deteriorate due to defective workmanship, within the specified warranty period.

1. Warranty Period: Manufacturer's standard but not less than 10 (Ten) years form date of Substantial Completion.

G. Risk-Analysis: In addition to Base Warranty, perform "Risk Analysis" establishing breakage possibility of glass units furnished. Recommend number of glass units which should be stocked by the Employer and provide same, but not less than one percent (1%) of the total project requirement of each size for each glass type, as selected by the Engineer.

H. Material Replacement: Material and labor covered under Contractor's Warranty shall include glass, metal and other accessory materials required to complete rectification/replacement works. Glass units shall be used from Employer's stock and shall be re-stocked to comply with requirements specified under "Risk Analysis" article, maintaining minimum stock required for a period not less than that of the specified Warranty Period.

PART 2 - PRODUCTS

2.1 ACCEPTABLE GLASS MANUFACTURERS

A. Primary manufacturers and glazing fabricators specified hereinafter are specified for the purposes of establishing minimum quality standards and performance requirements. Both international operating companies as entity will have to take full responsibility and guarantee of the whole façade; it can joint venture with a local installer, which will have to be submitted for approval. Product and Fabricators which are equal to, or better than those specified and which conform to the Engineer's design requirements and color selections, may be acceptable subject to Engineer's approval.

B. Primary Glass Manufacturers

1. Guardian, ME (UAE/KSA)/ Europe
2. Pilkington, ME (KSA)/ UK
3. Emirates Float Glass LLC, ME (UAE)
4. Arabian United Float Glass Company, ME (KSA)
5. Arcon, Germany.
6. PPG, USA.
7. Saint-Gobain, France.
8. EuroGlas, Europe.
9. Cardinal Glass, USA.
10. Trosch, Europe.
11. Asahi Glass Company, Europe.

Or Equivalent, as per Engineer's acceptance.

C. Glazing Fabricators/Assemblies/System Manufacturers

1. Gulf Glass Industries, ME.
2. Future Architectural Glass, ME.
3. Emirates Glass, ME.
4. White Aluminium Glass, ME.
   Or Equivalent, as per Engineer's acceptance.

D. Conditions to qualify "equivalent": Qualification exercise consists of assessment of technical adaptability of the design and the material, quality procedures maintained, minimum 10 (Ten) years of its usage in similar project and credibility of its performance record and other project references. All cost related to this exercise shall be borne by the proposer. The proposer shall agree to use above listed pre- approved system manufactures, should the proposed system be not acceptable to the Engineer, after the completion of qualification exercise.

2.2 PRIMARY FLOAT GLASS

A. Float Glass: ASTM C 1036, Type I (transparent glass, flat), Quality Q3 (glazing selects); class as indicated below.
   1. Class 1 (clear) unless otherwise indicated.
   2. Class 2 (tinted, heat-absorbing, and light-reducing) where indicated.

2.3 TEMPERED /TOUGHENED GLASS

A. Tempered Glass: ASTM C 1048; Type I (transparent glass, flat), quality Q3. Tempering shall be obtained using horizontal, tong free method and any linear distortion from tempering shall be horizontal and subject to Consultant's acceptance.

B. Heat strengthened glass: ASTM C1048 Kind HS. Heat strengthening shall be performed using horizontal tong-free method, and any linear distortion from tempering shall be horizontal and subject to Consultant's acceptance; surface compression shall not exceed 50 MPa.

C. Sizes and Cutting: Prior to tempering/toughening or heat treating, cut glass to required sizes as determined by accurate measurement of openings to be glazed, making allowance for required edge clearances. Cut and process edges in accordance with glass manufacturer's recommendations. Do not cut or treat edges in the field.

D. Tempered/Heat strengthened Glass: Provide glass tempered/heat strengthened to increase flexural strength 4-5/1-2.5 times its strength before treatment respectively.

E. All Tempered glass shall be Heat Soak tested, using two stage regime heat soaking method. Written warranties against nickel sulphide inclusions or statistical heat soaking in lieu of heat soaking shall not be accepted.

F. Edge dip shall not be more than 0.25mm.

G. Ensure that the toughening process does not produce iridescence, distortion, roll marks or ripples in the glass which are unacceptable to the Engineer. Such anticipated imperfections shall be demonstrated by the provision of full-size samples prior to commencement of glass production. The Engineer will examine the samples provided and reasonably advise what is acceptable and what is unacceptable. All glass produced for the Works shall comply with the acceptable samples as a minimum standard.

H. Take all reasonable measures to control the toughening process so as to reduce the occurrence of anisotropy (iridescences) at the time of manufacture. Provide full size samples for review and agreement to set an acceptable range of occurrence of anisotropy, glass shall be rejected if it does not fall within the range of agreed samples.

2.4 COATED GLASS

A. Pyrolytic-Coated Float Glass: ASTM C 1376, float glass with metallic-oxide coating applied by pyrolytic deposition process during initial manufacture, and complying with other requirements specified.

B. Sputter-Coated Float Glass: ASTM C 1376, float glass with metallic-oxide coating deposited by vacuum deposition process after manufacture, and heat treatment (if any) and complying with other requirements specified.

C. Ceramic-Coated Glass: Float glass with ceramic enamel frits and complying with ASTM C 1048, condition C, Type I, quality Q3 and other requirements specified.

2.5 INSULATING GLASS

A. Insulating Glass Unit- Comply with ASTM E2190 and IGMA TM-3000, certified by the Insulating Glass Certification Council (IGCC) or equivalent European standards BS EN 1279, consisting of 2 panes separated by a 12/16/20mm wide factory double time sealed, dehydrated and air/argon filled cavity space utilizing black (recommended for aesthetics) stainless steel/warm edge spacer, without metal edge banding. Seal color to be selected by consultant. Air/gas pressure within the glass cavity space shall suit atmospheric conditions at the location of the installation to prevent distortion of the installed units. Edges of glass shall be straight cut, free from nicks and other imperfections conducive to breakage. Units shall be manufactured in IGMA or equivalent European certified facility. Aris all edges of glass adjacent to structural silicone glazing.
   1. Insulated glass spacers shall be straight and evenly set into glass units. Reflective and low emissivity coatings shall be edge deleted over entire depth of spacer and edge seal at locations where silicone sealant is in contact with glass.
   2. Use 2 component silicone sealant for secondary seal of the insulating glass units at structural silicone glazing. Remove excess silicone seal from glass edges of outer lite.

B. Fabrication: Fabricate units at factory with sheets of glass hermetically sealed at all edges, on the dual seal principle using bendable type, perforated, hollow, stainless steel or warm edge spacer tubes and shall conform to ASTM E 774. A molecular sieve-type desiccant material shall be inserted into the spacer tube to allow for absorption of residual moisture during fabrication. Glass shall be hermetically sealed double-glazed units as indicated on drawings and as herein specified.
   1. Sealants: Primary sealant shall comprise polyisobutylene applied to the spacer tube. Secondary sealant shall comprise of high modulus structural silicone applied to the space tube. Secondary sealant shall comprise 2-part structural silicone, Sputter coating shall be deleted at edges to allow for glass-to-glass bonding of the primary and secondary sealants.
   2. Insulated glass spacers shall be straight and evenly set into glass units. Reflective and low emissivity coatings shall be edge deleted (for soft coated glass) over entire depth of spacer and edge seal at locations where silicone sealant is in

contact with glass.

3. Remove excess silicone seal from glass edges of outer lite.

C. Thermal Performance of Glazing Units: As indicated on the drawing " Glass Types Overviews" at its latest revision.

D. Low-e coated glass: All insulated glass units shall have low-E coating or silver performance coating on glass surface indicated.

E. Fritted Glass: Frit pattern as indicated on the Drawings and directed by the Engineer and to the approval of the Engineer/Employer. Frit on the outer face of the glass with 10 (ten) years of manufacturers Guaranty.

F. Tinted Glass: Color as indicated on drawings and to match approved sample.

G. Manufacture: Manufacturers of insulating glass units shall have been in the business of producing units of similar size and configuration for not less than 10 (ten) years.

H. Visual Quality Testing: Visual quality testing of insulated glass units of float glass shall be carried out at a distance of 1000mm from the viewed surface of the glass at a viewing angle which corresponds to the normal usage of the room and the following discrepancies, or better, shall be allowed per unit:
   1. Rebate zone (width 18mm - no limit on discrepancies, with the exception of mechanical damage to the edges).
      a. External shallow damage to the edge or conchoidal fractures, which do not affect the glass strength and which do not project beyond the width of the edge seal.
      b. Internal conchoidal fractures without loose shards, which are filled by the sealant.
      c. Unlimited spots or patches of residue, or scratches.
   2. Edge zone - area around edge with width (w)/ 10 or height (h)/ 10 respectively.
      a. Inclusions, bubbles, spots, stains, etc.
         1) Pane area less than or equal to 1m²: Maximum 4 cases, each less than or equal to 3mm diameter.
         2) Pane area greater than 1m²: Maximum 1 case, each less than or equal to 3mm diameter per meter of perimeter.
      b. Residues (spots) in the gas-filled cavity.
         1) Pane area less than or equal to 1m²: Maximum 4 cases, each less than or equal to 3mm diameter.
         2) Pane area greater than 1m²: Maximum 1 case, each less than or equal to 3mm diameter per meter of perimeter.
      c. Residues (patches) in the gas-filled cavity: Whitish grey or transparent - maximum 1 case less than or equal to 3cm².
      d. Scratches: Total of individual lengths; maximum 90mm. Individual length; maximum 30mm.
      e. Hair-line scratches: Not allowed in higher concentration.
   3. Main zone (remainder of unit inside the edge zone).
      a. Inclusions, bubbles, spots, stains, etc.
         1) Pane area less than or equal to 1m²: Maximum 2 cases, each less than or equal to 2mm diameter.
         2) Pane area greater than 1m² but less than or equal to 2m²: Maximum 3 cases, each less than or equal to 2mm diameter.
         3) Pane area greater than 2m²: Maximum 5 cases, each less than or equal

to 2mm diameter.

b. Scratches: Total of individual lengths; maximum 45mm. Individual length; maximum 15mm.

c. Hair-line scratches: Not allowed in higher concentration.

4. Main and edge zone combined:
   a. Maximum number of allowable discrepancies as in edge zone.
   b. Inclusions, bubbles, spots, stains, etc. of dimensions 0.5 - 1.0mm are allowable without any area related limitation, except when they appear in higher concentration. 'Higher concentration' means that at least 4 inclusions, bubbles, points, patches, etc. are located within a circle with a diameter of less than or equal to 20cm.
5. Visual inspection of the glass edges, edge seals and spacers shall be unhindered, prior to glazing.

2.6 LAMINATED GLASS

A. Laminated glass: Comply with ASTM C 1172 for kinds of laminated glass indicated and the following:
   1. Interlayer: Interlayer material of 1.52mm thick polyvinyl butyral sheets or as indicated on drawings, with a proven record of no tendency to bubble, discolor or lose physical and mechanical properties after laminating glass lites and installation.
   2. Interlayer shall be EVA (Ethyl Vinyl Acetate) or SGP (SentryGlas Plus): For laminated glass without continuous edge protection cap used in humid or exterior areas like exposed edge balustrade etc.
   3. Laminating Process: Fabricate laminated glass to produce glass free of foreign substances and air or gas pockets.

2.7 GLAZING SEALANTS

A. General: Provide products of type indicated, complying with the following requirements:
   1. Compatibility: Select glazing sealants that are compatible with one another and with other materials they will contact, including glass products, seals of insulating-glass units, and glazing channel substrates, under conditions of service and application, as demonstrated by sealant manufacturer based on testing and field experience.
   2. Suitability: Comply with sealant and glass manufacturers' written instructions for selecting glazing sealants suitable for applications indicated and for conditions existing at time of installation.
   3. Color of exposed glazing sealant: To approval.

B. Elastomeric Glazing Sealant Standard: Comply with ASTM C 920 and other requirements indicated for each liquid-applied, chemically curing sealant, including those referencing classifications for type grade, class and uses related to exposure and joint substrates.

C. Primer, cleaning materials and joint backing: Recommended by the sealant manufacturer to suit the various substrates and conditions of the Work, compatible with each other and with sealant.

D. Concealed Sealing Materials: All sealing materials concealed within the aluminium curtain wall system shall be silicone compatible with and adherent to each material it will be in contact with, as recommended by the manufacturer to fulfill performance requirements.

E. Sealant and gasket seals shall be non-bleeding, non-staining and capable of supporting their own weight. Sealants and gasket seals exposed to exterior shall also be UV resistant and ozone resistant. Sealants used in glazing systems shall also be capable of supporting specified loads associated with the glazing systems.

F. Where indicated, exposed weather seal locations in structural silicone applications shall be gaskets.

G. Sealant at exposed locations: ASTM C920, type which minimize silicone fluid leaching, one component, neutral cure, medium modulus, 'Silruf NB-SCS9000' by GE Silicones or Spectrum 3 or 'Spectrum 4' by Tremco Ltd., or 756 SMS' by Dow Corning Corporation or equivalent.

H. Structural silicone and heal bead for shop and field application (where shop application is not possible): ASTM C920, one component, '795' by Dow Corning Corporation, 'Utraglaze 4000' by GE Silicones, or 'Spectrum 2' by Tremco Ltd. or equivalent.

I. Sealant used in structural joints shall have adequate strength to retain insulating units to the metal framing under the design conditions.

J. Sealant used in contact with edges of laminated glass shall not cause damage or disfigurement or laminate interlayer of the laminated glass.

K. Air seal sealants: For back-pan seals; one-part elastomer sealant.

L. Sealant shall be from the same manufacturers for all work of this Section.

M. Sealant color: To be selected by the Engineer/Employer, not necessarily from the sealant manufacturer's standard color range.

2.8 GLAZING TAPES

A. Back-Bedding Mastic Glazing Tape: Preformed, butyl-based elastomeric tape with a solids content of 100 percent; non-staining and non-migrating in contact with nonporous surfaces; with or without spacer rod as recommended in writing by tape and glass manufacturers for application indicated; packaged on rolls with a release paper backing; and complying with ASTM C 1281 and AAMA 800 for products indicated below:
   1. AAMA 806.3 tape, for glazing applications in which tape is subject to continuous pressure.
   2. AAMA 807.3 tape, for glazing applications in which tape is not subject to continuous pressure.

B. Expanded Cellular Glazing Tape: Closed-cell, foam tape; factory coated with adhesive on both surfaces; packaged on rolls with release liner protecting adhesive; and complying with AAMA 800 for the following types:
   1. Type 1, for glazing applications in which tape acts as the primary sealant.
   2. Type 2, for glazing applications in which tape is used in combination with a full bead of liquid sealant.

2.10 GLAZING GASKETS

A. Lock-Strip Gaskets: Neoprene extrusions in size and shape indicated, fabricated into frames

with molded corner units and zipper lock strips, complying with ASTM C 542, black.

B. Dense Compression Gaskets: Molded or extruded gaskets of material indicated below, complying with standards referenced with name of elastomer indicated below, and of profile and hardness required to maintain watertight seal:
   3. EPDM, ASTM C 864.

C. Soft Compression Gaskets: Extruded or molded, closed-cell, integral-skinned gaskets of material indicated below; complying with ASTM C 509, Type II, black; and of profile and hardness required to maintain watertight seal:
   4. EPDM.

2.11 MISCELLANEOUS GLAZING MATERIALS

A. General: Provide products of material, size, and shape complying with referenced glazing standard, requirements of manufacturers of glass and other glazing materials for application indicated, and with a proven record of compatibility with surfaces contacted in installation.

B. Cleaners, Primers, and Sealers: Types recommended by sealant or gasket manufacturer.

C. Setting Blocks: Elastomeric material with a Shore A durometer hardness of 85 $\pm$ 5.

D. Spacers: Elastomeric blocks or continuous extrusions with a Shore A durometer hardness required by glass manufacturer to maintain glass lites in place for installation indicated.

E. Edge Blocks: Elastomeric material of hardness needed to limit glass lateral movement (side walking).

F. Cylindrical Glazing Sealant Backing: ASTM C 1330, Type C (closed-cell material), of size and density to control glazing sealant depth and otherwise produce optimum glazing sealant performance.

2.12 FABRICATION OF GLASS AND OTHER GLAZING PRODUCTS

C. Fabricate glass and other glazing products in sizes required to glaze openings indicated for Project, with edge and face clearances, edge and surface conditions, and bite complying with written instructions of product manufacturer and referenced glazing publications, to comply with system performance requirements.

D. Grind smooth and polish exposed glass edges.

2.13 REFERENCED STANDARDS

G. AAMA — American Architectural Manufacturer Association

H. AAMA 800 — Voluntary Specifications and Test Methods for Sealants

I. ASCE 7-10 — American Society of Civil Engineers: Minimum Design

	Loads for Buildings and Other Structures
J. ASTM C 509	Specification for Elastomeric Cellular Preformed Gasket and Sealing Material
K. ASTM C 542	Specification for Lock-Strip Gaskets
L. ASTM C 716	Specification for Installing Lock-Strip Gaskets and Infill Glazing Materials
M. ASTM C 864	Specification for Dense Elastomeric Compression Seal Gaskets, Setting Blocks, and Spacers
N. ASTM C 920	Specification for Elastomeric Joint Sealants
O. ASTM C 1021	Practice for Laboratories Engaged in the Testing of Building Sealants
P. ASTM C 1036	Specification for Flat Glass
Q. ASTM C 1048	Specification for Heat-Treated Flat Glass - Kind HS, FT Coated and Uncoated Glass
R. ASTM C 1172	Specification for Laminated Architectural Flat Glass
S. ASTM C 1281	Specification for Preformed Tape Sealants for Glazing Applications
T. ASTM C 1330	Specification for Cylindrical Sealant Backing for Use with Cold Liquid Applied Sealants
U. ASTM E 548	Guide for General Criteria Used for Evaluating Laboratory Competence
V. ASTM E 774	Specification for the Classification of the Durability of Sealed Insulating Glass Units
W. ASTM E 1300	Practice for Determining Load Resistance of Glass in Buildings
X. EN 673	Glass in building. Determination of thermal transmittance (U value). Calculation method
Y. EN 12150-1	Glass in building. Thermally toughened soda lime silicate safety glass. Definition and description
Z. EN 1279-1	Glass in building. Insulating glass units. Generalities, dimensional tolerances and rules for the system description
AA. 16 CFR 1201	Safety Standard for Architectural Glazing Materials BB.
IGMA TB-3001	Sloped Glazing Guidelines

CC. IGMA TM-3000 Glazing Guidelines for Sealed Insulating Glass Units DD.

NFPA 252 Fire Tests of Door Assemblies

EE. NFPA 257 Fire Test for Window and Glass Block Assemblies

FF. The minimum standards for products specified in this section shall be including as under but not limited to the following. Except as otherwise specified herein, perform work in accordance with specifications, codes and standards cited therein, and their latest editions, revisions, applicable addenda and supplements. Where there is conflict between the referenced standards the most stringent of the conditions/requirements shall be applicable.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine framing glazing, with Installer present, for compliance with the following:
   1. Manufacturing and installation tolerances, including those for size, square, and offsets at corners.
   2. Presence and functioning of weep system.
   3. Minimum required face or edge clearances.
   4. Effective sealing between joints of glass-framing members.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Clean glazing channels and other framing members receiving glass immediately before glazing. Remove coatings not firmly bonded to substrates.

3.3 GLAZING, GENERAL

A. Comply with combined written instructions of manufacturers of glass, sealants, gaskets, and other glazing materials, unless more stringent requirements are indicated, including those in referenced glazing publications.

B. Glazing channel dimensions, as indicated on Drawings, provide necessary bite on glass, minimum edge and face clearances, and adequate sealant thicknesses, with reasonable tolerances. Adjust as required by Project conditions during installation.

C. Protect glass edges from damage during handling and installation. Remove damaged glass from Project site and legally dispose of off Project site. Damaged glass is glass with edge damage or other imperfections that, when installed, could weaken glass and impair performance and appearance.

D. Apply primers to joint surfaces where required for adhesion of sealants, as determined by pre-construction sealant-substrate testing.

E. Install setting blocks in sill rabbets, sized and located to comply with referenced glazing publications, unless otherwise required by glass manufacturer. Set blocks in thin course of

compatible sealant suitable for heel bead.

F. Do not exceed edge pressures stipulated by glass manufacturers for installing glass lites.

G. Provide spacers for glass lites where the length plus width is larger than 1250mm as follows:
   1. Locate spacers directly opposite each other on both inside and outside faces of glass. Install correct size and spacing to preserve required face clearances, unless gaskets and glazing tapes are used that have demonstrated ability to maintain required face clearances and to comply with system performance requirements.
   2. Provide 3mm minimum bite of spacers on glass and use thickness equal to sealant width. With glazing tape, use thickness slightly less than final compressed thickness of tape.

H. Provide edge blocking, where indicated or needed to prevent glass lites from moving sideways in glazing channel, as recommended in writing by glass manufacturer and according to requirements in referenced glazing publications.

I. Set glass lites in each series with uniform pattern, draw, bow, and similar characteristics.

3.4 TAPE GLAZING

A. Position tapes on fixed stops so that, when compressed by glass, their exposed edges are flush with or protrude slightly above sightline of stops.

B. Install tapes continuously, but not necessarily in one continuous length. Do not stretch tapes to make them fit opening.

C. Where framing joints are vertical, cover these joints by applying tapes to heads and sills first and then to jambs. Where framing joints are horizontal, cover these joints by applying tapes to jambs and then to heads and sills.

D. Place joints in tapes at corners of opening with adjoining lengths butted together, not lapped. Seal joints in tapes with compatible sealant approved by tape manufacturer.

E. Do not remove release paper from tape until just before each glazing unit is installed.

F. Apply heel bead of elastomeric sealant.

G. Center glass lites in openings on setting blocks and press firmly against tape by inserting dense compression gaskets formed and installed to lock in place against faces of removable stops. Start gasket applications at corners and work toward centers of openings.

3.5 GASKET GLAZING (DRY)

A. Fabricate compression gaskets in lengths recommended by gasket manufacturer to fit openings exactly, with stretch allowance during installation.

B. Insert soft compression gasket between glass and frame or fixed stop so it is securely in place with joints miter cut and bonded together at corners.

C. Center glass lites in openings on setting blocks and press firmly against soft compression

gasket by inserting dense compression gaskets formed and installed to lock in place against faces of removable stops. Start gasket applications at corners and work toward centers of openings. Compress gaskets to produce a weather tight seal without developing bending stresses in glass. Seal gasket joints with sealant recommended by gasket manufacturer.

D. Install gaskets so they protrude past face of glazing stops.

3.6 SEALANT GLAZING (WET)

A. Install continuous spacers, or spacers combined with cylindrical sealant backing, between glass lites and glazing stops to maintain glass face clearances and to prevent sealant from extruding into glass channel and blocking weep systems until sealants cure. Secure spacers or spacers and backings in place and in position to control depth of installed sealant relative to edge clearance for optimum sealant performance.

B. Force sealants into glazing channels to eliminate voids and to ensure complete wetting or bond of sealant to glass and channel surfaces.

C. Tool exposed surfaces of sealants to provide a substantial wash away from glass.

3.7 LOCK-STRIP GASKET GLAZING

A. Comply with ASTM C 716 and gasket manufacturer's written instructions. Provide supplementary wet seal and weep system, unless otherwise indicated.

3.8 PROTECTION AND CLEANING

A. Protect exterior glass from damage immediately after installation by attaching crossed streamers to framing held away from glass. Do not apply markers to glass surface. Remove nonpermanent labels, and clean surfaces.

B. Protect glass from contact with contaminating substances resulting from construction operations, including weld splatter. If, despite such protection, contaminating substances do come into contact with glass, remove them immediately as recommended by glass manufacturer.

C. Examine glass surfaces adjacent to or below exterior concrete and other masonry surfaces at frequent intervals during construction, but not less than once a month, for build-up of dirt, scum, alkaline deposits, or stains; remove as recommended by glass manufacturer.

D. Remove and replace glass that is broken, chipped, cracked, abraded, or damaged in any way, including natural causes, accidents, and vandalism, during construction period.

E. Wash glass on both exposed surfaces in each area of Project not more than four days before date scheduled for inspections that establish date of Substantial Completion. Wash glass as recommended by glass manufacturer.

END OF SECTION 08 8000.01

SECTION 08 9100.01 - LOUVERS

PART 1 - GENERAL

Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.1 SECTION INCLUDES

A. This section includes the performance criteria, supply and installation of fixed louvers, bird screen, and structural support and attachment brackets for the project.

B. The work performed under this Section consists of the provision of all required materials, labor and equipment and the like necessary and/or required for the complete execution of all fixed louvers as required by this section, schedules, keynotes and drawings.

C. The louver to be furnished or supply includes the following:
   1. Standard fixed extruded aluminium weather louvers with bird screen; independent, Unitized and Stick alum. curtain wall integrated.
   2. Sandtrap louvers as indicated in the drawings.

D. Design to follow the architectural intent indicated in the drawings.

E. This section to be read in conjunction with the related wall types/ exterior wall system drawings submitted as a part of the facade tender package.

1.2 RELATED SECTIONS

A. This schedule is intended to be used as a helpful indication of the related sections within the Project Specifications. It is not necessarily comprehensive or complete and it is the Contractor's responsibility to ascertain all applicable sections required to understand the full scope of Works intended.

1.	Section 01 4500	-	Quality Control
2.	Section 03 3000	-	Cast-in-Place Concrete.
3.	Section 04 2200	-	Concrete Unit Masonry.
4.	Section 07 6200.01	-	Sheet Metal Flashing and Trim.
5.	Section 07 9200.01	-	Joint Sealants.
6.	Section 08 4000	-	Aluminium Works.
7.	Section 08 4413.01	-	Glazed Aluminium Curtain Wall.

1.3 DEFINATION

A. Louver Terminology: Definitions of terms for metal louvers contained in AMCA (Air Movement and Control Association) 501 apply to this Section unless otherwise defined in this Section or in referenced standards.

B. Drainable-Blade Louver: Louver with blades having gutters that collect water and drain it to channels in jambs and mullions, which carry it to bottom of unit and away from opening.

C. Horizontal Louver: Louver with horizontal blades (i.e., the axes of the blades are horizontal).

D. Vertical Louver: Louver with vertical blades (i.e., the axes of the blades are vertical).

E. Louver that provides specified wind-driven rain performance, as determined by testing according to AMCA 500-L.

1.4 REFERENCES

A. The minimum standards for products specified in this section shall be including as under but not limited to the following. Except as otherwise specified herein, perform work in accordance with specifications, codes and standards cited therein, and their latest editions, revisions, applicable addenda and supplements. Where there is conflict between the referenced standards the most stringent of the conditions/requirements shall be applicable.

B. Air Movement and Control Association International, Inc.
   1. AMCA 500-L-99 - Laboratory Methods of Testing Louvers for Rating.
   2. AMCA 501 - Application Manual for Louvers.

C. American Society for Testing and Materials.
   1. ASTM B209 - Standard specification for Aluminum and Aluminum Alloy Sheet and Plate.
   2. ASTM B211 - Standard specification for Aluminum & Aluminum Alloy bar, Rod and Wire.
   3. ASTM B221 - Standard specification for Aluminum & Aluminum Alloy Extruded bars, Rods, Wire, Profiles and Tubes.
   4. ASTM E90 - Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements.
   5. ASTM E84 - Standard Test Method for Surface Burning Characteristics of Building Materials

D. Architectural Aluminum Manufacturers Association
   1. AAMA 800 Voluntary Specifications and Test Methods for Sealants.
   2. AAMA 605.2 Voluntary Specifications for High Performance Organic Coatings on Aluminum Extrusions and Panels.
   3. AAMA TIR Metal Curtain Wall Fasteners.
   4. AAMA 612: Voluntary Specification, Performance Requirements, and Test Procedures for Combined Coatings of Anodic Oxide and Transparent Organic Coatings on Architectural Aluminum.
   5. AAMA 2604-05 Voluntary Specification, Performance Requirements And Test Procedures For High Performance Organic Coatings On Aluminum Extrusions And Panels.

E. The Aluminium Association Incorporated.
   1. Aluminum Standards and Data.
   2. Specifications and Guidelines for Aluminum Structures.

F. American Society of Civil Engineers.
   1. Minimum Design Loads for Buildings and Other Structures.

1.5 PERFORMANCE REQUIREMENTS

A. General: Engineer, fabricate, and install exterior louvers to withstand the effects of loads and stresses from wind and normal thermal movement without evidencing permanent deformation

of louver components including blades, frames, and supports; noise or metal fatigue caused by louver blade rattle or flutter; or permanent damage to fasteners and anchors.

B. Wind Loads: Provide systems, including anchorage, capable of withstanding wind-load design pressures calculated according to requirements of the local code or wind analysis report as applicable.
   1. Wind Loads: Refer to the available wind tunnel analysis or project wind report at its latest revision for the design loads.

C. Deflection of framing members:
   1. Deflection Normal to Wall Plane: Limited to 1/175 of clear span up to a length of 4.1m. Above 4.1m, the deflection is limited to 1/240+6.35. However, such deflection shall not influence the integrity of adjacent trade like bulkhead, wall connections etc.
   2. Deflection parallel to Wall Plane: Maximum allowable deflection for the louver blades to be 1/120 or 12.7mm across the weak axis, whichever is less.

D. Thermal Movements: Provide systems and its associated structure like anchoring structures, fasteners, frames etc. shall accommodate expansion and contraction movement due to surface temperature differential without causing buckling, failure of joint seals, excessive stress on structural elements, reduction of performance or other detrimental effects. Base engineering calculation on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.
   1. Temperature Change (Range): 70 deg. C, ambient; 100 deg. C material surfaces.
   2. Night time or lower range +10 deg. C.

E. Structural-Support Movement: Provide systems that accommodate structural movements including, but not limited to, sway and deflection.

F. Air-Performance, Water-Penetration, Air-Leakage and Wind-Driven Rain Ratings: Provide louvers complying with performance requirements indicated, as demonstrated by testing manufacturer’s stock units identical to those provided, except for length and width according to AMCA 500-L.

G. Seismic Performance: Louvers, including attachments to other construction, shall withstand the effects of earthquake motions as per local regulation, in accordance to the local code ASCE 7-10, as acceptable to the local authorities.

H. Seismic load – As per ASCE 7-16 as acceptable to the local authorities.

I. SMACNA Standard: Comply with recommendations in SMACNA's "Architectural Sheet Metal Manual" for fabrication, construction details, and installation procedures.

J. Delegated Design: Notwithstanding the drawings and other information provided, the Contractor shall be fully responsible for the design including comprehensive engineering analysis by an experienced façade professional engineer, detailing of the works, described herein, and shall provide a complete, and secure installation, which shall meet the specified performance requirements and the design criteria indicated in this specification.

K. Louver performance for all sub stations to comply with local requirements. Percentage free area as indicated in the drawings and confirmed by the Engineer.

1.6 SUBMITTALS:

A. Product Data: For each type of product indicated.

B. Shop drawings of louver units and accessories. Include plans, elevations, sections, and details showing profiles, angles, and spacing of louver blades; unit dimensions related to wall openings and construction; free areas for each size indicated; profiles of frames at jambs, heads, and sills; and anchorage details and locations.
   1. Show weep paths, gaskets, flashing, sealant, and other means of preventing water and sand intrusion.
   2. Show mullion profiles and locations.

C. Samples for initial selection in the form of manufacturer's color charts showing the full range of colors available for units with factory applied color finishes.

D. Samples for verification of each type of metal finish required, prepared on samples of same thickness and material indicated for final unit of Work. Where finishes involve normal color and texture variations, include sample sets showing the full range of variations expected.

E. Submit wind load and free air area calculations for review, including air performance and water penetration control certification from the Air Movement and Control Association (AMCA) or approved equivalent.

F. Product Test Reports: Based on evaluation of comprehensive tests performed according to AMCA 500-L by a qualified testing agency or by manufacturer and witnessed by a qualified testing agency, for each type of louver and showing compliance with performance requirements specified.

1.7 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as minimum.

B. For acceptance of any approved equal manufacturer and/or product, the Contractor shall comply with all requirements specified in Division 1 for the same and produce test certificates from UKAS or IAS accredited laboratory that the subject product is complying with the specified performance requirements. Test certificates from any other laboratories will not be acceptable.

C. Source Limitations: Obtain louvers and vents through one source from a single manufacturer where indicated to be of same type, design, or factory-applied color finish.

D. Single subcontract responsibility: Subcontract the work to a single firm that has had not less than five years experience in the design and manufacturing of work similar to that shown and required.

E. Performance Data: Provide AMCA and BSRIA test data as required to confirm that the louvers have the specified air and water performance characteristics.

F. Acoustical Performance (As applicable): Submit test reports to confirm that the louvres meet the specified STC and Noise Reduction requirements applicable for the project.

G. Fire Performance Characteristics (As applicable per local authority): Shall comply with ASTM E84 Class A/1 requirements.

H. Comply with most stringent of the local code or acceptable national/international building standard requirements for VOC’s limits.

I. Welding: Qualify procedures and personnel according to the following:
   1. AWS D1.2/D1.2M, "Structural Welding Code-Aluminum".

2. AWS D1.3/D1.3M, "Structural Welding Code-Sheet Steel".
3. AWS D1.6/D1.6M, "Structural Welding Code-Stainless Steel".

J. SMACNA (Sheet Metal and Air Conditioning Contractors' National Association) Standard: Comply with recommendations in SMACNA's "Architectural Sheet Metal Manual" for fabrication, construction details, and installation procedures.

K. Mockup: Prior to fabrication, erect mockups for each type of unit, configuration and finish required.

1. Locate mock-up in the location and of the size indicated or, if not indicated, as directed by the Engineer.
2. Mockup shall be complete with adjoining material and support system to the satisfaction of the Engineer. If the required standard is not achieved, the Contractor shall replace (at the contractor's cost) until the required standard is achieved as judged by the Engineer.
3. Obtain written approval of mock-up from the Engineer before starting fabrication.
4. Maintain mock-ups during construction in an undisturbed condition as a standard for judging the completed Work.
5. Mock up shall become part of finished work if approved by the Engineer at the time of Substantial Completion.

L. Professional Engineer Requirements: Drawings and structural calculations shall be signed and sealed by a professional Engineer licensed in place of the work.

1.8 DELIVERY, STORAGE AND HANDLING

A. Deliver louvers in the manufacturer's original protective packaging.

B. At the time of delivery all materials shall be visually inspected for damage. Any damaged boxes, crates, louver sections, etc. shall be noted on the receiving ticket and immediately reported to the shipping company and the material manufacturer.

C. Store louvers in a clean dry location and away from uncured masonry or concrete. Cover components with waterproof paper, tarpaulin or polyethylene sheeting in a manner to permit circulation of air.

D. If stored outdoors the material must be raised sufficiently off the ground to prevent it being flooded.

E. Where possible, deliver louvers as a component pre-installed at factory.

F. Provide removable protective covering over all aluminum to be exposed in the completed work.

G. Do not use adhesive papers or sprayed coatings which will become bonded when exposed to sunlight or weather.

H. Remove temporary protection after installation when acceptable to the Engineer. Do not leave coating residue on any surface.

I. Material shall be handled in accordance with sound material handling practices and in such a way as to minimize racking.

J. Louver sections should only be lifted and carried by the jambs. Heads, sills and blades are not to be used for lifting or hoisting louver sections.

1.9 PROJECT CONDITION

A. Field Measurements: Check actual louver openings by accurate field measurements before fabrication, and show recorded measurements on final shop drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

1.10 WARRANTY

A. Submit a Five (5) year written warranty against defects in design, materials, and workmanship. Warrant that work will be structurally sound and free from distortion and deformation under load.

B. Submit thermal setting coating manufacturer's warranty for (Twenty-Five) 25 years. Warrant that materials and finishes will not develop excessive fading or non-uniformity of color, and will not crack, peel, delaminate or otherwise deteriorate or corrode.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Subject to compliance with the requirements of the specifications of this Section, the following manufacturers are acceptable

1. Construction Specialties- ME/USA
2. Schuco, ME
3. Technal, ME
4. Gulf Extrusion, ME
5. White Aluminium Extrusion, ME
6. Emirates Extrusion, ME
7. Hunter Douglas Ltd- Luxalon- ME/UK
8. Quality Architectural Systems- ME

Or Equivalent, subjected to Engineer's acceptance.

Condition to qualify "equivalent": Qualification exercise consists of assessment of technical adaptability of the design and the material, quality procedures maintained, minimum 10 years of its usage in similar project and credibility of its performance record and other project references. All cost related to this exercise shall be borne by the proposer. The proposer shall agree to use above listed pre-approved system manufactures, should the proposed system is not acceptable to the Engineer, after the completion of qualification exercise.

2.2 MATERIALS

A. Aluminum: Conform to the requirements published in the Aluminum Association's "Aluminum Standards and Data", referenced ASTM standards and the following. All aluminum extrusions shall be manufactured to dimensional tolerances so as to eliminate any edge projection or misalignment at joints. Unless otherwise specified, provide alloy and temper as required to suit performance requirements and finishes indicated.
   1. Aluminium Sheet and Plate: ASTM B209 (ASTM B 209M), Alloy 1100, 3003 or 5005.

2. Aluminium Extruded Bars, Rods, Shapes and Tubes: ASTM B 221 (ASTM B 221M). Alloy 6063-T5, T6 OR 6061-T6 tempered.

B. Fasteners: Use types and sizes to suit unit installation conditions.
   1. Use tamper-resistant screws for exposed fasteners unless otherwise indicated.
   2. For fastening aluminum, use 300 series stainless-steel fasteners.
   3. For color-finished louvers, use fasteners with heads that match color of louvers.
   4. Do not use metals that are corrosive or incompatible with joined materials.

C. Bolts, screws, nuts, washers and other fasteners: Stainless steel of austenitic grade, 300 series for aluminum connections; galvanized steel elsewhere.

D. Anchors and Inserts of type, size, and material required for type of loading and installation indicated. Use nonferrous metal or hot-dip galvanized anchors and inserts for exterior installations and elsewhere as required for corrosion resistance. Use toothed steel or expansion bolt devices for drilled-in-place anchors.

E. Sealants installed in perimeter joints between louver frames and adjoining construction are as specified in Section 07 9200.

F. Joint backing, primers and cleaner materials: Recommended by the sealant manufacturer to suit the various substrates and conditions of the Work, compatible with each other and with sealant.

G. Acrylic based one component dielectric separation as protective coating, suitable for use in hot climate condition. Class A, according to ASTM E84-16.

H. Flashings, trims and close-off panels: 1.3mm thick sheet aluminum, prefinished to match louvers finish where exposed, mill finished otherwise.

I. Zinc rich primer (touch-up primer for galvanized steel): Zinc rich anti-corrosion coating containing 92-95% zinc in dried film providing electro-chemical protection on ferrous metals.

2.3 FABRICATION, GENERAL

A. Provide louver models, bird screens, structural supports and accessories as specified and/or shown on the drawings. Materials, sizes, depths, arrangements and material thickness to be as indicated or as required for optimal performance with respect to strength; durability; and uniform appearance.

B. Louvers to be mechanically assembled using stainless steel or aluminum fasteners.

C. Include supports, anchorage, and accessories required for complete assembly.

D. Assemble louvers in shop to minimize field splicing and assembly. Disassemble units as necessary for shipping and handling limitations. Clearly mark units for reassembly and coordinated installation.

E. Fabricate frames, including integral sills, to fit in openings of sizes indicated, with allowances made for fabrication and installation tolerances, adjoining material tolerances, and perimeter sealant joints.

F. Maintain equal louver blade spacing, including separation between blades and frames at head and sill, to produce uniform appearance.

G. Include supports, anchorages, and accessories required for complete assembly. Connections of anchorages and louvers should be developed invisibly where possible.

H. Provide sill extensions and loose sills made of same material as louvers where indicated or required for drainage to exterior and to prevent water penetrating to interior.

I. Where height of louver units exceeds fabrication and handling limitations, fabricate units to permit field-bolted assembly with close-fitting joints in jambs and mullions, reinforced with splice plates.

2.4 LOUVER MODELS

A. Type: High performance drainable blade type, fixed aluminium extruded weather louvers (Model: to be confirm by the Engineer) or as indicated on drawings.
   1. Material: Heads, sills and jambs to be one-piece structural aluminum members with integral caulking slot and retaining beads. Blades to be one piece aluminum extrusions with reinforcing bosses. Fixed Blades to be supported and lined up with heavy gauge extruded aluminum blade braces, positively interlocked to each blade and mechanically secured to aluminum structural. Material thickness to be as follows:
      a. Heads, sills and jambs: Minimum 2.06mm.
      b. Blades: Minimum 2.06mm.

   2. AMCA Performance: A 4’ x 4’ unit shall conform to the following and be licensed to bear the AMCA seal:

Free Area	-	(to be confirm by Engineer).
Free area velocity at the point of beginning water penetration	-	(to be confirm by Engineer).
Intake Pressure drop at the point of beginning water penetration	-	(to be confirm by Engineer).
Exhaust pressure drop at 1000 fpm free area velocity (305 m/min)	-	(to be confirm by Engineer).

2.5 BIRD SCREENS

A. All louvers to be furnished with stainless bird or insect screens.
   1. Finish & Color: As per approved sample.

B. Screens to be minimum 15.9mm mesh, minimum 1.27mm thick expanded and flattened stainless steel bird screen secured within minimum 1.40mm thick extruded aluminium frames. Frames to have mitered corners and corner locks.

C. Pressure drop to be coordinated with MEP as applicable.

2.6 ALUMINUM FINISHES

A. All aluminium visible surfaces exposed to the building interior and exterior: Aluminium surfaces shall be finished with Super Durable Powder Coating complying with AAMA 2604 and that are thermoset, free from VOCs, with minimum dry film thickness of 60 microns.
   1. Color and Finish: As indicated in the drawings and to match approved sample.

B. Prepare, pretreat and apply coating (including protective coat) to exposed metal surface to comply with AAMA 2604.

C. Concealed Aluminum Surfaces: Any aluminum sections which are completely concealed and therefore do not require the facing color shall be anodized to a minimum standard of AMMA 611 Class II.

D. Identification marks shall not appear on finished surfaces.

E. Adhesion: No removal of finish after 1.5mm cross hatching to base metal, impacting to the point of metal rupture, and subjecting to application and quick removal of cellophane tape.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of glazed curtain wall system.

B. Do not proceed with installation until unsatisfactory conditions have been corrected or accommodations acceptable.

3.2 PREPARATION

A. Coordinate setting drawings, diagrams, templates, instructions, and directions for installation of anchorages that are to be embedded in concrete or masonry construction. Coordinate delivery of such items to Project site.

B. Remove dust and other loose material from openings.

3.3 INSTALLATION

A. Locate and place louver units plumb, level, and at indicated alignment with adjacent work.

B. Use concealed anchorages where possible. Provide brass or lead washers fitted to screws where required to protect metal surfaces and to make a weather tight connection.

C. Form closely fitted joints with exposed connections accurately located and secured.

D. Provide perimeter reveals and openings of uniform width for sealants and joint fillers, as indicated.

E. Where aluminum will contact dissimilar metals, protect against galvanic action by painting contact surfaces with primer or by applying sealant or tape recommended by manufacturer for this purpose. Where aluminum will contact concrete or masonry, protect against corrosion by painting contact surfaces with acrylic based protective paint/coating.

F. Install concealed gaskets, flashings, joint fillers, and insulation, as louver installation progresses, where required to make louver joints weather tight. Comply with Division 07 Section "Joint Sealants" for sealants applied during installation of louver.

G. Repair finishes damaged by cutting, welding, soldering, and grinding operations required for fitting and jointing. Restore finishes so there is no evidence of corrective work. Return items that cannot be refinished in the field to the shop, make required alterations, and refinish entire unit, or provide new units.

3.4 TOLERANCES

A. Erection tolerances for frame assemblies relate to the structural grid of the building, and apply to each individual assembly as follows:
   1. Vertical position: $\pm$ 2 mm.
   2. Horizontal Position: $\pm$ 2 mm.
   3. Racking in elevation: 1 mm.
   4. Deviation from true plumb over full height: maximum 3 mm.
   5. Deviation from true straight in plane over full length of each building face: maximum 3 mm.

B. Tolerances shall not be cumulative.

3.5 ADJUSTMENT AND PROTECTION

A. Protect louvers from damage of any kind during construction period including use of temporary protective coverings where needed and approved by louver manufacturer. Remove protective covering at time of Substantial Completion.

B. Restore louvers damaged during installation and construction period, so that no evidence remains of correction work. If results of restoration are unsuccessful, as judged, remove damaged units and replace with new units.

C. Clean and touch up minor abrasions in finishes with air dried coating that matches color and gloss of, and is compatible with, factory applied finish coating.

3.6 CLEANING

A. Periodically clean exposed surfaces of louvers and vents that are not protected by temporary covering to remove fingerprints and soil during construction period. Do not let soil accumulate until final cleaning.

B. Before final inspection, clean exposed surfaces with water and a mild soap or detergent not harmful to finishes. Rinse surfaces thoroughly and dry.

END OF SECTION 08 9100.01

SECTION 11 8123.01 - FAÇADE CLEANING AND MAINTENANCE EQUIPMENTS

PART 1 - GENERAL

Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.01 SECTION INCLUDES

A. This section includes design, engineer, fabricate, supply, installation, testing, commissioning and maintenance of a suitable, workable and practical Façade Access Systems for the following areas/buildings; fully in compliance with the requirement of the relevant parts of specifications, contract drawings and all acceptable codes and regulations as detailed elsewhere in this section.
   1. Tower H32 façade, main entrance, event/ entertainment box and cylindrical lift glazing/ façade. Including, Skylight and standing seam roof.
   2. Building H30 and H27 façade, ground floor and entrance glazing/ façade.
   3. Free standing shading canopy cladding and glazing at ground floor level.
   4. For cleaning and maintenance strategy, refer to the BMU Zoning/ illustration, at the end of this specification.
   5. This section to be read in conjunction with the related BMU drawings submitted as a part of the tender package.

B. Delegated Design: The design responsibility of the comprehensive Facade Access Systems, including engineering analysis by a qualified engineer, using the performance requirements and design criteria indicated shall remain the sole responsibility of the Contractor.

C. All anchors, fixings, brackets, angles, attachments, electrical points, electrical wiring, water points, and water pipes, as required for a complete installation.

D. Coordination with work of other trades including setting out, and making penetrations for mechanical, electrical and other services

E. Shop drawings, calculations, engineering data, test reports, and including all necessary submissions and approvals required by the Employer / Engineer or with relevant authorities but not limited to Civil Defence / Municipality / Any relevant local Authority. The Façade Access Systems Sub-Contractor shall be responsible for providing any drawings, information, test results and calculations necessary to obtain all Local and Statutory Authority approvals.

F. Scheduling and monitoring of works and samples and mock-ups.

G. Testing and verification of design, components and total assembly. The façade access equipment Sub-Contractor shall include the payment of all government charges required for inspection permits, approval certificates, and other checking and certification costs to be paid to the local authority for the testing, certification and approval of each of the pieces of Facade Access Equipments.

H. CE certification to be provided for approval and relevant documents, prior to handover.

I. Certificate of origin to be provided for the following items and the Façade Access Systems sub contractor to take approval from the Engineer before proceeding for production.

   1. Electrical control panel wiring

2. Wire ropes
3. All Motors
4. Gear boxes
5. Hydraulic units
6. All major structural steel members

J. All materials, products, components and systems and the work executed with them are to comply with all applicable regulations and requirements. Where appropriate, works shall comply with requirements by Statutory Regulatory Inspectors / Façade Access Engineer's satisfaction.

K. The specification is to be read in conjunction with the tender drawings and the section "Cleaning and Maintenance Strategy" of document "Façade Tender Document Report at its latest revision.

1.02 RELATED SECTIONS

A. This schedule is intended to be used as a helpful indication of the related sections within the Project Specifications. It is not necessarily comprehensive or complete and it is the Contractor's responsibility to ascertain all applicable sections required to understand the full scope of Works intended.

B. Work to be coordinated and used in conjunction with this specification includes but is not restricted to:

1.	Section 01 4500	-	Quality Control
2.	Section 03 3000	-	Cast-in-Place Concrete
3.	Section 05 5000	-	Metal Fabrication
4.	Section 05 1200	-	Structural Steel Framing
5.	Section 05 1213	-	Architecturally Exposed Structural Steel
6.	Section 05 7500.01	-	Decorative Formed Metal
7.	Section 07 2100	-	Building Insulation
8.	Section 07 2500	-	Weather Barrier
9.	Section 07 6113.01	-	Standing Seam Sheet Metal Roofing
10.	Section 07 9200.01	-	Joint Sealants
11.	Section 08 8000.01	-	Glazing
12.	Section 08 4413.01	-	Glazed Aluminium Curtain Wall
13.	Section 09 9100	-	Painting.

1.03 GENERAL SCOPE OF WORK

A. Plan, supply, install and maintain all Façade Access Systems hereinafter called the BMU Systems, (for routine cleaning, maintenance of the building envelope. The Works to be carried out under this Section shall comprise the arrangement, engineering, submission of shop drawings, testing, fabrication, transportation, coordination, assembly, installation and testing and commissioning of all elements required for the successful operation of the BMU systems.

B. BMU system(s) equipment shall be new; factory fabricated and tested, robust construction and shall be fit for purpose in outdoor use in the place of Work and manufactured to relevant Standards stated herein.

C. The complete BMU installation shall comply with this specification and satisfy the testing and inspection by authorities having jurisdiction on the subject installation. Completion of the BMU work is subject to the Engineer's approval and on acceptance of the overall performance of the BMU and their complete installation as per the Contract requirements.

D. This system shall be designed and engineered to meet the requirements of the applicable standards, codes and related rules and to provide safe façade cleaning and maintenance access. All systems including electronic, electrical, hydraulic and mechanical components shall be suitable for use in service conditions at the place of the Work.

E. All BMU systems shall be designed by contractor in coordination with the structural steel specialist to the approval of the Engineer. Design shall be in accordance with all codes mentioned in this specification as well with the design parameters provided in the drawings and specification but not limited to weight, carrying weight, path, distance from the façade and structural elements, load, connection points, speed.

F. Work of this Section shall meet the requirements of governing codes and authorities having jurisdiction and shall comply with properly engineered loading and safety criteria for the intended use.

G. Coordinate design of system to ensure and demonstrate that all aspects of the BMU works are coordinated with adjacent interfacing works as acceptable to Engineer, and ensure that loadings imposed by the BMU system(s) while operational and/or stationary shall in no way cause failure and/or deterioration to the building's finishing materials. Co-ordination with other sections of work for the following but not limited to:
   1. Placement of the anchor plate (on the concrete upstand) for BMU tracks supports.
   2. Loading and location of bracket / support structure for BMU tracks.
   3. Civil / structural steel / water proofing systems.
   4. Facade systems, includes restraint socket locations.
   5. Power and Water point locations for the complete system.
   6. Clear passage for the movement of cleaning devices.

H. Include for supports to carry all structural reaction, impact and uplift loads imposed by the equipments.

I. Electrical and mechanical installation in association with the system.

1.04 DESIGN CRITERIA

A. The BMU systems shall be able to access all parts of the building envelope below the highest element of the buildings, including recesses, soffits/ceiling and architectural features etc., for cleaning works along with light façade rendering maintenance and glass / façade panel replacements wherein it could not be done from inside or within the building.
   1. Equipped with GRU (glass replacement unit) for lifting the glass panels; contractors' responsibility to coordinate and ensure the required GRU capacity and design.

B. Facade cleaning system shall be designed to withstand 45 km/h wind velocities while being used for normal operations, shall be fully operational at wind velocities up to 80 km/h, and shall withstand 160 km/h wind velocity, or that indicated on wind pressure study whichever is more stringent when in their secured stored positions. Refer to design loads and wind pressure studies or wind tunnel reports, as applicable.

C. The exposed areas subjected to wind pressure shall be the total areas of all portions of the exposed parts with no shielding effect of one element by another where the distance between elements is four times or more than the smaller projected area of the windward element.

D. Strictly comply with National and International Codes such as EN1808.

E. The BMU system are required to be designed for a 25 year life and so as to require an absolute minimum of maintenance.

F. The strategy must enable the operative staff to be tied safely to safety lines and safety anchor points at all times. All machine driven equipment must have safety features to avoid collision or lose of control even in case of power failure or mechanical failure. Manual drive must be allowed for all equipment and these shall allow for the operative staff to descend safely.

G. All structural members and assemblies shall be designed and certified with a minimum 4:1 safety factor based on ultimate strength and normal operating conditions unless greater requirements are indicated or specified. All stresses and deflections are limited in accordance with governing codes and regulations. Structural calculations to be prepared and certified by a Licensed Professional Engineer registered in or Equivalent as per EN1808.

H. Design suspension cables/ropes with a minimum 12:1 safety factor.

I. Locks, braking or other safety devices shall be provided for securing the equipments.

J. Locate anchorages to suit suspension equipment that will be used on the building with respect to items such as reach, rigging, spacing, roof edge condition, Lower floor clearances and similar items.

K. Design all anchor components to provide adequate attachment to the building and suited to current window cleaning/suspended maintenance practices. Ensure compatibility with industry standard equipment.

L. Ensure all anchor components conform to proper engineering principles and have been designed and engineered by a professional Engineer licensed/registered in the place of Work, qualified in the design of window cleaning/suspended maintenance equipment, its application, and safety requirements.

M. Design and engineer building maintenance unit structural steel supports, related work necessary to provide a complete installation. Ensure that the attachment to the building imparted to load carrying members is within allowable limits and that no member is overstressed. Loads imposed on the structure shall be submitted to the Engineer for review.

N. BMU Restraint Anchor will be required on all external façade. Restraints are required at intermittent levels (i.e., at every 20meters, first at 40meters from ground level) and spaced a maximum of 2.0 – 3.0 meters apart horizontally. (for levels verify tender drawings)
   1. Pull-out design load of minimum 1kN, if not specified.
   2. Lateral design load 1kN, if not specified, in any direction parallel to the façade
   3. Loads above may be concurrent.
   4. Above stated loading and forces to be further coordinated, provided and confirm by the BMU supplier as per the system design in-line with EN 1808.

O. Fall arrest/ safety line system or access rail system, anchors and equipment supports to comply with the requirements as per BS EN 795 and BS 7883.

P. All the structural steel or aluminum parts constantly exposed to weather shall be PVDF coated or hot dip galvanized and painted to comply with corrosive marine environment requirements. All coatings / paintings to be warranted for a minimum of 10 years from coating / painting Manufacturer.

Q. Provide assemblies to allow for thermal movements resulting from the following maximum change (range) in ambient and surface temperatures. Base engineering calculations on surface temperatures of materials due to both solar heat gain and nighttime heat lost.
   1. Temperature Change (Range): 70°C, ambient; 100°C, material surfaces.
   2. Night time or lower range +10 deg. C

R. Electrical Requirements:
   1. The electrical equipment shall be watertight and suitable for Marine environment at Hot-aired Desert climates.
   2. Cable glands shall be provided to ensure sealed passage is obtained.
   3. All hanging / loose cables to be provided with cable grips or restrainers or socks to avoid any damage.
   4. The BMU, tracks, other permanent metal features shall be earthed to protect against lightning damage.
   5. A sealed box control box positioned on the platform / carriage encloses the motor contacts/hydraulic circuits used to operate various movements using push buttons: Lifting, lowering, translation, tilting, etc. as applicable.
   6. The equipment shall be protected against overloads and short-circuits in compliance with the standards in force.
   7. Motors shall be protected against overloads and short circuits in compliance with current standards (fuses, thermo magnetic shunts). The control and monitoring system shall be supplied at maximum of 48 volts
   8. All electrical equipment shall have phase protection.
   9. All terminals shall be identified with the same markings as shown on the electrical diagrams.

S. All apparatus and equipment shall be so designed and constructed that they operate satisfactorily and without any deleterious effect for prolonged and continuous periods in the applicable climatic conditions and ambient temperature conditions.

1.05 SUBMITTALS

A. All Technical Submittal, Material Submittal, certificates and calculations shall bear the stamp and signature of a professional Engineer licensed/registered to design structures in the place of work.

B. Provide a Detailed Technical Submittal, Material Submittal and Shop Drawings: Comprehensive shop drawings showing complete layout and configuration with all necessary detailed dimensions of the complete Façade/window cleaning and maintenance BMU system including all components and accessories. Shop drawings shall include installation and rigging instructions and all necessary Restrictive and Non-Restrictive Working Usage Notes and General Safety Notes, and the extent of all surfaces to be cleaned by each BMU system. Detailed drawings shall show all components of each BMU system mentioned above but not limited to:
   1. Provide setting drawings, templates, and directions for installation of anchor bolts and other anchorages to be installed by others.
   2. Show assembly, including layout of roof top system.
   3. Principal parts of the systems.
   4. Reactions at points of support.
   5. Weights of principal components.
   6. Location of circuit breaker, switchboard panel or disconnect switch.
   7. Control functions and operational description.
   8. Dimensions of each unit.
   9. Compliance statement comparing project specification and proposed by the supplier.
   10. Inspection Test Plan / Quality Control Plan-in line with this complete specification.
   11. Design guarantee certificate (Professional Indemnity).

12. Guarantee letter confirming the access to the entire proposed external envelope.
13. All required detailed shop drawings related to façade access equipments.

C. Product Data: Manufacturer's published catalogue containing physical properties and characteristics of window washing equipment, with all modifications and options to be provided on this Project identified, as well as installation instructions and recommendations.
   1. Extent of information and samples required shall be sufficient to demonstrate to the Engineer and Employer that the proposed system and materials are in compliance with the Contract requirements. However, the Engineer can require the Contractor to submit any additional or supplementary information to substantiate the performance of the proposed system or material.

D. Calculations: Calculations shall be submitted to prove the performance and structural design of the BMU systems proposed. Manufacturer's catalogues, test certificates, and installation method statements, material samples of all relevant materials shall be submitted to the Engineer.

E. Provide method statement for installation & erection of complete façade access system and other attachments. Do not start installation unless the contractor reviews the method statement.
   1. Method statement must include the following details and clarify each description:
      a) Setting out & control points.
      b) Cross reference of identification of elements.
      c) Details of equipment necessary to lift elements and components onto the building and to hold line and level them once they have been positioned on the building.
      d) Details of points, which are susceptible to damage or should not be loaded during installation.
      e) Details of isolation & packing.
      f) The tolerance to be achieved in location of equipment.
      g) All fixing requiring specific torques.
      h) Detail of protection.

F. Operation and Maintenance Data: Submit complete operating and maintenance data, including parts list.

G. Submit test data from independent qualified testing laboratory indicating conformance with Specifications of all anchors which shall be used on the Project.

H. Provide proof of manufacturer's specific Product liability insurance (products and completed operations) for all aspects of the design and installation of all components and against product/system failure.

I. Letter of Conformance: Submit letter of conformance.

J. CE qualified certification.

1.06 CODES AND STANDARDS

A. Design, material, fabrication, examination, testing, inspection, certification, documentation and operation shall conform to the local regulations, British, Euro-code, American or equivalent standards, OSHA standards and specifications as applicable.

1.07 REFERENCES

A. The Facade Access design shall comply with international codes, good industry practice guides and local authority requirements. Compliance with European Standards for access equipments (EN 1808) and compliance with British Standards on steel and aluminium engineering are jointing is required. Comply with the following standards and regulations:

1. ANSI/IWCA 1-14.1: Window Cleaning Safety Standard.
2. ANSI Z359.1: Safety requirements for personal fall arrest systems, subsystems and components.
3. AWS D1.2/D1.2M: Structural Welding Code-Aluminum; 2003.
4. AWS D1.1/D1.1M: Structural Welding Code-Steel; 2006.
5. EN 1808: Safety Requirements on Suspended Access Equipment.
6. EN 795: Protection against falls from a height: Anchor Devices-Requirements and testing.
7. ASTM A276: Standard Specification for Stainless Steel Bars and Shapes, 2006.
8. ASTM A492: Standard Specification for Stainless Steel Rope, 2004.
9. ASTM B 221: Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes; 2006.
10. ASTM A123/A123M-02: Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products; 2002.
11. EN 982: Safety requirements for fluid power systems and their components-Hydraulics.
12. BS 6037: Code of Practice for Permanently Installed Suspended Access Equipment.
13. BS 2830: Suspended access equipment (suspended chairs, traditional steeplejack's seats, work cages, cradles and platforms) for use in the building, engineering construction, steeple jack and cleaning industries.
14. BS 3032: High Tensile Steel Shackles.
15. BS 5845: Specification for permanent anchors for industrial safety belts and harnesses.
16. BS 5950: Steelwork Design.
17. BS 6858: Manually operated positioning devices and associated anchorage lines for use with industrial safety belts and harnesses.
18. BS 7883: Code of practice for application and use of anchor devices conforming to EN 795.
19. BS 8118: Structural Use of Aluminium.
20. BS EN 353: Personal protective equipment against falls from a height: Guided fall arrester.
21. BS EN 354-365: Specification for industrial safety belts, harnesses and safety lanyards.
22. BS EN 360: Personal protective equipment against falls from a height: Retractable type fall arrester.
23. BS EN 361: Personal protective equipment against falls from a height: Full body harnesses.

B. The minimum standards for products specified in this section shall be including as under but not limited to the following. Except as otherwise specified herein, perform work in accordance with specifications, codes and standards cited therein, and their latest editions, revisions, applicable addenda and supplements. Where there is conflict between the referenced standards the most stringent of the conditions/requirements shall be applicable.

C. All updates and amendments of applicable Standards and Regulations current at the date of tender return are to be complied with. Where a standard or regulation is updated during the course of the works, the BMU Sub-Contractor shall notify the Engineers, formally in writing.

1.08 QUALITY ASSURANCE

A. Manufacturer: Work of this Section shall be executed by a qualified firm specializing in the design, fabrication and installation of window cleaning/suspended maintenance systems similar in quality

and scope to that required on this Project having a minimum of 15 years documented production and installation experience.

1. Manufacturer must be able to show employment of experienced personnel, physical facilities, established quality control procedures, management capability sufficient to produce system without causing Project delay.
2. Manufacturer shall submit list of at least three representative systems three years old or older, with date of installation and Engineer's and building Employer's names, addresses and phone numbers.

B. Installer Qualifications: Engage an experienced Installer with minimum five years' experience in installation of systems in Middle East similar to this Project and who is an authorized representative of the equipment manufacturer for both installation and maintenance of the type of units required for this Project, and whose installations have resulted in construction with a record of successful in-service performance.
   1. Welder: Submit for Engineer's information, evidence in compliance with AWS D1.1, that each welder has satisfactorily passed qualification tests for welding processes involved and has undergone re-certification if welder has not welded for more than six months.

C. Quality Control: Contractor shall establish and maintain quality control to ensure compliance with Contract requirements.
   1. Maintain records of quality control for all construction operations required under this Section. Furnish copy of these records, as well as corrective action taken, to Employer at time of issue of taking over certificate. Record shall include welder AWS certification.
   2. Provide Employer's personnel instruction in the operation and maintenance of window washing equipment.
   3. All welds shall be visually inspected by an accredited test laboratory. A record of welders responsible for major welds both on and off-site shall be kept on site. Certification shall be provided for all welders following requirements stipulated in BS1295 and if specified, carry out approval tests to BS 4871: Pt.1 or BS 4872: Pt. 1 as applicable.
      a. Non destructive testing:
         1) Full penetration butt welds
            a) Ultrasonic or Radiographic test, 100% of all butt welds.
         2) Fillet weld
            a) Visual inspection, 100% of the primary site welds and 25% of factory welds or secondary site welds.
            b) Ultrasonic or Radiographic test, 25% of the primary site welds and 10% of factory welds or secondary site welds
         3) Approved methods of non-destructive testing:
            a) Ultrasonic test as per ANSI D1-1/6/F
            b) Radiographic testing as per ASTM E94 and ASTM E747.
            c) Magnetic Particle testing as per ASTM E709.
            d) Penetration testing as per ASTM E165.
         4) Destructive tests on samples:
            a) Traverse tensile test as per BS 18
            b) Traverse bend test as per BS 1639
            c) Side Bend Tests
            d) Macro Examination
         5) Acceptance criteria: as per ANSI D1-1/6/C

D. All tests shall be recorded for references. Records shall include location of welds test samples, types of test, date of test, inspection or testing personnel's information, and test result. The test records shall be submitted for records / make available for inspections.

E. All welds that fail the tests shall be rectified and re-tested If this is not possible, the metal section shall be partially or entirely replaced at no additional cost. The corrective actions shall be proposed, approved, executed and documented for further records.

F. In the event that a failure is observed, an additional 10 weld locations (locations to be confirmed by Employer / Engineer's Site Representative / Inspector) to be tested at the Façade Access Sub-Contractor's own cost.

G. Factory test of manufactured equipments: The tests that shall take place before shipping/ site delivery, should include minimum proof load test, impact resistance test, etc. These tests shall be repeated if any change or modification in the manufacturing process or materials takes place prior to dispatch from factory.

H. Anchors and Embeds Site Pull-out Testing: Testing to be carried out in accordance with ASTM E488, on site, by personnel from independent testing laboratory. During embed testing, test samples shall be loaded based on their final installed bracket group configuration, where applicable. Where the ratio of shear-to-tensile force or tensile-to-shear force ratio falls in excess of 1:8, incline testing setup will be required for the proposed bracket/embed configuration.

I. Non destructive field testing: Minimum 2% or 10 numbers of each type of bracket load configuration but at least 20 randomly selected out-of tolerance looking pieces to be tested on site. Should there be any non-compliance of test results, additional tests shall be performed per Engineer's requirement.
   1. 1.5x design load: No permanent deformation of the anchor or chips and failure of the substrate material.
   2. 2x design load: No failure of the anchor, no chips or failure of the substrate material.
   3. 3x design load: No failure of the structural substrate material.
   4. Failure: (only for laboratory samples) apply load until structural failure of either the anchor/embed or the surrounding substrate material.

J. Report to be submitted illustrated with pictures. No anchors used for testing shall be used on project installation.

K. Where requirements of governing codes, regulations, laws, and rules required by authorities having jurisdiction conflict with these Specifications and are mandatory, they shall be followed the same as if fully specified herein.

1.09 OPERATION AND MAINTENANCE DATA

A. Submit 3 sets of operation and maintenance data. Include description of the system, method of operation and control, motor control system, and special or non-standard features provided as per EN1808. Including in these manuals a detailed rescue plan.

B. Provide parts catalogues with complete list of equipment, replacement parts with equipment description and identifying numbers. Provide legible schematic wiring diagrams covering electrical equipment installed, including changes made in final work as per EN1808.

C. Provide proof of manufacturers' specific liability insurance (product and completed operations) for all aspects of the design and installation of all component and against product/system failure.

1.010 SPARE PARTS, TOOLS AND ACCESSORIES

A. Supply two new complete sets of tools complete with lockable tool boxes for the proper operation, routine maintenance and adjustments of all components and accessories.

B. All spare parts, tools and maintenance equipment shall be provided in original packing, clearly labeled and referenced. Provide complete schedules of spare parts and maintenance equipment clearly cross-referenced with the 'as-built' record drawings and with operation and maintenance instruction manuals.

C. All spare parts, tools and maintenance equipment shall be as recommended by the equipment manufacturer and shall be subject to the final approval of the Engineer.

1.011 WARRANTY

A. Upon completion of the BMU system handover to Employer, the BMU Sub Contractor shall provide a warranty for a period of 1 year (from date of Project Practical Completion Date). The warranty signed & countersigned by the supplier / Installer and the contractor to be provided for the following:
   1. Comprehensive Warranty Maintenance for the complete system-02 Year.
   2. Structural Warranty for the complete system, Repair & replacement of defects in materials and workmanship-10 Years.
   3. Availability of spare parts for a minimum of-10 Years.

B. During warranty period, any component found defective due to faulty workmanship or manufacturing defect should be replaced free of cost to the Employer. Components found damaged due to misuse or incorrect handling or worn out due to normal wear & tear should be replaced at extra charges. The standard list showing prices of all Spare parts to be submitted in advance for approval.
   1. Any defects occurring during the stated warranty period shall be made to the same quality as that required in the Technical Specifications by the BMU Sub-Contractor with no additional cost.
   2. The supplier shall provide servicing within 3 working days during the warranty period. The supplier shall provide free of cost regular maintenance inspections during this period, every six months.
   3. Spare parts shall be available within maximum 8 working days including delivery for a period of 10 years.
   4. All maintenance works shall be conducted by BMU's supplier's direct staff.

C. Warranty date starts from the practical completion date.

1.012 DEFECT LIABILITY PERIOD

A. Provide Employer with 24 months maintenance Defect Liability period, at no additional cost, commencing from the issue of taking over certificate.

PART 2 - PRODUCTS

2.01 ACCEPTABLE MANUFACTURERS

A. The system shall be from a single source overseas firm which has at least 15 years proven track record in the type of work required by this section and which is acceptable to the manufacturers of the primary materials.

B. For Roof Car Twin Track BMU System and Davits System:
   1. Transwill Engineering L.L.C, UAE
   2. XS Platforms, UAE.
   3. Or Equivalent.

C. Condition to qualify “equivalent”: Qualification exercise consists of assessment of technical adaptability of the design and the material, quality procedures maintained, minimum 10 years of its usage in similar project and credibility of its performance record and other project references. All cost related to this exercise shall be borne by the proposer. The proposer shall agree to use above listed pre-approved system manufactures, should the proposed system is not acceptable to the Engineer, after the completion of qualification exercise.

D. The installer shall be a firm of established reputation, which has been approved (sole within the region), trained and Authorized by the manufacturer for the proper installation of the specified equipment.

2.02 FABRICATION REQUIREMENT-GENERAL

A. Components shall be constructed of heat-treated aluminum alloy, stainless steel, or hot-dip galvanized structural steel. Dissimilar metals, when used, shall be protected against electrolytic action. All connectors shall be Type 316 stainless steel, or stainless steel with equivalent or greater corrosion protection and strength, unless otherwise noted. Certified welders shall perform all welding work and they shall examine all welds by non-destructive testing.

B. Components in contact with façades and the platform casters shall be non-marking and scuff resistant material.

C. All carbon steel assemblies and components shall be hot-dip galvanized, 610 g/m² coating, in accordance with ASTM A123 and ASTM A153. Galvanize after fabrication. Clean and touch-up steel surfaces with 3 coats of zinc rich primer where burned by field welding or where damaged. In addition, steel shall be finished where exposed to view as specified herein.

D. Exterior finish of all roof carriages and equipment and all other factory painted assemblies shall be weather, UV and salt resistant, exterior quality machinery enamel; colors as directed by the Engineer.

E. Stainless steel assemblies shall be of natural color; mill finish.

F. All hardware, clips, bolts, nuts, washers, etc. shall be stainless steel, Type 316.

G. All electrical equipment/controls shall have phase protection and environmental protection.

2.03 DESCRIPTION OF EQUIPMENTS

A. The project is divided into various zones to explain the breakup of systems used to cover the facades for cleaning as well as occasional maintenance. For zone details, refer to the Subsection- “3.09 REFERENCE SKETCH” at the end of this specification.

B. The equipment’s should not be limited to:

   1. H-32 MAIN TOWER FAÇADE [Refer to sketch/ illustration attached at the end of this specification]

a. For cleaning and maintenance – Centralized fixed mast and 3~4 stage telescopic jib BMU on pedestal.
   1) Machine/Jib function to access all areas and elevations of the façade.
   2) Compact as possible and should not be visible in parking position, contractor to coordinate and agree with the Engineer for the same.
   3) Integrated with the GRU for the required capacity.
   4) Design to allow for smooth operation, and clear cradle movement through the roof trellis/ shading and structure.
   5) Coordinate with MEP and structures for pedestal/ platform and electrical requirement respectively.
   6) Refer to the latest BMU DD/TD package for layout and details.
   7) For Tower Podium and Event/ Entertainment Box façade cleaning and maintenance strategy/ equipment's, refer to the zoning illustration at the end of this specification.

2. H30 Building Façade, for both the buildings [Refer to sketch/ illustration attached at the end of this specification].
   a. For cleaning and ~~minor~~ maintenance – Customized heavy duty monorail system capable of carrying 2-men pantograph power cradle for the cleaning and maintenance of the recessed façade and cladding features, integrated with GRU.
      1) Not visible in parking position.
      2) Refer to the latest BMU DD/TD package for layout and details.
3. H27 Building Façade, for all three building [Refer to sketch/ illustration attached at the end of this specification].
   a. For cleaning – Standard monorail system capable of carrying 2-men extended platform type power cradle for the cleaning of the recessed façade and cladding features.
   b. For glass replacement, provide temporary adjustable bracket and beam arrangement for roof trellis/ shading periphery primary steel, to lift the glass. Contractor to ensure structural coordination with the Engineer.
   c. Not visible in parking position.
   d. Refer to the latest BMU DD/TD package for layout and details

2.04 RESTRAINT SOCKET SYSTEM

A. SS 316-Intermittent tie-back system is required to prevent the suspended cradle from swinging. Spacing's are to be as per EN 1808.

B. Any restraints and brackets to be mounted directly on façade shall be provided by BMU Sub Contractor and installed by the Façade Contractor. The Façade Sub-contractor shall design connections of the BMU socket and carry out installation of the restraint plates for the cradle wire rope tiebacks.

C. Testing of the restraint system shall be carried out by the Façade Sub-contractor. The complete systems are to be designed all as per EN 1808.

D. Provide tie-in lanyards for intermittent stabilization of each of the platform's suspension ropes complete with stainless steel devices for attachment to the building anchors.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Examine surfaces and areas upon which the work of this Section depends, verify site dimensions. Report defects of work prepared by other trades and other unsatisfactory site conditions, which would cause defective installation of products, or cause latent defects in workmanship and function. Commencement of work shall imply acceptance of prepared work. Do not proceed with installation of equipment until unsatisfactory conditions have been corrected.

3.02 PREPARATION

A. Provide templates for anchor bolts and other items encased in concrete or below finished surfaces in sufficient time so as not to delay the Work.

3.03 DELIVERY, STORAGE AND HANDELING

A. Deliver materials in manufacturer's original, unopened, undamaged containers with identification labels intact. Store materials protected from exposure to harmful weather conditions and at temperature and humidity conditions recommended by material manufacturer.

B. The BMU Contractor has full responsibility to deliver the equipment, materials, attachments, documentation, etc to site without any damage, in appropriate packaging that protects from impact and harmful weather.

C. Storage location and conditions shall be coordinated with the main contractor. Provide protective barriers and aim to minimize possible accidental damage.

3.04 INSTALLATION

A. The BMU Sub-contractor shall undertake a survey of the preceding Main Contractor, Façade or other related Sub-contractor’s works including the concrete base structure, plinths, slabs, steel structure, embeds, interfacing of the primary brackets (such as position, dimensions and suitability of any work) and other relevant systems.

B. Report defects of works by other trades and other unsatisfactory site conditions, which would cause defective installation of products, or cause latent defects in workmanship and function. Inspect the provided electric power and water points, record any discrepancies and coordinate for rectification if such required.

C. The BMU Sub-contractor shall coordinate with the Main Contractor if such works are wrongly installed or otherwise not suitable. The Main Contractor shall propose methods of rectification works where necessary to ensure timely delivery of installation. Deviations shall be recorded in the form of drawings.

D. Commencement of work shall imply acceptance of prepared work.

E. The BMU Contractor shall have no claim or right of action against the Employer / Engineer / Main contractor arising from work executed or to be executed by others.

F. Install equipment in accordance with approved shop drawings and manufacturer’s instructions under supervision of qualified structural Engineer, licensed/registered at the Place of the Work.

G. Site welding is not allowed. Use pre-fabricated brackets and steel support work and bolted connections, serrated washers, nylon lock Nuts, etc. for site installation.

H. Coordinate with electrical power receptacle to ensure compatibility with BMU system requirements. Power points shall be located in accordance with the Drawings as appropriate but not greater than every 20~30 meters. Outlets shall experience no more than 3% voltage drop under full load current.

I. Water points shall be provided in accordance with the Drawings as appropriate but not greater than every 20~30 meters.

3.05 FIELD QUALITY CONTROL

A. Verify installed products are installed correctly and function properly; Check the Torque, etc as per approved quality control plan or inspection test plan and adjust to ensure satisfactory operation.

B. Conduct full live load and operational tests, after completion of the installation, under maximum design live loading conditions over a selected range of the building surfaces, in accordance with applicable standards as per approved quality control plan or inspection test plan.

C. The manufacturer shall make available to the user the following information on one or more durable signs or plates mounted on the inward side of the platform.
   1. Name and address of the manufacturer
   2. Designation of series or type Serial number (if existing)
   3. Year of construction.
   4. Rated load of the platform and the maximum number of persons

D. At a time mutually agreeable to all parties, allow 1 full day to conduct operational demonstrations for the Employer and/or the Employer's representative, after completion of the operational tests.

E. Repair or replace any components and correct all deficiencies observed as a result of these tests and demonstrations, and retest to assure compliance with the Contract Documents.

3.06 TESTING AND COMMISSIONING

A. Upon completion of the installation, prior to issue of taking over certificate, Contractor shall employ, at his own expense, a qualified structural Engineer, licensed/registered at the Place of the Work, to inspect and certify in writing the completion and safety of the BMU system installation and operation as specified.

B. The BMU systems shall be load tested, commissioned, and witnessed to the satisfaction of the Engineer on the performance as specified. Contractor shall submit a testing and commissioning schedule two months before the actual testing and commissioning works for the Engineer’s approval.

C. Test Reports: Certified test reports showing compliance with specified performance characteristics and physical properties by Government. Approved Third Party Authority-TUV or Equivalent as per EN1808.

D. The building maintenance equipment supplier shall include in the tender the following costs for witnessing of testing and commissioning of equipment by the Engineer’s and Employer’s representatives. A third party testing & commissioning as per Approved ITP to be carried out

through approved safety Engineer's (TUV or equivalent), approved by Municipality according to EN standards before commissioning & handing over subject to Engineer's approval

E. Submit project record ("as-built") drawings showing actual installed locations and configuration, and record specifications documenting all changes to original design criteria and other specification requirements.

F. Appropriate temporary protection of the installed equipment and its parts shall be provided by the BMU Sub Contractor to reduce the risk of accidental damage after installation prior to building handover.

G. Test also includes Restraint Socket load test from an independent certified third-party testing and commissioning agency.

3.07 EMPLOYERS STAFF TRAINING

A. Contract shall include for training Employer's staff by Building Maintenance Equipment's qualified technical personnel in approved training schedule regarding operation and proper use of all systems including the installation. Contractor shall provide the training schedule for approval three months prior to completion of works

B. Supplier / manufacturer to provide Clear training with Training Manual and Log book (containing all details of daily inspections requirements, persons who are using, duration of usage, etc.) as approved in prior by the Engineer. Supplier should ensure the operators are aware about what to do during any / all possible emergency situations and also provide certification for the personnel trained for usage of such equipments at the end of specific training sessions, during handover.

C. Contractor shall provide adequate training of the Employer's staff until they are fully familiar with the operation and maintenance of the complete installation. Any and all Keys / passwords for operation & maintenance of façade cleaning equipment to be handed over prior to training and a signature from relevant authorities to be obtained. Training shall include lectures supported by all major field demonstrations.

D. Building Maintenance Equipment training attendance certificates shall be issued by the Building Maintenance Equipment supplier to each of the Employer's trained personnel upon completion of training to certify that they are permitted/qualified to use the equipment under the training program provided by the Contractor.

3.08 MAINTENANCE

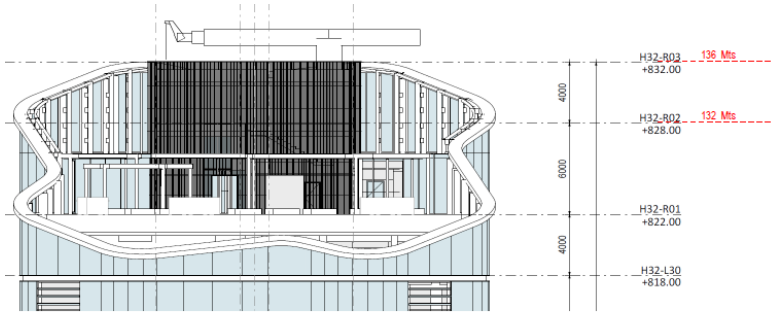
A. After completion of the installation and acceptance, a maintenance and call back service team for the equipment shall be provided for a period of Defects Liability Period (2 years).

B. This service shall include
   1. Furnishing complete service and maintenance of the system and components during the Defect Liability Period.
   2. Examining semi-monthly, clean, adjusts and lubricate all equipment as per manufacturer's instructions.
   3. Repairing or replacing parts whenever required. Use parts produced by the manufacturer of the original equipment.
   4. Maintaining locally, an adequate stock of parts for replacement or emergency purposes, and have qualified installation personnel available to ensure the fulfilment of this maintenance service without unreasonable loss of time.

5. Performing maintenance work using competent personnel, under the supervision of the system manufacturer.
6. Maintenance service shall not be assigned or transferred to any agent or sub-contractor without prior written consent of the Employer.

3.09 REFERENCE SKETCH



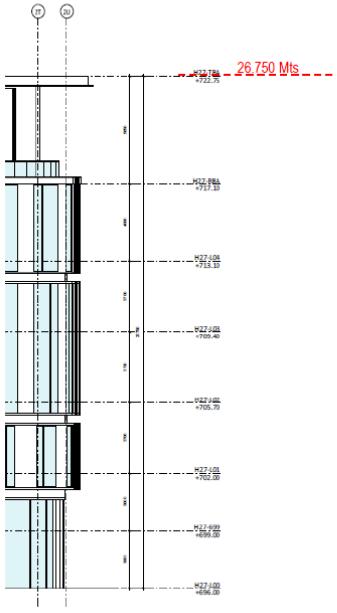


H32 BMU Location & Height

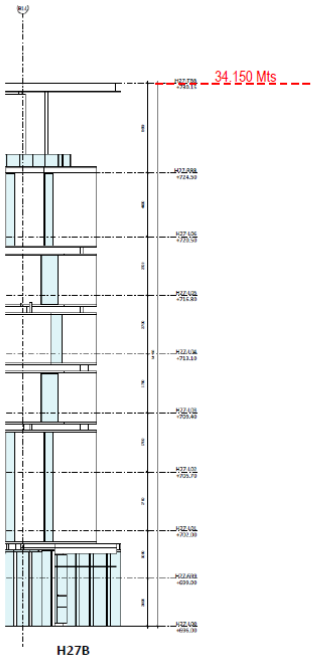
For 5-Star Hotel Buildings –H32

BMU Type	Centralized multistage telescopic arm BMU ~~with luffing function,~~ and fixed vertical mast, for the cleaning and replacement of the glass façade. Integrated with GRU.
Main supporting structure	Roof slab, on pedestals.
Cradle	2 men cradle, and accessories to replace the façade glass.
GRU capacity	360~400 kgs Approx., considering 2x3.6 m as the max. glass on the façade (SSH&WEA to note).
Total load	Contractor to assess (for cradle & lifting arrangement, SWL, tools etc.).
Transverse movement	N/A, fixed machine.
Max outreach	25~26 Mts (Approx.)
Mast height	5 Mts (Approx.)
Restraint socket	At every 2 m approx. horizontally, and vertically first at 40 m and rest at every 20 m interval from 40 m.
Code compliance	BS EN 1808 and related.
Finish	Architectural, color as per approval.
Parking	At roof, and not visible in the elevation.
Others	Refer to the drawing slides for more info. on building height, roof layout, machine location etc.





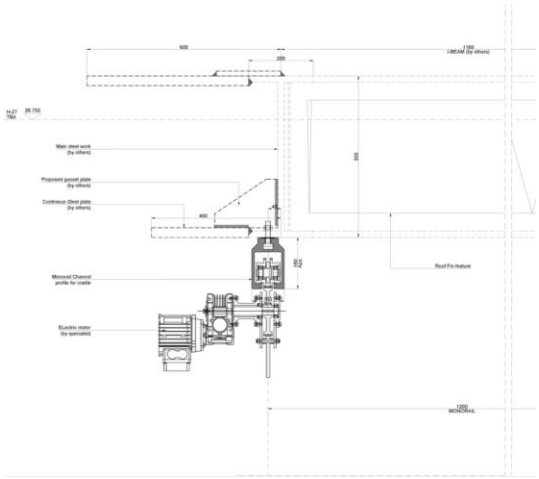
H27A
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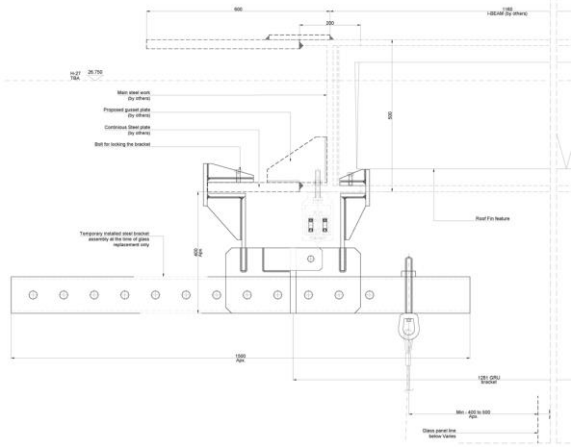


H27B
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H27C
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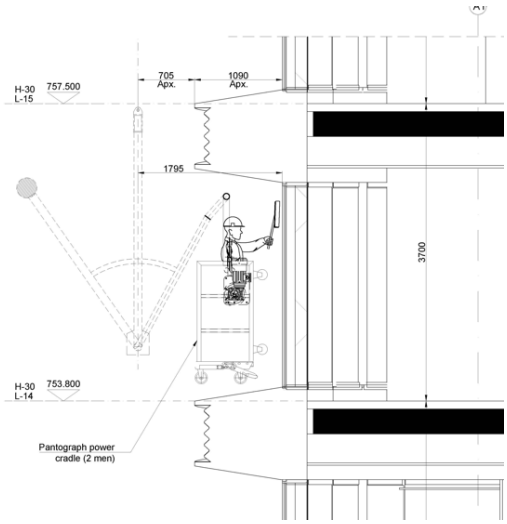




For Apartment Buildings_H27 (A,B & C)

BMU Type		Standard monorail capable of cleaning and replacement (activity) of the glass façade. Independent GRU hook, fixed to the roof canopy steel structure.
Main supporting structure		Continuous steel beam of the roof canopy.
Cradle		2 men power cradle. **Extended platform type**
GRU capacity	**For(H27 A) building height 26.750 Mts**	360~400 kgs Approx., considering 2x3.6 m as the max. glass on the façade (SSH&WEA to note).
	For(H27 B) building height 34.150 Mts	360~400 kgs Approx., considering 2x3.6 m as the max. glass on the façade (SSH&WEA to note).
	For(H27 C) building height 37.850 Mts	360~400 kgs Approx., considering 2x3.6 m as the max. glass on the façade (SSH&WEA to note).
Total load		Contractor to assess (for power cradle & lifting arrangement, SWL, tools, wire and cable etc.).
Transverse movement		Electrical.
Rail length	**For(H27 A) building height 26.750 Mts**	122 Mts
	For(H27 B) building height 34.150 Mts	122 Mts
	For(H27C) building height 37.850 Mts	122 Mts
Restraint socket		N/A
Code compliance		BS EN 1808 and related.
Finish		Architectural, color as per approval.
Parking		At roof, and not visible. Including monorail set-up.
Others		Refer to the drawing slides for more info. on building height, recess etc.

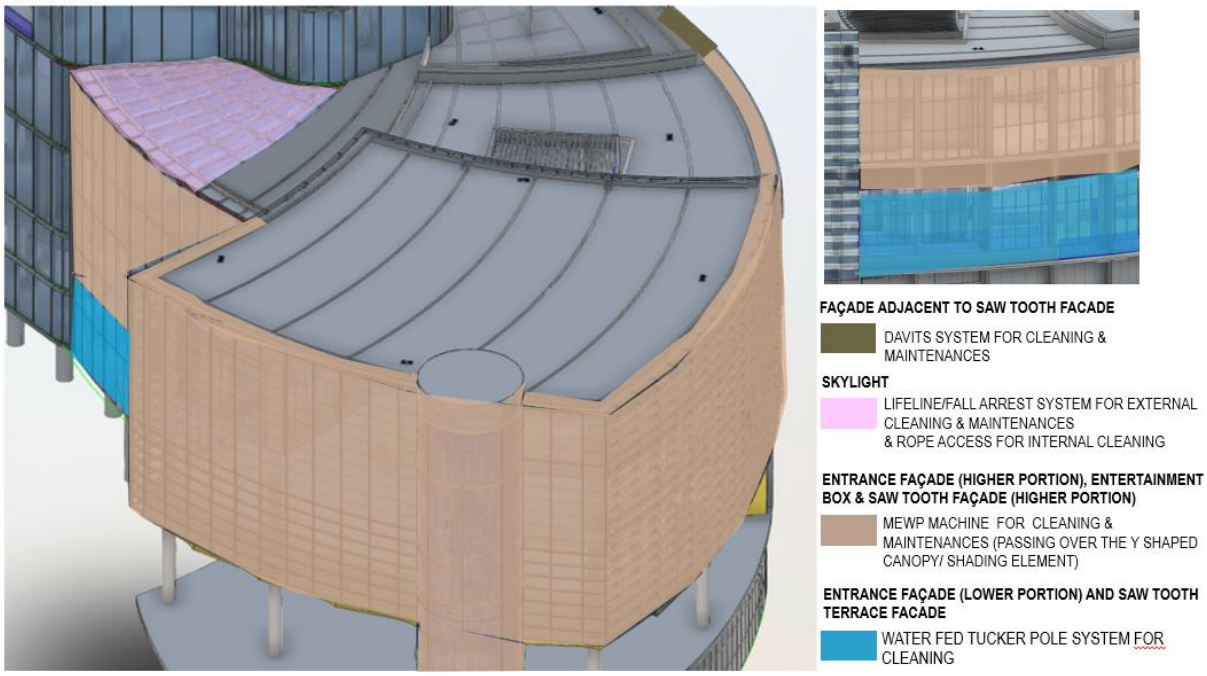




For 4-Star Hotel Buildings –H30

BMU Type		Custom made monorail capable of cleaning and replacement of the recessed glass façade. Integrated with GRU.
Main supporting structure		Continuous steel beam of the roof canopy.
Cradle		2 men pantograph power cradle, and accessories to replace the recessed glass.
GRU capacity	**For building height 68.165 Mts**	400~450 kgs Approx., considering 2.3x3.5 m as the max. glass on the façade (SSH&WEA to note).
	For building height 43.875 Mts	400~450 kgs Approx., considering 2.3x3.5 m as the max. glass on the façade (SSH&WEA to note).
Total load		Contractor to assess (for pantograph cradle & lifting arrangement, SWL, tools, wire and cable etc.).
Transverse movement		Electrical.
Rail length	**For building height 68.165 Mts**	124 Mts
Rail length	**For building height 43.875 Mts**	120 Mts
Restraint socket		N/A
Code compliance		BS EN 1808 and related.
Finish		Architectural, color as per approval.
Parking		At roof, and not visible. Including monorail set-up.
Others		Refer to the drawing slides for more info. on building height, recess etc.

BMU AT PODIUM



END OF SECTION 11 8123.01